THE GROMOV-HAUSDORFF DISTANCE BETWEEN ULTRAMETRIC SPACES: ITS STRUCTURE AND COMPUTATION*

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Abstract. The Gromov-Hausdorff distance (d_{GH}) provides a natural way of quantifying the dissimilarity between two given metric spaces. It is known that computing d_{GH} between two finite metric spaces is NP-hard, even in the case of finite ultrametric spaces which are highly structured metric spaces in the sense that they satisfy the so-called *strong triangle inequality*. Ultrametric spaces naturally arise in many applications such as hierarchical clustering, phylogenetics, genomics, and even linguistics. By exploiting the special structures of ultrametric spaces, (1) we identify a one parameter family $\{d_{\mathrm{GH}}^{(p)}\}_{p\in[1,\infty]}$ of distances defined in a flavor similar to the Gromov-Hausdorff distance on the collection of finite ultrametric spaces, and in particular $d_{\mathrm{GH}}^{(1)} = d_{\mathrm{GH}}$. The extreme case when $p = \infty$, which we also denote by u_{GH}, turns out to be an ultrametric on the collection of ultrametric spaces. Whereas for all $p \in [1,\infty)$, $d_{\mathrm{GH}}^{(p)}$ yields NP-hard problems, we prove that surprisingly u_{GH} can be computed in polynomial time. The proof is based on a structural theorem for u_{GH} established in this paper; (2) inspired by the structural theorem for u_{GH}, and by carefully leveraging properties of ultrametric spaces, we also establish a structural theorem for d_{GH} when restricted to ultrametric spaces. This structural theorem allows us to identify special families of ultrametric spaces on which d_{GH} is computationally tractable. These families are determined by properties related to the doubling constant of metric space. Based on these families, we devise a fixed-parameter tractable (FPT) algorithm for computing the exact value of d_{GH} between ultrametric spaces. We believe ours is the first such algorithm to be identified.

1 Introduction and main results

Edwards [14] and Gromov [17] independently introduced a notion nowadays called the *Gromov-Hausdorff distance* d_{GH} for comparing metric spaces. This distance enjoys many pleasing mathematical properties: if we let $\mathcal{M}$ denote the collection of all compact metric spaces, then modulo isometry, $(\mathcal{M}, d_{\mathrm{GH}})$ is a complete and separable metric space [35, Proposition 43], with rich pre-compact classes [18]. It has also recently been proved that

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this space is geodesic [20, 9]. This distance has been widely used in differential geometry [35], as a model for shape matching procedures [29, 5], and applied algebraic topology [8] for establishing stability properties of invariants.

Despite admitting many lower bounds which can be computed in polynomial time [8, 30], computing d_{GH} itself between arbitrary finite metric spaces leads to solving certain generalized quadratic assignment problems [30] which have been shown to be NP-hard [39, 40, 2]. In fact, in [40] Schmiedl proved the following stronger result (see however [28] for the case of point sets on the real line where the authors describe a poly time approximation algorithm):

Theorem 1 ([40, Corollary 3.8])**.** *The Gromov-Hausdorff distance cannot be approximated within any factor less than 3 in polynomial time, unless $\mathcal{P} = \mathcal{NP}$.*

The proof of this result reveals that the claim still holds even in the case of *ultrametric spaces*. An ultrametric space (X, d_X) is a metric space which satisfies the *strong triangle inequality*:

$$\forall x, x', x'' \in X, \text{ one has } d_X(x, x') \leq \max\left(d_X(x, x''), d_X(x'', x')\right).$$

In this paper, we will henceforth use u_X instead of d_X to represent an ultrametric. Ultrametric spaces appear in many applications: they arise in statistics as a geometric encoding of *dendrograms* [21, 7], in taxonomy and phylogenetics [41] as representations of phylogenies, and in linguistics [38]. In theoretical computer science, ultrametric spaces arise as building blocks for the probabilistic approximation of finite metric spaces [4].

In many of the aforementioned applications (including phylogenetics), in order to characterize the difference between relevant objects, it is important to compare ultrametric spaces via meaningful metrics. This is one of the main motivations behind our study of the Gromov-Hausdorff distance between ultrametric spaces.

Being a well understood and highly structured type of metric spaces, we are particularly interested in exploiting possible advantages associated to either *restricting* or *adapting* d_{GH} to the collection $\mathcal{U}^{\mathrm{fin}}$ of all finite ultrametric spaces. In this paper, we provide positive answers to the following two questions naturally arising from trying to bypass/overcome the hardness result in Theorem 1:

(Q1) Is there any suitable variant of the Gromov-Hausdorff distance on the collection of finite ultrametric spaces which can be approximated/computed in polynomial time?

(Q2) Is there any subcollection of ultrametric spaces on which the Gromov-Hausdorff distance can be approximated/computed in polynomial time?

In this paper we provide positive answers to these two questions and in the course of answering these questions, we establish structural theorems for both d_{GH} and a suitable *ultrametric* variant u_{GH} which in each case allow us to convert the problem of comparing two given spaces into instances of the problem on strictly smaller spaces.

Related work The Gromov-Hausdorff ultrametric, which we denote by u_{GH}, on the collection $\mathcal{U}$ of compact ultrametric spaces was first introduced by Zarichnyi [43] in 2005 as an ultrametric counterpart to d_{GH}. Moreover, the author proved that $(\mathcal{U}, u_{\mathrm{GH}})$ is a complete but not separable (ultra) metric space, where $\mathcal{U}$ denotes the collection of all compact ultrametric spaces. u_{GH} was further studied by Qiu in [37] where the author established several characterizations of u_{GH} similar to the classical ones for d_{GH} (cf. [6, Chapter 7]) such as those arising via the notions of ε-isometry and (ε, δ)-approximation. Qiu has also found a suitable version of Gromov's pre-compactness theorem for $(\mathcal{U}, u_{\mathrm{GH}})$.

Phylogenetic tree shapes (unlabled rooted trees) are closely related to ultrametric spaces. In [11], Colijn and Plazzotta studied a metric between tree shapes to compare evolutionary trees of influenza. In [27], Liebscher studied a class of metrics analogous to d_{GH} between unrooted phylogenetic trees. In [26], Lafond et al. extended different types of metrics on phylogenetic trees to metrics between tree shapes via optimization over permutations of labels. They studied the computational aspect of these metric extensions. In particular, they proved that computing the extension of the path distance is NP-complete via a similar argument used for proving that approximating d_{GH} between merge trees in NP-complete [2]. Moreover, they devised an FPT algorithm which computes the extension of the so-called Robinson-Foulds distances. Their FPT algorithm is a recursive algorithm comparing subtrees of nodes at each iteration, which is of similar flavor to our algorithms (Algorithms 2 and 4) for computing d_{GH} between ultrametric spaces.

In [15], Touli and Wang devised FPT algorithms for the computation of the interleaving distance d_{I} between merge trees [32]. Since any finite ultrametric space can be naturally represented by a merge tree (see for example [16]) it turns out that d_{I} between ultrametric spaces as merge trees is a 2-approximation of d_{GH} between the ultrametric spaces (see [31, Corollary 6.13]). Thus, one could potentially adapt the algorithm from [15] for computing a 2-approximation for d_{GH} between ultrametric spaces, which is FPT. In this paper we obtain essentially the same time complexity for the *exact* computation of d_{GH} (see Remark 49) via algorithms specifically tailored for ultrametric spaces.

1.1 Our results

In this section, we summarize our main results obtained in the course of answering the two major questions mentioned above.

1.1.1 Polynomial time computable variant of d_{GH}

Let (X, d_X) and (Y, d_Y) be two metric spaces. A *correspondence* R between the underlying sets X and Y is any subset of $X \times Y$ such that the images of R under the canonical projections

$p_X : X \times Y \to X$ and $p_Y : X \times Y \to Y$ are full: $p_X(R) = X$ and $p_Y(R) = Y$. Then, the Gromov-Hausdorff distance d_{GH} between X and Y is defined as follows [29]:

$$d_{\mathrm{GH}}(X,Y) := \frac{1}{2} \inf_{R} \sup_{(x,y),(x',y') \in R} |d_X(x,x') - d_Y(y,y')|, \tag{1}$$

where the infimum is taken over all correspondences R between X and Y.

We modify Equation (1) to obtain a one-parameter family of related quantities:

Definition 2 (p-Gromov-Hausdorff distance)**.** *Given $p \in [1,\infty)$ and any two metric spaces X and Y, we define a quantity $d_{\mathrm{GH}}^{(p)}(X,Y)$ which we call the p-Gromov-Hausdorff distance as follows:*

$$d_{\mathrm{GH}}^{(p)}(X,Y) := 2^{-\frac{1}{p}} \inf_{R} \sup_{(x,y),(x',y') \in R} \left| (d_X(x,x'))^p - (d_Y(y,y'))^p \right|^{\frac{1}{p}}. \tag{2}$$

In this way, as p increases, the discrepancy between large distance values is more heavily penalized. It turns out that for each $p \in [1,\infty)$, $d_{\mathrm{GH}}^{(p)}$ is a metric on the collection $\mathcal{U}$ of all compact ultrametric spaces and this justifies the name "p-Gromov-Hausdorff distance". Moreover, we will later show as a consequence of Theorem 1 the following as one of our motivations of considering $d_{\mathrm{GH}}^{(p)}$ (recall that $\mathcal{U}^{\mathrm{fin}}$ below denotes the collection of all *finite* ultrametric spaces):

Corollary 3. *For each $p \in [1,\infty)$ and for any $X, Y \in \mathcal{U}^{\mathrm{fin}}$, $d_{\mathrm{GH}}^{(p)}(X,Y)$ cannot be approximated within any factor less than $3^{\frac{1}{p}}$ in polynomial time, unless $\mathcal{P} = \mathcal{NP}$.*

Note that the factor $3^{\frac{1}{p}}$ approaches 1 as $p \to \infty$. This suggests us considering $d_{\mathrm{GH}}^{(\infty)} := \lim_{p\to\infty} d_{\mathrm{GH}}^{(p)}$, which could potentially be a computationally tractable quantity. Before stating our computational result for $d_{\mathrm{GH}}^{(\infty)}$, it is worth mentioning that $d_{\mathrm{GH}}^{(\infty)}$ turns out to be an ultrametric on $\mathcal{U}$. Moreover, it actually coincides with the Gromov-Hausdorff ultrametric defined by Zarichnyi [43], which we denote by u_{GH}; see Equation (11) for the definition of u_{GH} and see Theorem 29 for the equivalence between u_{GH} and $d_{\mathrm{GH}}^{(\infty)}$. In this paper, we will hence use the two notations u_{GH} and $d_{\mathrm{GH}}^{(\infty)}$ interchangeably.

One of our main contributions in the paper is the following structural characterization of u_{GH}. This structural result eventually leads to a polynomial time computable algorithm for computing u_{GH} which we will state later.

Theorem 4 (Structural theorem for u_{GH})**.** *For any $X, Y \in \mathcal{U}^{\mathrm{fin}}$ one has that*

$$u_{\mathrm{GH}}(X,Y) = \min \left\{ t \geq 0 : \left(X_{\mathfrak{c}(t)}, u_{X_{\mathfrak{c}(t)}}\right) \text{ is isometric to } \left(Y_{\mathfrak{c}(t)}, u_{Y_{\mathfrak{c}(t)}}\right) \right\}.$$

Here $\left(X_{\mathfrak{c}(t)}, u_{X_{\mathfrak{c}(t)}}\right)$ is the *t-closed quotient* of X where x and x' are identified if $u_X(x,x') \leq t$ (cf. Definition 10). In terms of dendrograms (cf. Section 2.2), intuitively, $X_{\mathfrak{c}(t)}$ is obtained by coarsening the dendrogram of X *below* height t. See Figure 1 for an illustration of both the closed quotient operation and Theorem 4.

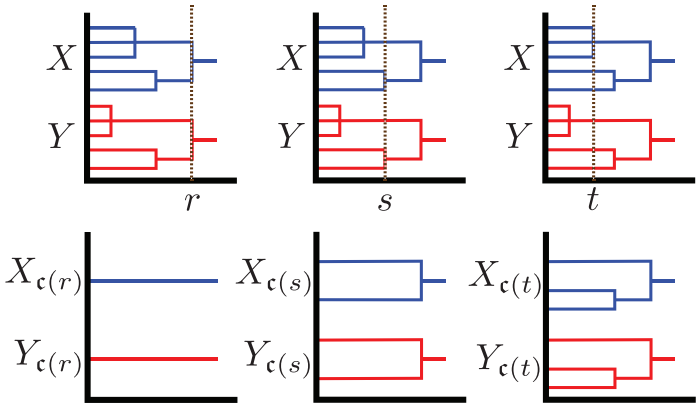


Figure 1: **Illustration of Theorem 4.** We represent two ultrametric spaces X and Y as dendrograms (see Theorem 15 for more details) on the first row. Imagine that we move a vertical dotted line from *right to left* ($r > s > t$) to obtain successive quotient spaces (shown on the second row) according to the parameter indicated by the line, as described in Definition 10. It is obvious from the figure that ($\cong$ denotes isometry) $X_{\mathfrak{c}(r)} \cong Y_{\mathfrak{c}(r)}$, $X_{\mathfrak{c}(s)} \cong Y_{\mathfrak{c}(s)}$, $X_{\mathfrak{c}(t)} \cong Y_{\mathfrak{c}(t)}$, and that t is the minimum value such that the resulting quotient spaces are isometric. Thus, $u_{\mathrm{GH}}(X, Y) = t$.

Based on Theorem 4, we devise an algorithm for computing u_{GH} as follows. For a finite ultrametric space X, the isometry type of $X_{\mathfrak{c}(t)}$ only changes finitely many times along $0 \leq t < \infty$. In fact, the set of all ts when $X_{\mathfrak{c}(t)}$ changes its isometry type is exactly the *spectrum* $\mathrm{spec}(X) := \{u_X(x, x') : x, x' \in X\}$ of X. Then, in order to compute $u_{\mathrm{GH}}(X, Y)$, we simply check whether $X_{\mathfrak{c}(t)} \cong Y_{\mathfrak{c}(t)}$, starting from the largest t and progressively scanning all possible ts until reaching the smallest t in $\mathrm{spec}(X) \cup \mathrm{spec}(Y)$; the smallest t such that $X_{\mathfrak{c}(t)} \cong Y_{\mathfrak{c}(t)}$ will be $u_{\mathrm{GH}}(X, Y)$. Since ultrametric spaces can be regarded as weighted trees (cf. Section A.1), determining whether two ultrametric spaces are isometric is equivalent to determining whether two weighted trees are isomorphic, which can be achieved in polynomial time on the number of vertices involved.

We prove that computing u_{GH} can be done in time $O(n \log(n))$ where n is the maximum of the cardinalities of X and Y (cf. Theorem 32 and Remark 33). See Section 4 for the pseudocode (cf. Algorithm 1) of the algorithm described above and a detailed complexity analysis, and also see Appendix B for an extension of u_{GH} to the case of ultra-dissimilarity spaces. We also remark that our computational results regarding the determination of u_{GH} (between finite ultrametric spaces) can be interpreted as providing a novel computationally tractable instance of the well known quadratic assignment problem (cf. Remark 34).

In the end, we summarize our complexity results in Figure 2.

1.1.2 Polynomial time computable family with respect to d_{GH}

Inspired by the usefulness of Theorem 4 for u_{GH}, we exploit special properties of ultrametric spaces and establish a structural theorem for d_{GH} between ultrametric spaces (Theorem 5).

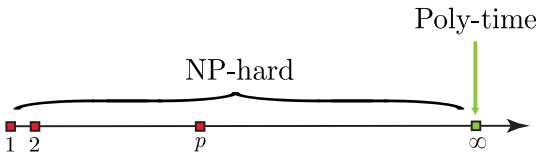


Figure 2: **Complexity of $d_{\mathrm{GH}}^{(p)}$.** Computing/approximating $d_{\mathrm{GH}}^{(p)}$ is NP-hard for each $p \in [1, \infty)$ whereas computing $d_{\mathrm{GH}}^{(\infty)}$ can be done in polynomial time.

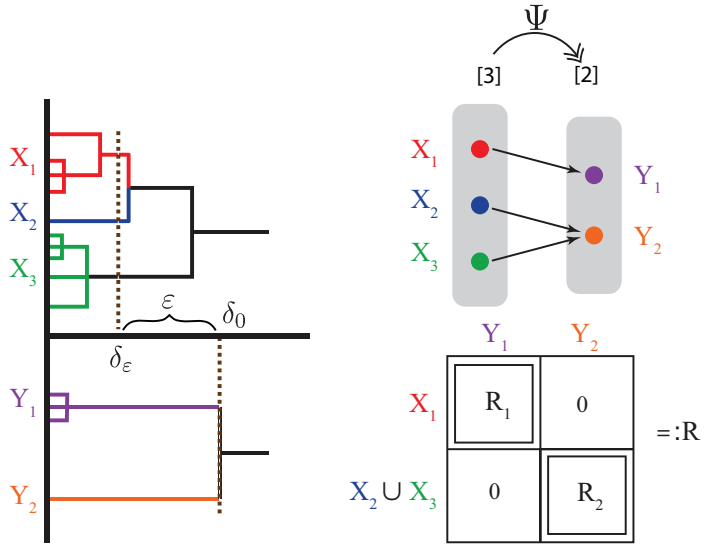


Figure 3: **Illustration of the claim in Theorem 5.** Note that here δ_0 is a shorthand notation for $\delta_0(Y) = \mathrm{diam}(Y)$ and similarly, $\delta_\varepsilon := \delta_\varepsilon(Y)$.

See also Figure 3 for an illustration of Theorem 5.

Below, $X_{\mathfrak{o}(t)}$ denotes the *t-open partition* of X where x and x' belong to the same block if $u_X(x, x') < t$ (cf. Definition 12). Intuitively speaking, this partition is obtained by slicing the dendrogram of X (cf. Section 2.2 and Figure 5) at a height sufficiently close to but strictly smaller than t. We call any correspondence R between X and Y with 1−distortion (cf. Section 3) bounded above by $\varepsilon \geq 0$ an *ε-correspondence* (see Definition 18). Given a metric space X and $\varepsilon \geq 0$, we let $\delta_\varepsilon(X) := \mathrm{diam}(X) - \varepsilon$, where $\mathrm{diam}(X)$ denotes the diameter of X. With this notation, $\delta_0(X) = \mathrm{diam}(X)$. Sometimes we will write $\mathrm{diam}(X)$ instead of $\delta_0(X)$ in order to avoid cumbersome superscripts/subscripts.

Theorem 5 (Structural theorem for d_{GH})**.** *Let $X, Y \in \mathcal{U}^{\mathrm{fin}}$ and $\varepsilon \geq 0$ be such that*

$$|\mathrm{diam}(X) - \mathrm{diam}(Y)| \leq \varepsilon < \mathrm{diam}(Y). \tag{3}$$

Consider the following open partitions

$$X_{\mathfrak{o}(\delta_\varepsilon(Y))} := \{X_i\}_{i=1}^{N_X} \quad \textit{and} \quad Y_{\mathfrak{o}(\delta_0(Y))} := \{Y_j\}_{j=1}^{N_Y}.$$

Then, there exists an ε-correspondence between X and Y if and only if:

(1) there exists a surjection $\Psi : [N_X] \twoheadrightarrow [N_Y]$ and, with this surjection,

(2) for every $j \in [N_Y]$ there exists an ε-correspondence between $\left(X_{\Psi^{-1}(j)}, u_X|_{X_{\Psi^{-1}(j)} \times X_{\Psi^{-1}(j)}}\right)$ and $\left(Y_j, u_Y|_{Y_j \times Y_j}\right)$ where for each $j \in [N_Y]$, $X_{\Psi^{-1}(j)} \coloneqq \bigcup_{i \in \Psi^{-1}(j)} X_i$.

Remark 6 (Interpretation of Equation (3))**.** *Note that for any correspondence R between X and Y, the relations $|\mathrm{diam}(X) - \mathrm{diam}(Y)| \leq \mathrm{dis}_1(R) \leq \max(\mathrm{diam}(X), \mathrm{diam}(Y))$ always hold (here $\mathrm{dis}_1(R)$ denotes the 1-distortion of R; see Section 3 for its definition and see Proposition 24 for the proof of these inequalities.). Therefore, (1) If $|\mathrm{diam}(X) - \mathrm{diam}(Y)| > \varepsilon$, then there exists no ε-correspondence between X and Y; (2) If $\max(\mathrm{diam}(X), \mathrm{diam}(Y)) \leq \varepsilon$, then every correspondence R between X and Y is an ε-correspondence. In this way, in order to analyze existence of ε-correspondence we only need to consider the case when*

$$|\mathrm{diam}(X) - \mathrm{diam}(Y)| \leq \varepsilon < \max(\mathrm{diam}(X), \mathrm{diam}(Y)). \tag{4}$$

Therefore, Equation (3) is simply an asymmetric variant of Equation (4).

Remark 7. *In the course of proving Theorem 5 (cf. Section 5.1), we actually establish the following result: under the assumption that a surjection $\Psi : [N_X] \twoheadrightarrow [N_Y]$ and an ε-correspondence R_j between $X_{\Psi^{-1}(j)}$ and Y_j for each $j \in [N_Y]$ as above all exist, the set*

$$R \coloneqq \bigcup_{j \in [N_Y]} R_j$$

is an explicit ε-correspondence between X and Y. This fact will be used in Algorithms 2 and 4.

The structural theorem for d_{GH} is 'anatomically' similar to the structural theorem for u_{GH} in that, in some sense, it converts the problem related to comparing two spaces into smaller problems related to comparing subspaces. This naturally suggests considering a *divide-and-conquer strategy* for devising a recursive algorithm (Algorithm 2) for (asserting the existence of and) finding an ε-correspondence between two given ultrametric spaces. It turns out that the recursive algorithm performs many repetitive computations, so we further improve this strategy via a dynamic programming (DP) idea to obtain a more efficient algorithm (Algorithm 4). See Section 5 for a detailed description of both algorithms.

One key factor which will influence the complexity of either the recursive or the DP algorithm is the *size* of the subproblems. By exploiting the inherent tree-like structure of ultrametric spaces, we identify in Definition 35 the first (ε, γ)-growth condition (FGC) which suitably quantifies the structural complexity of ultrametric spaces and thus controls the size of the subproblems in our recursive algorithm. When two ultrametric spaces satisfy the FGC for some fixed parameters, the recursive algorithm (Algorithm 2) is proved to run in polynomial time (Theorem 37).

A similar but more general second (ε, γ)-growth condition (SGC) is identified in Definition 42 for the DP algorithm (Algorithm 4). If we denote by $\mathcal{U}_2(\varepsilon, \gamma)$ the collection

of all finite ultrametric spaces satisfying the second (ε,γ)-growth condition, then for any $X,Y\in\mathcal{U}_2(\varepsilon,\gamma)$, we can determine whether $d_{\mathrm{GH}}(X,Y)\leq\frac{\varepsilon}{2}$ in time $O\left(n^2\log(n)2^{\gamma}\gamma^{\gamma+2}\right)$, where $n:=\max(\#X,\#Y)$ (cf. Theorem 46); and under the assumption $2d_{\mathrm{GH}}(X,Y)\leq\varepsilon$, we can compute the exact value of $d_{\mathrm{GH}}(X,Y)$ in time $O\left(n^4\log(n)2^{\gamma}\gamma^{\gamma+2}\right)$ (cf. Theorem 47). In particular, this implies that our DP algorithm is *fixed-parameter tractable* (FPT) with respect to the parameters given in the SGC.

Based on our algorithms for computing d_{GH} between ultrametric spaces, we further establish an FPT algorithm to additively approximate d_{GH} between arbitrary doubling (non necessary ultra) metric spaces which are themselves quantitatively close to being ultrametric spaces (cf. Corollary 57). One of the key observations leading to this approximation algorithm is the 'transfer' of the doubling condition on a metric space into the satisfaction of the second growth condition by its corresponding single-linkage ultrametric space (cf. Lemma 54).

Implementations The GitHub repository [1] provides implementations of some of our algorithms as well as an experimental demonstration.

1.2 Organization of the paper

In Section 2 we review facts about ultrametric spaces and dendrograms, and introduce the quotient operations mentioned above. In Section 3 we discuss the p-Gromov-Hausdorff distance $d_{\mathrm{GH}}^{(p)}$ and connect $d_{\mathrm{GH}}^{(\infty)}$ with the Gromov-Hausdorff ultrametric u_{GH}. In Section 4 we prove Theorem 4 and provide details of an algorithm (Algorithm 1) for computing u_{GH}. In Section 5 we prove Theorem 5 and discuss how to utilize Theorem 5 for devising algorithms (Algorithms 2 and 4) computing d_{GH}. In Appendix A we specify the data structure for ultrametric spaces used in algorithms throughout the paper. In Appendix B we provide details for generalizing u_{GH} to the so-called ultra-dissimilarity spaces. Some proofs are relegated to Appendix C.

List of Symbols

The list below describes several symbols that will be used in the article.

d_{GH}	The Gromov-Haussdorff distance; page 17.
u_{GH}	The Gromov-Haussdorff ultrametric; page 17.
$d_{\mathrm{GH}}^{(p)}$	p-Gromov-Hausdorff distance; page 14.
dis_p	p-distortion; page 13.
Λ_p	Absolute p-difference; page 13.
$\mathcal{U}$	The collection of all compact ultrametric spaces.
$\mathcal{U}^{\mathrm{fin}}$	The collection of all finite ultrametric spaces.

(X, u_X) An ultrametric space.

$[x]^X_{\mathfrak{c}(t)}$ The equivalence class of the closed equivalence relation; page 10.

$[x]^X_{\mathfrak{o}(t)}$ The equivalence class of the open equivalence relation; page 11.

$\left(X_{\mathfrak{c}(t)}, u_{X_{\mathfrak{c}(t)}}\right)$ The t-closed quotient of the ultrametric space X; page 10.

$X_{\mathfrak{o}(t)}$ The t-open partition of the ultrametric space X; page 11.

$[x]^{\theta_X}_t$ The block in $\theta_X(t)$ containing x; page 12.

$\#X$ The cardinality of set X.

$\delta_\varepsilon(X)$ A convenient symbol representing $\delta_\varepsilon(X) \coloneqq \operatorname{diam}(X) - \varepsilon$; page 6.

$\delta^+_{2\varepsilon}(X)$ A convenient symbol representing $\delta^+_{2\varepsilon}(X) \coloneqq \max(\delta_{2\varepsilon}(X), 0) = \max(\operatorname{diam}(X) - 2\varepsilon, 0)$; page 26.

$\mathcal{U}_1(\varepsilon, \gamma)$ The collection of all finite ultrametric spaces satisfying the first (ε, γ)-growth condition; page 23.

$\mathcal{U}_2(\varepsilon, \gamma)$ The collection of all finite ultrametric spaces satisfying the second (ε, γ)-growth condition; page 28.

$\operatorname{sep}(X)$ The separation of X; page 32.

$\operatorname{spec}(X)$ The spectrum of a metric space X: $\operatorname{spec}(X) \coloneqq \{d_X(x, x') : x, x' \in X\}$; page 4.

θ_X The dendrogram corresponding to the ultrametric space X; page 11.

$B_{(\varepsilon)}$ The set of ε-maximal unions of closed balls of the ball B; page 26.

T_X The weighted rooted tree corresponding to X and also its associated TDS; page 41.

V_X The set of all closed balls in the ultrametric space X; page 26.

2 Ultrametric spaces

Ultrametric spaces, as defined in the introduction, are metric spaces which satisfy the strong triangle inequality. The following basic properties of ultrametric spaces are direct consequences of the strong triangle inequality.

Proposition 8 (Basic properties of ultrametric spaces)**.** *Let X be an ultrametric space. Then, X satisfies the following basic properties:*

1. *(*Isosceles triangles*) Any three distinct points $x, x', x'' \in X$ constitute an isosceles triangle, i.e., two of $u_X(x, x'), u_X(x, x'')$ and $u_X(x', x'')$ are the same and are greater than the rest.*

2. *(*Center of closed balls*) Let* $B_t(x) := \{x' \in X : u_X(x,x') \leq t\}$ *denote the closed ball centered at* $x \in X$ *with radius* $t \geq 0$*. Then, for any* $x' \in B_t(x)$ *we have that* $B_t(x') = B_t(x)$*.*

3. *(*Relation between closed balls*) For any two closed balls* B *and* B' *in* X*, if* $B \cap B' \neq \emptyset$*, then either* $B \subseteq B'$ *or* $B' \subseteq B$*.*

4. *(*Cardinality of spectrum*) Suppose* X *is a finite space. Then,* $\#\mathrm{spec}(X) \leq \#X$*.*

Proof. The first three items are well known (and easy to prove) and we omit their proof. As for the fourth item, see for example [19, Corollary 3]. □

Next, we introduce two important notions for ultrametric spaces: quotient operations and dendrograms.

2.1 Quotient operations

There are two special equivalence relations on ultrametric spaces whose respectively induced quotient operations will be helpful in revealing the structure of both u_{GH} and d_{GH}.

A 'closed' equivalence relation For any ultrametric space (X, u_X), we introduce a relation $\sim_{\mathfrak{c}(t)}$ on X such that $x \sim_{\mathfrak{c}(t)} x'$ iff $u_X(x,x') \leq t$. Due to the strong triangle inequality, $\sim_{\mathfrak{c}(t)}$ is an equivalence relation which we call the *closed equivalence relation*. For each $x \in X$ and $t \geq 0$, denote by $[x]^X_{\mathfrak{c}(t)}$ the equivalence class of x under $\sim_{\mathfrak{c}(t)}$. We abbreviate $[x]^X_{\mathfrak{c}(t)}$ to $[x]_{\mathfrak{c}(t)}$ whenever the underlying set is clear from the context. Consider the set $X_{\mathfrak{c}(t)} := \{[x]_{\mathfrak{c}(t)} : x \in X\}$ of all $\sim_{\mathfrak{c}(t)}$ equivalence classes.

Remark 9 (Relationship with closed balls)**.** *Note that for each* $x \in X$*, the equivalence class* $[x]^X_{\mathfrak{c}(t)}$ *satisfies* $[x]^X_{\mathfrak{c}(t)} = \{x' \in X : u_X(x,x') \leq t\}$*. This implies that* $[x]_{\mathfrak{c}(t)}$ *coincides with the closed ball* $B_t(x) := \{x' \in X : u_X(x,x') \leq t\}$*. We will henceforth use both notation* $[x]_{\mathfrak{c}(t)}$ *and* $B_t(x)$ *to represent closed balls interchangeably.*

Now, we introduce a function $u_{X_{\mathfrak{c}(t)}} : X_{\mathfrak{c}(t)} \times X_{\mathfrak{c}(t)} \to \mathbb{R}_{\geq 0}$ as follows:

$$u_{X_{\mathfrak{c}(t)}}\left([x]_{\mathfrak{c}(t)}, [x']_{\mathfrak{c}(t)}\right) := \begin{cases} u_X(x,x') & \text{if } [x]_{\mathfrak{c}(t)} \neq [x']_{\mathfrak{c}(t)} \\ 0 & \text{if } [x]_{\mathfrak{c}(t)} = [x']_{\mathfrak{c}(t)}. \end{cases} \tag{5}$$

It is clear that $u_{X_{\mathfrak{c}(t)}}$ is an ultrametric on $X_{\mathfrak{c}(t)}$.

Definition 10 (t-closed quotient)**.** *For any ultrametric space* (X, u_X) *and* $t \geq 0$*, we call* $\left(X_{\mathfrak{c}(t)}, u_{X_{\mathfrak{c}(t)}}\right)$ *the* t-closed quotient of X*.*

For each $t \geq 0$, the t-closed quotient gives rise to a map which we call the *t-closed quotient operator* $Q_{\mathfrak{c}(t)} : \mathcal{U}^{\mathrm{fin}} \to \mathcal{U}^{\mathrm{fin}}$ sending $X \in \mathcal{U}^{\mathrm{fin}}$ to $X_{\mathfrak{c}(t)} \in \mathcal{U}^{\mathrm{fin}}$.

An 'open' equivalence relation Given an ultrametric space X and $t > 0$, let $\sim_{\mathfrak{o}(t)}$ be such that $x \sim_{\mathfrak{o}(t)} x'$ if $u_X(x, x') < t$. Due to the strong triangle inequality again, $\sim_{\mathfrak{o}(t)}$ is an equivalence relation on X which we call the *open equivalence relation*. Its difference with the closed equivalence relation $\sim_{\mathfrak{c}(t)}$ is that we now require a *strict* inequality for defining the equivalence relation. Denote by $[x]^X_{\mathfrak{o}(t)}$ the equivalence class of $x \in X$ under $\sim_{\mathfrak{o}(t)}$. We will use the simpler notation $[x]_{\mathfrak{o}(t)}$ instead of $[x]^X_{\mathfrak{o}(t)}$ when the underlying set is clear from the context.

Remark 11 (Relation with open and closed balls)**.** *In the same way that $[x]_{\mathfrak{c}(t)}$ is the closed ball centered at x with radius t, when $t > 0$ $[x]_{\mathfrak{o}(t)} = \{x' \in X : u_X(x, x') < t\}$ is actually the open ball centered at x with radius t. If X is finite, then each open ball is actually a closed ball: for any open ball $[x]_{\mathfrak{o}(t)}$, we have $[x]_{\mathfrak{o}(t)} = [x]_{\mathfrak{c}(t')}$, where $t' := \operatorname{diam}([x]_{\mathfrak{o}(t)})$.*

Now in analogy with Equation (5), we introduce an ultrametric $u_{X_{\mathfrak{o}(t)}}$ on $X_{\mathfrak{o}(t)}$ as follows:

$$u_{X_{\mathfrak{o}(t)}}\left([x]_{\mathfrak{o}(t)}, [x']_{\mathfrak{o}(t)}\right) := \begin{cases} u_X(x, x') & \text{if } [x]_{\mathfrak{o}(t)} \neq [x']_{\mathfrak{o}(t)} \\ 0 & \text{if } [x]_{\mathfrak{o}(t)} = [x']_{\mathfrak{o}(t)}. \end{cases} \tag{6}$$

Definition 12 (t-open quotient)**.** *For any ultrametric space (X, u_X) and any $t > 0$, we call $(X_{\mathfrak{o}(t)}, u_{X_{\mathfrak{o}(t)}})$ the t-open quotient of X. When $t = 0$, by definition we let $\sim_{\mathfrak{o}(0)} := \sim_{\mathfrak{c}(0)}$ and let $(X_{\mathfrak{o}(0)}, u_{X_{\mathfrak{o}(0)}}) := (X, u_X)$ be the 0-open quotient of X.*

Given a finite set X, a set $P = \{B_1, \ldots, B_n\}$ of non-empty subsets of X is called a *partition* of X if $\bigcup_{i=1}^n B_i = X$ and $B_i \cap B_j = \emptyset$ if $i \neq j$. It is well known that any equivalence relation on a given set induces a partition of that set. For the open equivalence relation, instead of the metric $u_{X_{\mathfrak{o}(t)}}$ we will mainly focus on the partition induced by $\sim_{\mathfrak{o}(t)}$, i.e., the partition $\left\{[x]_{\mathfrak{o}(t)} : x \in X\right\}$. We call this partition the *t-open partition* of X.

Example 13 (t-closed and open quotients when $t =$ diameter)**.** *Let X be a finite ultrametric space with at least two points and let $\delta := \operatorname{diam}(X)$. Then, $X_{\mathfrak{c}(\delta)} = \{X\}$ is the one point space whereas $\#X_{\mathfrak{o}(\delta)} > 1$. Indeed, for any point $x \in X$, $X = [x]_{\mathfrak{c}(\delta)}$ and thus $X_{\mathfrak{c}(\delta)} = \{X\}$; since X is finite and $\#X \geq 2$, there exist $x, x' \in X$ such that $u_X(x, x') = \operatorname{diam}(X) = \delta$, then $[x]_{\mathfrak{o}(\delta)} \neq [x']_{\mathfrak{o}(\delta)}$ and thus $\#X_{\mathfrak{o}(\delta)} \geq \#\{[x]_{\mathfrak{o}(\delta)}, [x']_{\mathfrak{o}(\delta)}\} > 1$.*

2.2 Dendrograms

One essential mental picture to evoke when thinking about ultrametric spaces is that of a *dendrogram* (see Figure 4). To proceed with the definition of dendrograms, we first introduce some related terminology.

Partitions Given any finite set X and a partition $P = \{B_1, \ldots, B_n\}$ of X, we call each $B_i \in P$ a *block* of P. We denote by $\operatorname{Part}(X)$ the collection of all partitions of X. Given two partitions $P_1, P_2 \in \operatorname{Part}(X)$, we say that P_1 is a *refinement* of P_2, or equivalently, that P_2 is *coarser* than P_1, if every block in P_1 is contained in some block in P_2.

Definition 14 (Dendrograms, [7])**.** *A dendrogram θ_X over a finite set X is any function $\theta_X : [0, \infty) \to \operatorname{Part}(X)$ satisfying the following conditions:*

(1) $\theta_X(0) = \{\{x_1\}, \ldots, \{x_n\}\}$.

(2) For any $s < t$, $\theta_X(s)$ *is a refinement of* $\theta_X(t)$.

(3) There exists $t_X > 0$ *such that* $\theta_X(t_X) = \{X\}$.

(4) For any $r \geq 0$, *there exists* $\varepsilon > 0$ *such that* $\theta_X(r) = \theta_X(t)$ *for* $t \in [r, r + \varepsilon]$.

There exists a close relationship between dendrograms and ultrametric spaces. Fix a finite set X, by $\mathcal{U}(X)$ denote the collection of all ultrametrics over X and by $\mathcal{D}(X)$ denote the collection of all dendrograms over X. We define a map $\Delta_X : \mathcal{U}(X) \to \mathcal{D}(X)$ by sending u_X to a dendrogram θ_X as follows via the closed quotient: given $t \geq 0$, we let $\theta_X(t) \coloneqq X_{\mathfrak{c}(t)} = \{[x]_{\mathfrak{c}(t)} : x \in X\}$. It turns out that the map Δ_X is bijective. In fact, the inverse $\Upsilon_X : \mathcal{D}(X) \to \mathcal{U}(X)$ of Δ_X is the following map: for any dendrogram θ_X, $u_X \coloneqq \Upsilon(\theta_X)$ is defined by $u_X(x, x') \coloneqq \inf\{t \geq 0 : [x]_t^{\theta_X} = [x']_t^{\theta_X}\}$ for any $x, x' \in X$, where $[x]_t^{\theta_X} \in \theta_X(t)$ denotes the block containing x. It turns out that $[x]_t^{\theta_X}$ coincides with the equivalence class $[x]_{\mathfrak{c}(t)}$ of the closed equivalence relation with respect to $u_X = \Upsilon_X(\theta_X)$. Hence, we also use either $[x]_{\mathfrak{c}(t)}^X$ or $[x]_{\mathfrak{c}(t)}$ to represent the block containing x in a dendrogram θ_X at level t. We summarize our discussion above into the following theorem.

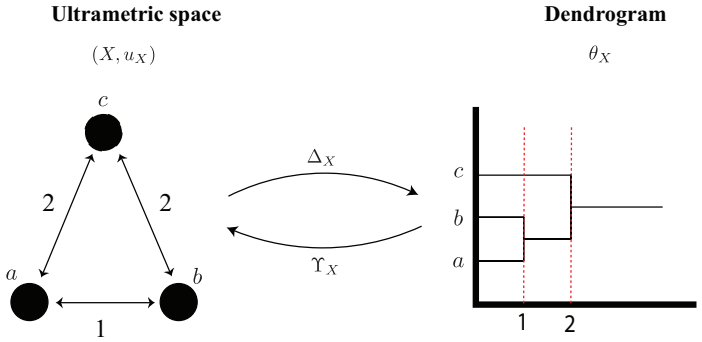


Figure 4: **Transforming ultrametric spaces into dendrograms.**

Theorem 15 (Dendrograms as ultrametric spaces, [7, Theorem 9])**.** *Given a finite set* X, *then* $\Delta_X : \mathcal{U}(X) \to \mathcal{D}(X)$ *is bijective with inverse* $\Upsilon_X : \mathcal{D}(X) \to \mathcal{U}(X)$.

Theorem 15 above establishes that dendrograms and ultrametric spaces are equivalent concepts – a point of view which helps to formulate subsequent ideas in this paper.

Example 16 (t-closed/open quotient in terms of dendrograms)**.** *It is helpful to understand both the open and closed* t*-quotients by viewing ultrametric spaces as dendrograms: both* t*-quotients simply forget the details of a given dendrogram strictly below scale* t. *Whereas the* t*-open equivalence relation retains the partition information at scale* t, *the* t*-closed equivalence does not. See Figure 5 for an illustration.*

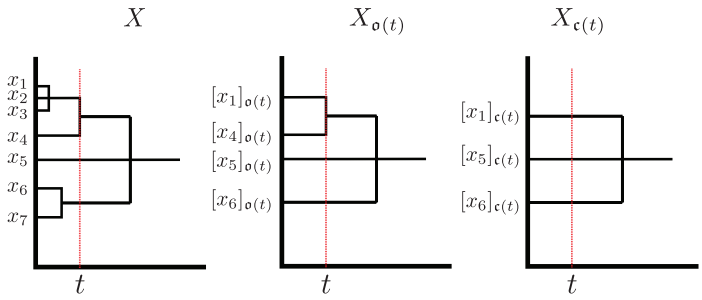


Figure 5: **Illustration of open and closed equivalence relations.** The leftmost figure is a dendrogram representing a 7-point ultrametric space X. The middle figure is the dendrogram corresponding to $X_{\mathfrak{o}(t)}$ whereas the rightmost figure is the dendrogram corresponding to $X_{\mathfrak{c}(t)}$.

3 Gromov-Hausdorff distances between ultrametric spaces

For convenience, we adopt the following notation to represent the *absolute p-difference* (for $p \in [1, \infty]$) between two non-negative numbers $a, b \in \mathbb{R}_{\geq 0}$:

$$\Lambda_p(a,b) \coloneqq |a^p - b^p|^{\frac{1}{p}}, \qquad \text{for } p \in [1,\infty);$$
$$\Lambda_\infty(a,b) \coloneqq \begin{cases} \max(a,b), & a \neq b \\ 0, & a = b \end{cases}, \qquad \text{for } p = \infty.$$

Note that $\Lambda_1(a,b) = |a-b|$ is the usual Euclidean distance and that $\lim_{p\to\infty} \Lambda_p(a,b) = \Lambda_\infty(a,b)$. In particular, we have the following obvious characterization of Λ_∞:

Proposition 17. *For any $a, b \geq 0$, we have that*

$$\Lambda_\infty(a,b) = \inf\{c \geq 0 : a \leq \max(b,c) \text{ and } b \leq \max(a,c)\}. \tag{7}$$

Proof. We have the following two cases.

1. If $a = b$, then $\Lambda_\infty(a,b) = 0 \leq c$.
2. If $a \neq b$, we assume without loss of generality that $a > b$. Then, $a \leq \max(b,c)$ implies that $\Lambda_\infty(a,b) = \max(a,b) = a \leq c$.

□

Now, given $p \in [1,\infty]$ and two ultrametric spaces (X, u_X) and (Y, u_Y), for any non-empty subset $S \subseteq X \times Y$, we define its *p-distortion* with respect to u_X and u_Y as follows:

$$\mathrm{dis}_p(S, u_X, u_Y) \coloneqq \sup_{(x,y),(x',y') \in S} \Lambda_p(u_X(x,x'), u_Y(y,y')). \tag{8}$$

We abbreviate $\mathrm{dis}_p(S, u_X, u_Y)$ to $\mathrm{dis}_p(S)$ whenever clear from the context. In particular, for a map $\varphi : X \to Y$, we define its p-distortion by

$$\mathrm{dis}_p(\varphi) \coloneqq \mathrm{dis}_p(\mathrm{graph}(\varphi)) = \sup_{x,x' \in X} \Lambda_p\big(u_X(x,x'), u_Y(\varphi(x),\varphi(x'))\big),$$

where $\mathrm{graph}(\varphi) \coloneqq \{(x,\varphi(x)) \in X \times Y : x \in X\}$.

Recall from the introduction that a correspondence R between the underlying sets X and Y is any subset of $X \times Y$ such that the images of R under the canonical projections $p_X : X \times Y \to X$ and $p_Y : X \times Y \to Y$ are full: $p_X(R) = X$ and $p_Y(R) = Y$. The following terminology regarding correspondences will be used frequently in later sections.

Definition 18 (ε-correspondence)**.** *Let R be any correspondence between X and Y. For any $\varepsilon \geq 0$ and $p \in [1,\infty]$, we call R an (ε,p)-correspondence if $\mathrm{dis}_p(R) \leq \varepsilon$. When $p = 1$, we call an $(\varepsilon,1)$-correspondence simply an ε-correspondence.*

Example 19. *Let $X = \{x_1,x_2\}$ and $Y = \{y_1,y_2\}$ be a pair of two-point spaces. Assume two ultrametrics u_X and u_Y on X and Y, respectively, such that $u_X(x_1,x_2) = 1$ and $u_Y(y_1,y_2) = 2$. Let $R \coloneqq \{(x_1,y_1),(x_2,y_2)\}$. Then, R is a correspondence between X and Y. For any $p \in [1,\infty]$, it is clear that*

$$\mathrm{dis}_p(R,u_X,u_Y) = \Lambda_p\big(u_X(x_1,x_2),u_Y(y_1,y_2)\big) = \begin{cases} |1^p - 2^p|^{\frac{1}{p}} & p \in [1,\infty) \\ 2 & p = \infty \end{cases}.$$

Now, for any $p \in [1,\infty]$, we define $d_{\mathrm{GH}}^{(p)}$ as follows, which is an extension of Equation (2) defined only for $p \in [1,\infty)$:

$$d_{\mathrm{GH}}^{(p)}(X,Y) \coloneqq 2^{-\frac{1}{p}} \inf_R \mathrm{dis}_p(R), \tag{9}$$

where the infimum is taken over all correspondences between X and Y. Here we adopt the convention that $\frac{1}{\infty} = 0$. Note that $d_{\mathrm{GH}}^{(1)} = d_{\mathrm{GH}}$, and as a consequence of our convention

$$d_{\mathrm{GH}}^{(\infty)}(X,Y) = \inf_R \mathrm{dis}_\infty(R).$$

As already mentioned in the introduction, when only considering *finite* ultrametric spaces, we easily have the following property of the family $\{d_{\mathrm{GH}}^{(p)}\}_{p\in[1,\infty]}$:

Proposition 20. *Given any $X,Y \in \mathcal{U}^{\mathrm{fin}}$, the function $p \mapsto d_{\mathrm{GH}}^{(p)}(X,Y)$ is continuous and increasing with respect to $p \in [1,\infty]$. In particular, $\lim_{p\to\infty} d_{\mathrm{GH}}^{(p)}(X,Y) = d_{\mathrm{GH}}^{(\infty)}(X,Y)$.*

Remark 21. *Note that $d_{\mathrm{GH}}^{(p)}(X,Y)$ in Equation (9) is actually well-defined for any two metric spaces X and Y, i.e., X and Y are not restricted to be ultrametric spaces. See Section 3.3 for alternative definitions of $d_{\mathrm{GH}}^{(p)}(X,Y)$ on classes of metric spaces larger than $\mathcal{U}$.*

Example 22 (Distance to the one point space)**.** *Since there exists a unique correspondence $R_* \coloneqq X \times *$ between a given finite set X and the one point space $*$, we have for each $p \in [1,\infty]$ that*

$$d_{\mathrm{GH}}^{(p)}(X,*) = 2^{-\frac{1}{p}}\mathrm{dis}_p(R_*) = 2^{-\frac{1}{p}}\mathrm{diam}(X).$$

3.1 Computing $d_{\mathrm{GH}}^{(p)}$ is NP-hard when $p \in [1, \infty)$

Given an ultrametric space (X, u_X) and any positive real number α, the function $(x, x') \mapsto (u_X(x, x'))^\alpha$ is still an ultrametric on X so that the space $(X, (u_X)^\alpha)$ is still an ultrametric space. We let $S_\alpha(X)$ denote the ultrametric space $(X, (u_X)^\alpha)$. Then, we have the following transformation between $d_{\mathrm{GH}}^{(p)}$ and d_{GH} for all $p \in [1, \infty)$. The proof of the following result is relegated to Appendix C.

Proposition 23. *Given $1 \leq p < \infty$ and any two ultrametric spaces X and Y, one has*

$$d_{\mathrm{GH}}^{(p)}(X, Y) = \left(d_{\mathrm{GH}}(S_p(X), S_p(Y))\right)^{\frac{1}{p}}.$$

Conversely,

$$d_{\mathrm{GH}}(X, Y) = \left(d_{\mathrm{GH}}^{(p)}\left(S_{\frac{1}{p}}(X), S_{\frac{1}{p}}(Y)\right)\right)^p.$$

Therefore, solving an instance of $d_{\mathrm{GH}}^{(p)}$ is equivalent to solving an instance of d_{GH}. Since it is NP-hard to compute d_{GH} between finite ultrametric spaces [40], it is also NP-hard to compute $d_{\mathrm{GH}}^{(p)}$ between finite ultrametric spaces for every $p \in [1, \infty)$.

Moreover, combining Proposition 23 with Theorem 1, we have the following more precise statement:

Corollary 3. *For each $p \in [1, \infty)$ and for any $X, Y \in \mathcal{U}^{\mathrm{fin}}$, $d_{\mathrm{GH}}^{(p)}(X, Y)$ cannot be approximated within any factor less than $3^{\frac{1}{p}}$ in polynomial time, unless $\mathcal{P} = \mathcal{NP}$.*

Proof. Suppose otherwise that there exist finite ultrametric spaces X and Y such that $d_{\mathrm{GH}}^{(p)}(X, Y)$ can be approximated within a factor $c^{\frac{1}{p}} < 3^{\frac{1}{p}}$ in polynomial time. Then, Proposition 23 implies that one can approximate $\left(d_{\mathrm{GH}}(S_p(X), S_p(Y))\right)^{\frac{1}{p}}$ within a factor $c^{\frac{1}{p}} < 3^{\frac{1}{p}}$ in polynomial time. This implies that one can approximate $d_{\mathrm{GH}}(S_p(X), S_p(Y))$ within a factor $c < 3$ in polynomial time which contradicts with Theorem 1. □

As mentioned in the introduction, the factor $3^{\frac{1}{p}}$ in Corollary 3 approaches 1 as $p \to \infty$. This suggests that $d_{\mathrm{GH}}^{(\infty)}$ could be a computationally tractable quantity; later in Section 4.2 we will illustrate this point.

3.2 An estimate of $d_{\mathrm{GH}}^{(p)}$ via diameters of input spaces

The following result shows how the Gromov-Hausdorff distance interacts with the diameters of the input spaces.

Proposition 24 ([30, Theorems 3.3 and 3.4])**.** *For any finite metric spaces X and Y, let R be a correspondence between them. Then, we have*

$$|\mathrm{diam}(X) - \mathrm{diam}(Y)| \leq \mathrm{dis}_1(R) \leq \max(\mathrm{diam}(X), \mathrm{diam}(Y)).$$

In particular,

$$\frac{1}{2}|\mathrm{diam}(X) - \mathrm{diam}(Y)| \leq d_{\mathrm{GH}}(X, Y) \leq \frac{1}{2}\max(\mathrm{diam}(X), \mathrm{diam}(Y)).$$

Invoking Proposition 23, we immediately have the following analogue to the second part of Proposition 24 for $d_{\mathrm{GH}}^{(p)}$:

Proposition 25. *For any $p \in [1,\infty)$ and finite ultrametric spaces X and Y, we have that*

$$2^{-\frac{1}{p}}\Lambda_p(\mathrm{diam}(X),\mathrm{diam}(Y)) \leq d_{\mathrm{GH}}^{(p)}(X,Y) \leq 2^{-\frac{1}{p}}\max(\mathrm{diam}(X),\mathrm{diam}(Y)).$$

Note that $\lim_{p\to\infty} 2^{-\frac{1}{p}}\Lambda_p(\mathrm{diam}(X),\mathrm{diam}(Y)) = \Lambda_\infty(\mathrm{diam}(X),\mathrm{diam}(Y))$. When $\mathrm{diam}(X) \neq \mathrm{diam}(Y)$, $\lim_{p\to\infty} 2^{-\frac{1}{p}}\Lambda_p(\mathrm{diam}(X),\mathrm{diam}(Y)) = \max(\mathrm{diam}(X),\mathrm{diam}(Y))$. Then, by continuity of $d_{\mathrm{GH}}^{(p)}$ with respect to $p \in [1,\infty]$ (cf. Proposition 20), we have the following property for $d_{\mathrm{GH}}^{(\infty)}$:

Proposition 26. *Let X and Y be any two finite ultrametric spaces with different diameters, then*

$$d_{\mathrm{GH}}^{(\infty)}(X,Y) = \max(\mathrm{diam}(X),\mathrm{diam}(Y)).$$

This proposition indicates that, when X and Y have different diameters, $d_{\mathrm{GH}}^{(\infty)}(X,Y)$ is determined by the diameter values of the input spaces, which suggests that $d_{\mathrm{GH}}^{(\infty)}$ is more rigid than other $d_{\mathrm{GH}}^{(p)}$ when $p < \infty$ and as a consequence, $d_{\mathrm{GH}}^{(\infty)}$ exhibits a distinct computational behavior in comparison to $d_{\mathrm{GH}}^{(p)}$ when $p < \infty$.

Example 27 (Distance between homothetic spaces)**.** *Let X be a finite ultrametric space and let $\lambda > 0$. Then, for any $p \in [1,\infty]$ we have*

$$d_{\mathrm{GH}}^{(p)}((X,u_X),(X,\lambda\cdot u_X)) = 2^{-\frac{1}{p}}\Lambda_p(\lambda,1)\cdot\mathrm{diam}(X).$$

Indeed, since $\mathrm{diam}(X,\lambda\cdot u_X) = \lambda\cdot\mathrm{diam}(X)$, by Proposition 25 and Proposition 26 we have that

$$d_{\mathrm{GH}}^{(p)}((X,u_X),(X,\lambda\cdot u_X)) \geq 2^{-\frac{1}{p}}\Lambda_p(\lambda\cdot\mathrm{diam}(X),\mathrm{diam}(X)) = 2^{-\frac{1}{p}}\Lambda_p(\lambda,1)\cdot\mathrm{diam}(X).$$

For the converse, consider the identity correspondence $R_{\mathrm{id}} \coloneqq \{(x,x) \in X\times X : x \in X\}$. Then,

$$d_{\mathrm{GH}}^{(p)}((X,u_X),(X,\lambda\cdot u_X)) \leq 2^{-\frac{1}{p}}\mathrm{dis}_p\big(R_{\mathrm{id}},u_X,\lambda\cdot u_X\big) = 2^{-\frac{1}{p}}\Lambda_p(\lambda,1)\cdot\mathrm{diam}(X).$$

3.3 $d_{\mathrm{GH}}^{(\infty)}$ exactly coincides with the Gromov-Hausdorff ultrametric u_{GH}

Recall that on a metric space Z, the *Hausdorff distance* d_{H}^{Z} between two subsets $A,B \subseteq Z$ is defined as

$$d_{\mathrm{H}}^{Z}(A,B) := \max\left(\sup_{a\in A}\inf_{b\in B} d_Z(a,b), \sup_{b\in B}\inf_{a\in A} d_Z(a,b)\right).$$

Given two metric spaces X and Y, we say a map $\varphi : X \to Y$ is an *isometric embedding* from X to Y if for every $x,x' \in X$, $d_X(x,x') = d_Y(\varphi(x),\varphi(x'))$. We usually use the symbol $\hookrightarrow$ (instead of $\to$) to represent isometric embeddings. Then, the Gromov-Hausdorff distance can be characterized as follows:

Theorem 28 (Duality formula for d_{GH}, [6, Theorem 7.3.25])**.** *The Gromov-Hausdorff distance d_{GH} between two compact metric spaces X and Y satisfies the following:*

$$d_{\mathrm{GH}}(X,Y) = \inf d_{\mathrm{H}}^{Z}(\varphi_X(X), \varphi_Y(Y)), \tag{10}$$

where the infimum is taken over all metric spaces Z and isometric embeddings $\varphi_X : X \hookrightarrow Z$ and $\varphi_Y : Y \hookrightarrow Z$.

In fact, Equation (10) is the original definition of the Gromov-Hausdorff distance given by Gromov in [17]. In the spirit of Equation (10), Zarichnyi defines in [43] the *Gromov-Hausdorff ultrametric*, which we denote by u_{GH}, between *compact* ultrametric spaces X and Y as follows:

$$u_{\mathrm{GH}}(X,Y) := \inf d_{\mathrm{H}}^{Z}(\varphi_X(X), \varphi_Y(Y)), \tag{11}$$

where the infimum is taken over all *ultrametric spaces* Z and isometric embeddings $\varphi_X : X \hookrightarrow Z$ and $\varphi_Y : Y \hookrightarrow Z$. It turns out that u_{GH} agrees with $d_{\mathrm{GH}}^{(\infty)}$ as defined by Equation (9).

Theorem 29 (Duality formula for $d_{\mathrm{GH}}^{(\infty)}$)**.** *Given two* compact *ultrametric spaces X and Y, we have that*

$$d_{\mathrm{GH}}^{(\infty)}(X,Y) = u_{\mathrm{GH}}(X,Y). \tag{12}$$

See Appendix C for the proof.

Zarichnyi proved in [43] that u_{GH} (and thus $d_{\mathrm{GH}}^{(\infty)}$) is an ultrametric on the collection of all compact ultrametric spaces. Similar metric properties also hold for $d_{\mathrm{GH}}^{(p)}$, for $p \in [1,\infty)$; see the following remark.

Remark 30 (Duality formula for $d_{\mathrm{GH}}^{(p)}$)**.** *An alternative formulation similar to Equation (12) via the Hausdorff distance exists for $d_{\mathrm{GH}}^{(p)}$ for every $p \in [1,\infty]$. For $p \in [1,\infty)$, we call a metric space X a* p-metric space *if it satisfies the* p-triangle inequality*:*

$$\forall x, x', x'' \in X,\ d_X(x,x') \leq \left(d_X^p(x,x'') + d_X^p(x'',x')\right)^{\frac{1}{p}}.$$

In that case, we refer to d_X as a p-metric. Note that the above inequality reduces to the strong triangle inequality as p approaches ∞. So we also will refer to the strong triangle inequality as the ∞-triangle inequality and an ultrametric space as an ∞-metric space. Then, for any $p \in [1,\infty]$ and any two compact p-metric spaces, we have that

$$d_{\mathrm{GH}}^{(p)}(X,Y) = \inf d_{\mathrm{H}}^{Z}(\varphi(X), \varphi_Y(Y)),$$

where the infimum is taken over all p-metric spaces *Z and isometric embeddings $\varphi_X : X \hookrightarrow Z$ and $\varphi_Y : Y \hookrightarrow Z$. Moreover, $d_{\mathrm{GH}}^{(p)}$ is actually a p-metric on the collection of all isometry classes of compact metric spaces and in particular, is a p-metric on the collection of all isometry classes of compact p-metric spaces. See [31, Theorem 4.2] for more details.*

4 Structural results for u_{GH} and computational implications

In this section, we first prove our central observation regarding u_{GH}, the structural theorem for u_{GH} (Theorem 4). Then, we utilize this theorem to devise a poly-time algorithm for computing u_{GH}. We remark that the distance u_{GH} as well as Theorem 4 and Algorithm 1 can be extended to the so-called ultra-dissimilarity spaces, which can be regarded as generalization of ultrametric spaces. See Appendix B for details.

4.1 Proof of Theorem 4

Recall the statement of the structural theorem for u_{GH}:

Theorem 4 (Structural theorem for u_{GH})**.** *For any $X, Y \in \mathcal{U}^{\mathrm{fin}}$ one has that*

$$u_{\mathrm{GH}}(X,Y) = \min\left\{t \geq 0 : \left(X_{\mathfrak{c}(t)}, u_{X_{\mathfrak{c}(t)}}\right) \text{ is isometric to } \left(Y_{\mathfrak{c}(t)}, u_{Y_{\mathfrak{c}(t)}}\right)\right\}.$$

Proof. We first prove the following weaker version of Theorem 4 (with inf instead of min):

$$u_{\mathrm{GH}}(X,Y) = \inf\left\{t \geq 0 : \left(X_{\mathfrak{c}(t)}, u_{X_{\mathfrak{c}(t)}}\right) \cong \left(Y_{\mathfrak{c}(t)}, u_{Y_{\mathfrak{c}(t)}}\right)\right\}. \tag{13}$$

Suppose first that $X_{\mathfrak{c}(t)} \cong Y_{\mathfrak{c}(t)}$ for some $t \geq 0$, i.e. there exists an isometry $f_t : X_{\mathfrak{c}(t)} \to Y_{\mathfrak{c}(t)}$. Define

$$R_t := \left\{(x,y) \in X \times Y : [y]^Y_{\mathfrak{c}(t)} = f_t\left([x]^X_{\mathfrak{c}(t)}\right)\right\}.$$

That R_t is a correspondence between X and Y is clear since: f_t is bijective, every $x \in X$ belongs to exactly one block in $X_{\mathfrak{c}(t)}$ and every $y \in Y$ belongs to exactly one block in $Y_{\mathfrak{c}(t)}$. For any $(x,y),(x',y') \in R_t$, if $u_X(x,x') \leq t$, then we already have $u_X(x,x') \leq \max(t, u_Y(y,y'))$. Otherwise, if $u_X(x,x') > t$, then we have $[x]^X_{\mathfrak{c}(t)} \neq [x']^X_{\mathfrak{c}(t)}$. Since f_t is bijective, we have that $[y]^Y_{\mathfrak{c}(t)} = f_t([x]^X_{\mathfrak{c}(t)}) \neq f_t([x']^X_{\mathfrak{c}(t)}) = [y']^Y_{\mathfrak{c}(t)}$. Then,

$$u_Y(y,y') = u_{Y_{\mathfrak{c}(t)}}\left([y]^Y_{\mathfrak{c}(t)}, [y']^Y_{\mathfrak{c}(t)}\right) = u_{X_{\mathfrak{c}(t)}}\left([x]^X_{\mathfrak{c}(t)}, [x']^X_{\mathfrak{c}(t)}\right) = u_X(x,x').$$

Therefore, $u_X(x,x') \leq \max(t, u_Y(y,y'))$. Similarly, $u_Y(y,y') \leq \max(t, u_X(x,x'))$. Then, by Equation (7), $\Lambda_\infty(u_X(x,x'), u_Y(y,y')) \leq t$ and thus $\mathrm{dis}_\infty(R_t) \leq t$. This implies that

$$u_{\mathrm{GH}}(X,Y) \leq \inf\left\{t \geq 0 : X_{\mathfrak{c}(t)} \cong Y_{\mathfrak{c}(t)}\right\}.$$

Conversely, let R be any correspondence between X an Y and let $t := \mathrm{dis}_\infty(R)$. Consider any $(x,y),(x',y') \in R$ such that $[x']^X_{\mathfrak{c}(t)} = [x]^X_{\mathfrak{c}(t)}$ (i.e. $u_X(x',x) \leq t$). We define a map $f_t : X_{\mathfrak{c}(t)} \to Y_{\mathfrak{c}(t)}$ as follows: for each $[x]^X_{\mathfrak{c}(t)} \in X_{\mathfrak{c}(t)}$, suppose $y \in Y$ is such that $(x,y) \in R$, then we let $f_t\left([x]^X_{\mathfrak{c}(t)}\right) := [y]^Y_{\mathfrak{c}(t)}$.

We claim that the map f_t is well-defined. Indeed, if $[x]^X_{\mathfrak{c}(t)} = [x']^X_{\mathfrak{c}(t)}$ and $y, y' \in Y$ are such that $(x, y), (x', y') \in R$, then

$$\Lambda_\infty(u_X(x, x'), u_Y(y, y')) \leq \mathrm{dis}_\infty(R) = t.$$

This implies that $u_Y(y', y) \leq t$ which is equivalent to $[y]^Y_{\mathfrak{c}(t)} = [y']^Y_{\mathfrak{c}(t)}$. Similarly, there is a well-defined map $g_t : Y_{\mathfrak{c}(t)} \to X_{\mathfrak{c}(t)}$ sending $[y]^Y_{\mathfrak{c}(t)} \in Y_{\mathfrak{c}(t)}$ to $[x]^X_{\mathfrak{c}(t)}$ whenever $(x, y) \in R$. It is clear that g_t is the inverse of f_t and thus f_t is bijective. Now suppose that $u_{X_{\mathfrak{c}(t)}}\left([x]^X_{\mathfrak{c}(t)}, [x']^X_{\mathfrak{c}(t)}\right) = s > t$, which implies that $u_X(x, x') = s$. Let $y, y' \in Y$ be such that $(x, y), (x', y') \in R$. Then, since $\Lambda_\infty(u_X(x, x'), u_Y(y, y')) \leq \mathrm{dis}_\infty(R) = t < s$, $u_Y(y, y')$ is forced to be equal to $u_X(x, x') = s$. Therefore,

$$u_{Y_{\mathfrak{c}(t)}}\left(f_t\left([x]^X_{\mathfrak{c}(t)}\right), f_t\left([x']^X_{\mathfrak{c}(t)}\right)\right) = u_{Y_{\mathfrak{c}(t)}}\left([y]^Y_{\mathfrak{c}(t)}, [y']^Y_{\mathfrak{c}(t)}\right) = s = u_{X_{\mathfrak{c}(t)}}\left([x]^X_{\mathfrak{c}(t)}, [x']^X_{\mathfrak{c}(t)}\right).$$

This proves that f_t is an isometry and thus

$$u_{\mathrm{GH}}(X, Y) \geq \inf\left\{t \geq 0 : X_{\mathfrak{c}(t)} \cong Y_{\mathfrak{c}(t)}\right\}.$$

Since X and Y are finite, for each $t \geq 0$, there exists $\varepsilon > 0$ such that for all $s \in [t, t+\varepsilon]$ $X_{\mathfrak{c}(t)} \cong X_{\mathfrak{c}(s)}$ and $Y_{\mathfrak{c}(t)} \cong Y_{\mathfrak{c}(s)}$. This implies that the infimum in Equation (13) is attained and thus we obtain the claim. □

Remark 31. *Theorem 4 actually holds for compact ultrametric spaces; see [31] for details.*

4.2 A poly-time algorithm for computing u_{GH}

In Algorithm 1 below we provide pseudocode for computing u_{GH} and in Theorem 32 we prove that Algorithm 1 runs in time $O(n^2)$; see also Remark 33 for details about improving this time complexity to $O(n \log(n))$.

Recall that the spectrum $\mathrm{spec}(X)$ of the metric space X is the set of values defined by $\mathrm{spec}(X) := \{u_X(x, x') : \forall x, x' \in X\}$. The pseudocode for the function **ClosedQuotient** implementing the closed quotient operation is given in Algorithm 6 in Appendix A. The function **is_iso** determines whether two ultrametric spaces are isometric, for which we adapt the algorithm in [3, Example 3.2].

Algorithm 1 uGH(X, Y)

```
spec ← sort(spec(X) ∪ spec(Y), 'descend')
for i = 1 : length(spec) do
    t = spec(i)
    if ∼ is_iso(ClosedQuotient(X, t), ClosedQuotient(Y, t)) then return spec(i − 1)
    end if
end for
return 0
```

Complexity analysis of Algorithm 1 Let $n := \max(\#X, \#Y)$. By Proposition 8, $\#\mathrm{spec}(X) \leq \#X$. Then,

$$\#\mathrm{spec} = \#\left(\mathrm{spec}(X) \cup \mathrm{spec}(Y)\right) = O(n).$$

Thus, it takes time $O(n\log(n))$ to construct and to sort the sequence $\mathrm{spec} := \mathrm{spec}(X) \cup \mathrm{spec}(Y)$ (cf. Lemma 70). Now, for each $t \in \mathrm{spec}$, we need time $O(n)$ for running Algorithm **ClosedQuotient** (Algorithm 6) with input (X, t) and (Y, t).

Following Appendix A and Lemma 71, since $\max(\#X_{\mathfrak{c}(t)}, \#Y_{\mathfrak{c}(t)}) \leq \max(\#X, \#Y) = O(n)$, the function **is_iso** with input $(X_{\mathfrak{c}(t)}, Y_{\mathfrak{c}(t)})$ runs in time $O(n)$ as well (cf. Lemma 71).

Thus, the time complexity associated to computing $u_{\mathrm{GH}}(X, Y)$ via Algorithm 1 is

$$O(n\log(n)) + \mathrm{length}(\mathrm{spec}) \cdot O(n) = O(n\log(n)) + O(n) \cdot O(n) = O(n^2).$$

In this way we have proved the following theorem.

Theorem 32 (Time complexity of Algorithm **uGH** (Algorithm 1))**.** *Let X and Y be finite ultrametric spaces. Then, algorithm* **uGH**(X, Y) *(Algorithm 1) runs in time $O\left(n^2\right)$, where $n := \max(\#X, \#Y)$.*

Remark 33 (Acceleration via binary search)**.** *By replacing the for-loop over $i = 1 :$ length(spec) in* **uGH** *(Algorithm 1) with binary search, the total complexity will drop to $O(n\log(n)) + O(\log(n)) \cdot O(n) = O(n\log(n))$.*

Remark 34 (A novel poly time solvable instance of the quadratic assignment problem)**.** *Given $p \in [1, \infty]$ and two finite ultrametric spaces $X = \{x_1, \ldots, x_{n_X}\}$ and $Y = \{y_1, \ldots, y_{n_Y}\}$, we formulate the computation of $d_{\mathrm{GH}}^{(p)}(X, Y)$ as the following* generalized[1] *bottleneck quadratic assignment problem (*GQBAP_p*) as in [30, Remark 3.4] (cf. Equation (9)):*

$$\mathrm{GQBAP}_p(\mathbf{a}, \mathbf{b}) := 2^{\frac{1}{p}}\, d_{\mathrm{GH}}^{(p)}(X, Y) = \min_{R} \max_{i,j,k,l} \Lambda_p(a_{ik}, b_{jl})\, R_{ij}\, R_{kl},$$

where $\mathbf{a} = (a_{ik})$ is an $n_X \times n_X$ matrix such that $a_{ik} := u_X(x_i, x_k)$, $\mathbf{b} = (b_{jl})$ is an $n_Y \times n_Y$ matrix such that $b_{jl} := u_Y(y_j, y_l)$, and R is a correspondence, which is regarded as (R_{ij}), an $n_X \times n_Y$ matrix such that $R_{ij} \in \{0, 1\}$ and

1. $\sum_{i=1}^{n_X} R_{ij} \geq 1$ *for all j;*
2. $\sum_{j=1}^{n_Y} R_{ij} \geq 1$ *for all i.*

Then, by Corollary 3 and Theorem 32, whereas solving GQBAP_p *is NP-hard for each $p \in [1, \infty)$, the problem* GQBAP_∞ *can be solved in time $O(n\log(n))$ where $n := \max(n_X, n_Y)$.*

In general, quadratic assignment problems are NP-hard [34]. This includes instances such as GQBAP_p *in the case when $p < \infty$. However, by the above 'cost' matrices of the form of $(\Lambda_\infty(a_{ik}, b_{jl}))$ yield computationally tractable instances.*

[1] Here, 'generalized' refers to the fact that we are allowing matchings more general than permutations.

5 Structural results for d_{GH} and computational implications

In this section, we first prove the structural theorem for d_{GH} (Theorem 5), and then develop efficient algorithms for computing d_{GH} based on Theorem 5.

5.1 Proof of Theorem 5

Recall the structural theorem for the Gromov-Hausdorff distance:

Theorem 5 (Structural theorem for d_{GH})**.** *Let $X, Y \in \mathcal{U}^{\mathrm{fin}}$ and $\varepsilon \geq 0$ be such that*

$$|\mathrm{diam}(X) - \mathrm{diam}(Y)| \leq \varepsilon < \mathrm{diam}(Y). \tag{3}$$

Consider the following open partitions

$$X_{\mathfrak{o}(\delta_\varepsilon(Y))} := \{X_i\}_{i=1}^{N_X} \quad \text{and} \quad Y_{\mathfrak{o}(\delta_0(Y))} := \{Y_j\}_{j=1}^{N_Y}.$$

Then, there exists an ε-correspondence between X and Y if and only if:

(1) there exists a surjection $\Psi : [N_X] \twoheadrightarrow [N_Y]$ and, with this surjection,

(2) for every $j \in [N_Y]$ there exists an ε-correspondence between $\left(X_{\Psi^{-1}(j)}, u_X|_{X_{\Psi^{-1}(j)} \times X_{\Psi^{-1}(j)}}\right)$ and $\left(Y_j, u_Y|_{Y_j \times Y_j}\right)$ where for each $j \in [N_Y]$, $X_{\Psi^{-1}(j)} := \bigcup_{i \in \Psi^{-1}(j)} X_i$.

Proof. First suppose that there exists an ε-correspondence R between X and Y. Then, we define a map $\Psi : [N_X] \to [N_Y]$ as follows: for any $i \in [N_X]$, pick an arbitrary $x \in X_i$ and assume that $(x, y) \in R$ for some $y \in Y$; further assume that $y \in Y_j$ for some $j \in [N_Y]$, then we let $\Psi(i) := j$. Now, we verify that this map Ψ is well-defined, i.e., Ψ is independent of choice of $x \in X_i$ and choice of $(x, y) \in R$. For any $i \in [N_X]$ and $x, x' \in X_i$, we have by assumption that $u_X(x, x') < \delta_\varepsilon(Y)$. Suppose $y, y' \in Y$ are such that $(x, y), (x', y') \in R$. Then,

$$u_Y(y, y') \leq u_X(x, x') + \mathrm{dis}_1(R) \leq u_X(x, x') + \varepsilon < \delta_\varepsilon(Y) + \varepsilon = \mathrm{diam}(Y).$$

Therefore, there exists a common $j \in [N_Y]$ such that both $y \in Y_j$ and $y' \in Y_j$. This implies that Ψ is well-defined. Since R is a correspondence, Ψ must be surjective. Then, for each $j \in [N_Y]$, we define the set

$$R_j := R \cap (X_{\Psi^{-1}(j)} \times Y_j)$$

It can be checked that R_j is a correspondence between $X_{\Psi^{-1}(j)}$ and Y_j. Moreover, $\mathrm{dis}_1(R_j) \leq \mathrm{dis}_1(R) \leq \varepsilon$ for each $j \in [N_Y]$. Therefore, for each $j \in [N_Y]$, R_j is an ε-correspondence between $X_{\Psi^{-1}(j)}$ and Y_j.

Conversely, suppose that there exist a surjection $\Psi : [N_X] \twoheadrightarrow [N_Y]$ and for each $j \in [N_Y]$ an ε-correspondence R_j between $X_{\Psi^{-1}(j)}$ and Y_j. Then, define $R := \bigcup_{j \in [N_Y]} R_j$. It is clear that R is a correspondence between X and Y because

$$p_X\left(\bigcup_{j \in [N_Y]} R_j\right) = \bigcup_{j \in [N_Y]} p_X(R_j) = \bigcup_{j \in [N_Y]} X_{\Psi^{-1}(j)} = X$$

and

$$p_Y\left(\bigcup_{j\in[N_Y]} R_j\right) = \bigcup_{j\in[N_Y]} p_Y(R_j) = \bigcup_{j\in[N_Y]} Y_j = Y,$$

where $p_X : X \times Y \to X$ and $p_Y : X \times Y \to Y$ are the canonical coordinate projections. Given any $(x, y), (x', y') \in R$, suppose $(x, y) \in R_j$ and $(x', y') \in R_{j'}$ for some $j, j' \in [N_Y]$. Then, we verify that $|u_X(x, x') - u_Y(y, y')| \leq \varepsilon$ in the following two cases:

1. if $j = j'$, then $|u_X(x, x') - u_Y(y, y')| \leq \mathrm{dis}_1(R_j) \leq \varepsilon$;
2. if $j \neq j'$, then x and x' belong to different blocks of $X_{\mathfrak{o}(\delta_\varepsilon(Y))}$, and y and y' belong to different blocks of $Y_{\mathfrak{o}(\delta_0(Y))}$. Then, $u_X(x, x') \geq \delta_\varepsilon(Y) = \mathrm{diam}(Y) - \varepsilon$ and $u_Y(y, y') = \delta_0(Y) = \mathrm{diam}(Y)$. So, $u_X(x, x') \geq u_Y(y, y') - \varepsilon$. By the assumption that $|\mathrm{diam}(X) - \mathrm{diam}(Y)| \leq \varepsilon$, we have that $u_X(x, x') \leq \mathrm{diam}(X) \leq \mathrm{diam}(Y) + \varepsilon = u_Y(y, y') + \varepsilon$. Therefore, $|u_X(x, x') - u_Y(y, y')| \leq \varepsilon$.

Therefore, $\mathrm{dis}_1(R) \leq \varepsilon$ and thus R is an ε-correspondence between X and Y. This also proves Remark 7. □

5.2 Algorithms for computing d_{GH} based on Theorem 5

The main goal of this section is to develop an efficient algorithm for computing the exact value of d_{GH} between ultrametric spaces. To achieve the goal, we first consider the following decision problem:

Decision Problem GHDU-dec (d_{GH} distance computation between finite ultrametric spaces)

Inputs: Finite ultrametric spaces X and Y, as well as $\varepsilon \geq 0$.

Question: Is there an ε-correspondence between X and Y?

5.2.1 Strategy for solving GHDU-dec

Base cases for Problem GHDU-dec Proposition 24 shows how the Gromov-Hausdorff distance d_{GH} interacts with the diameters of the input spaces. This theorem then implies that **GHDU-dec** is solved immediately in the following two base cases:

Base Case 1: If $|\mathrm{diam}(X) - \mathrm{diam}(Y)| > \varepsilon$, then there exists no ε-correspondence between X and Y.

Base Case 2: If $\max(\mathrm{diam}(X), \mathrm{diam}(Y)) \leq \varepsilon$, then every correspondence R between X and Y is an ε-correspondence.

Base Case 1 justifies our assumption that $|\mathrm{diam}(X) - \mathrm{diam}(Y)| \leq \varepsilon$ in Theorem 5 since otherwise we would be in one of the two base cases. Note that the situation when one of the two spaces is the one point space will automatically fall in either of the above two base cases.

Application of Theorem 5 Suppose that we are given two ultrametric spaces X and Y and $\varepsilon \geq 0$ not falling in either of the two base cases mentioned above. This implies that one of $\mathrm{diam}(X)$ or $\mathrm{diam}(Y)$ must be strictly larger than ε.

Suppose $\mathrm{diam}(Y) > \varepsilon$ (otherwise we swap the roles of X and Y) and apply the open partition operation to X and Y to obtain $X_{\mathfrak{o}(\delta_\varepsilon(Y))} := \{X_i\}_{i\in[N_X]}$ and $Y_{\mathfrak{o}(\delta_0(Y))} := \{Y_j\}_{j\in[N_Y]}$. Here we use the same notation as in Theorem 5 that for each $i \in [N_X]$, X_i denotes an open equivalence class $[x_i]_{\mathfrak{o}(\delta_\varepsilon(Y))}$ for some $x_i \in X$ and similarly for notation Y_j.

If there is no surjection from $[N_X]$ to $[N_Y]$, i.e., $N_X < N_Y$, then we conclude from Theorem 5 that there is no ε-correspondence between X and Y. Otherwise, for each surjection $\Psi : [N_X] \twoheadrightarrow [N_Y]$ and for each $j \in [N_Y]$, we solve one instance of the decision problem **GHDU-dec** with input $(X_{\Psi^{-1}(j)}, Y_j, \varepsilon)$. If for some surjection Ψ, there exist ε-correspondences R_j between $X_{\Psi^{-1}(j)}$ and Y_j for all $j \in [N_Y]$, then the union of all R_js is an ε-correspondence between X and Y (cf. Remark 7). Otherwise, by Theorem 5 again, there exists no ε-correspondence between X and Y.

For each pair $(X_{\Psi^{-1}(j)}, Y_j)$ as described above, it is easy to see that $\#X_{\Psi^{-1}(j)} < \#X$ and $\#Y_j < \#Y$. So, if we repeatedly apply the open partition operation as in Theorem 5, we will eventually reduce the problem to one of the two base cases.

5.2.2 A recursive algorithm

From the analysis above we identify a recursive algorithm **FindCorrRec** (Algorithm 2) which takes as input two ultrametric spaces X and Y and a parameter $\varepsilon \geq 0$. If there exists an ε-correspondence between X and Y, then $\mathbf{FindCorrRec}(X, Y, \varepsilon)$ returns such an ε-correspondence. If there exists no ε-correspondence, $\mathbf{FindCorrRec}(X, Y, \varepsilon)$ returns 0.

Complexity analysis To analyze the complexity of this recursive algorithm, we need to control the size of subproblems, i.e., the sizes of the blocks of the partitions produced by the open equivalence relations. The following structural condition on ultrametric spaces serves this purpose.

Definition 35 (First (ε,γ)-growth condition)**.** *For $\varepsilon \geq 0$, and $\gamma > 1$, we say that an ultrametric space (X, u_X) satisfies the* first (ε,γ)-growth condition *(FGC) if for all $x \in X$, and $t \geq \varepsilon$,*

$$\#[x]_{\mathfrak{c}(t)} \leq \gamma \cdot \#[x]_{\mathfrak{o}(t-\varepsilon)}.$$

Note that on the left-hand side of the inequality above we consider a 'closed' equivalence class whereas on the right-hand side we consider an 'open' equivalence class. We denote by $\mathcal{U}_1(\varepsilon,\gamma)$ the collection of all finite ultrametric spaces satisfying the first (ε,γ)-growth condition. See Figure 6 for an illustration and Remark 36 for an interpretation.

Algorithm 2 FindCorrRec(X, Y, ε)

BoolSwap $\leftarrow$ FALSE
if $\operatorname{diam}(X) > \operatorname{diam}(Y)$ **then**
    Swap X and Y; BoolSwap $\leftarrow$ TRUE
end if

Phase 1 – Base Cases

if $\max(\operatorname{diam}(X), \operatorname{diam}(Y)) \leq \varepsilon$ **then**
    $R \leftarrow \operatorname{ones}(\#X, \#Y)$
    if *BoolSwap* **then**
        Transpose R
    end if
    return R
end if
if $|\operatorname{diam}(X) - \operatorname{diam}(Y)| > \varepsilon$ **then return** 0
end if

Phase 2 – Recursion

$\{X_i\}_{i\in[N_X]} = \mathbf{OpenPartition}(X, \delta_\varepsilon(Y))$
$\{Y_j\}_{j\in[N_Y]} = \mathbf{OpenPartition}(Y, \delta_0(Y))$
for *Each surjection* $\Psi : [N_X] \twoheadrightarrow [N_Y]$ **do**
    for $j \in [N_Y]$ **do**
        $R_j \leftarrow \mathbf{FindCorrRec}\left(X_{\Psi^{-1}(j)}, Y_j, \varepsilon\right)$
        if $((R_j\,! = 0)\,\forall j)$ **then**
            $R \leftarrow \bigcup_{j=1}^{N_Y} R_j$
            if *BoolSwap* **then**
                Transpose R
            end if
            return R
        end if
    end for
end for
return 0

Remark 36 (Interpretation of the FGC)**.** *The main idea behind the first (ε, γ)-growth condition is that for each $t > 0$ we want to have some degree of control over both the cardinalities of and the number of descendants of each $[x]_{\mathfrak{c}(t)}$ in $X_{\mathfrak{c}(t)}$, where we say that $[x']_{\mathfrak{o}(s)}$ is an (open)* descendant *of $[x]_{\mathfrak{c}(t)}$, or conversely that $[x]_{\mathfrak{c}(t)}$ is a (closed)* ancestor *of $[x']_{\mathfrak{o}(s)}$, if $[x']_{\mathfrak{o}(s)} \subseteq [x]_{\mathfrak{c}(t)}$.*

More precisely, we write explicitly the $(t-\varepsilon)$-open partition of $[x]_{\mathfrak{c}(t)}$ as follows: $[x]_{\mathfrak{c}(t)} = \sqcup_{i=1}^{N} [x_i]_{\mathfrak{o}(t-\varepsilon)}$ for some $x_i \in [x]_{\mathfrak{c}(t)}$, $i = 1, \ldots, N$.

First, we note that $[x]_{\mathfrak{c}(t)} = [x_i]_{\mathfrak{c}(t)}$ for each $i = 1, \ldots, N$ and thus the FGC implies

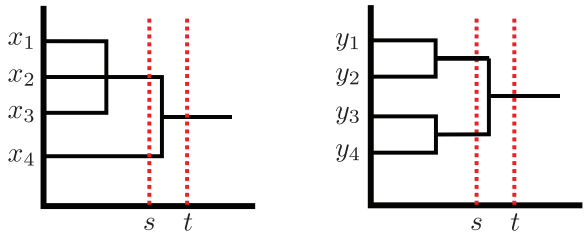


Figure 6: **Illustration of Definition 35.** X and Y are two 4-point ultrametric spaces. Suppose $s = t - \varepsilon$ for some $\varepsilon > 0$. It is easy to see that $Y \in \mathcal{U}_1(\varepsilon, 2)$. Since $2\,\#[x_4]_{\mathfrak{o}(s)} = 2 < 4 = \#[x_4]_{\mathfrak{o}(t)}$, it is easy to see that, in contrast, $X \notin \mathcal{U}_1(\varepsilon, 2)$. This example illustrates that the FGC prevents a given equivalence class in $X_{\mathfrak{o}(t)}$ from containing most of the points of X and thus its dendrogram will tend to split 'evenly'.

that

$$\#[x_i]_{\mathfrak{o}(t-\varepsilon)} \geq \frac{\#[x_i]_{\mathfrak{c}(t)}}{\gamma} = \frac{\#[x]_{\mathfrak{c}(t)}}{\gamma}.$$

This means that each descendant at scale $t-\varepsilon$ of a given block $[x]_{\mathfrak{c}(t)}$ contains at least a fixed proportion $\frac{1}{\gamma}$ of the number of points in its ancestor $[x]_{\mathfrak{c}(t)}$.

Moreover, we have

$$\#[x]_{\mathfrak{c}(t)} = \sum_{i=1}^{N} \#[x_i]_{\mathfrak{o}(t-\varepsilon)} \geq \frac{N}{\gamma} \#[x]_{\mathfrak{c}(t)}.$$

Therefore $N \leq \gamma$, which implies that each $[x]_{\mathfrak{c}(t)}$ has at most γ many descendants at scale $t - \varepsilon$.

By invoking the master theorem [12] we now prove the following theorem which provides an upper bound on the complexity of Algorithm 2. See Section C.2.1 for its proof.

Theorem 37 (Time complexity of Algorithm **FindCorrRec** (Algorithm 2))**.** *Fix some $\varepsilon \geq 0$ and $\gamma \geq 2$. Then, for any $X, Y \in \mathcal{U}_1(\varepsilon, \gamma)$, **FindCorrRec**$(X, Y, \varepsilon)$ (Algorithm 2) runs in time $O\left(n^{(\gamma+1)\log_{\mathrm{b}(\gamma)}\gamma}\right)$, where $n := \max(\#X, \#Y)$ and $\mathrm{b}(\gamma) := \frac{\gamma^2}{\gamma^2-1}$.*

Under the FGC, our recursive algorithm **FindCorrRec** (Algorithm 2) exhibits time complexity $O\left(n^{(\gamma+1)\log_{\mathrm{b}(\gamma)}\gamma}\right)$. Since the exponent of n depends on γ, this is only partially satisfactory. In other words, Algorithm 2 is not yet fixed-parameter tractable, a notion which requires the exponent to be independent of the parameters involved. This motivates us to further examine and improve Algorithm 2 in order to develop an FPT algorithm. Note that in the for-loop over surjections in Algorithm 2, for different surjections $\Psi_1, \Psi_2 : [N_X] \twoheadrightarrow [N_Y]$, there could be some $j_0 \in [N_Y]$ such that $\Psi_1^{-1}(j_0) = \Psi_2^{-1}(j_0)$. This would result in repetitive computations of **FindCorrRec** $\left(X_{\Psi_1^{-1}(j_0)}, Y_{j_0}, \varepsilon\right)$. With the goal of eliminating such repetitions, in the next section we devise a *dynamic programming algorithm* which eventually turns out to be FPT.

5.2.3 A dynamic programming algorithm

In this section, we introduce a dynamic programming algorithm **FindCorrDP** for solving the decision problem **GHDU-dec** for which we provide pseudocode in Algorithm 4. To proceed with the description of Algorithm 4, we first introduce some notation.

We let V_X denote the set of all closed balls in X. For each closed ball $B \in V_X$, recall the notation $\delta_{2\varepsilon}(B) = \operatorname{diam}(B) - 2\varepsilon$. We further cap this number by zero and let $\delta_{2\varepsilon}^+(B) := \max(\delta_{2\varepsilon}(B), 0)$. We then write the $\delta_{2\varepsilon}^+(B)$-open partition of B as:

$$B_{\mathfrak{o}(\delta_{2\varepsilon}^+(B))} := \left\{ [x_i]^B_{\mathfrak{o}(\delta_{2\varepsilon}^+(B))} \right\}_{i=1}^{N_B}$$

where $x_i \in B$ for each $i = 1, \dots, N_B$. Here recall that for any $t \geq 0$, $[x_i]^B_{\mathfrak{o}(t)}$ denotes the equivalence class of x_i in the ultrametric space $(B, u_X|_{B\times B})$ under the open equivalence relation $\sim_{\mathfrak{o}(t)}$ (cf. Section 2.1). For notational simplicity, we let $B_i := [x_i]^B_{\mathfrak{o}(\delta_{2\varepsilon}^+(B))}$ for each $i = 1, \dots, N_B$. By standard properties of ultrametric spaces, it can be checked that each $[x]^B_{\mathfrak{o}(\delta_{2\varepsilon}^+(B))} = [x]^X_{\mathfrak{o}(\delta_{2\varepsilon}^+(B))}$ is actually a closed ball in X. Then, $B_i \in V_X$ for all $i = 1, \dots, N_B$. Note that for any $I \subseteq \{1, \dots, N_B\}$, $\operatorname{diam}(\bigcup_{i\in I} B_i) \leq \operatorname{diam}(B)$. If the equality is achieved, we call $\bigcup_{i\in I} B_i$ an *ε-maximal union of closed balls* of B. Denote by $B_{(\varepsilon)}$ the set of all ε-maximal unions of closed balls *in* B. Then, define a new set $V_X^{(\varepsilon)} := \bigcup_{B\in V_X} B_{(\varepsilon)}$ by replacing each $B \in V_X$ with the set $B_{(\varepsilon)}$. We use the notation U^X to represent a generic element in $V_X^{(\varepsilon)}$. See Figure 7 for an illustration of $V_X^{(\varepsilon)}$.

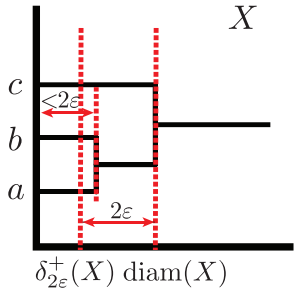


Figure 7: **Illustration of $V_X^{(\varepsilon)}$ from Section 5.2.3.** Note that $V_X = \{\{a\}, \{b\}, \{c\}, \{a, b\}, \{a, b, c\}\}$. For the ball $B := \{a, b, c\}$, we have $B_{(\varepsilon)} = \{\{a, c\}, \{b, c\}, \{a, b, c\}\}$. For other balls B' in V_X, we have $B'_{(\varepsilon)} = \{B'\}$. For example, $\{a, b\}_{\mathfrak{o}(\delta_{2\varepsilon}^+(\{a,b\}))} = \{\{a\}, \{b\}\}$, so $\{a, b\}_{(\varepsilon)} = \{\{a, b\}\}$. Therefore, $V_X^{(\varepsilon)} = \{\{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$.

Remark 38. *The value $\delta_{2\varepsilon}^+(X)$ originates from Theorem 5 as a lower bound for $\delta_\varepsilon(Y)$: when $|\operatorname{diam}(X) - \operatorname{diam}(Y)| \leq \varepsilon$ and $\operatorname{diam}(Y) > \varepsilon$, we have that $\delta_\varepsilon(Y) = \operatorname{diam}(Y) - \varepsilon \geq \max(\operatorname{diam}(X) - 2\varepsilon, 0) = \delta_{2\varepsilon}^+(X)$. This inequality results in the following favorable property of $V_X^{(\varepsilon)}$: for any $U^X \in V_X^{(\varepsilon)}$ and $B^Y \in V_Y$, if $|\operatorname{diam}(U^X) - \operatorname{diam}(B^Y)| \leq \varepsilon$ and $\operatorname{diam}(B^Y) > \varepsilon$, then each block of the open partition $U^X_{\mathfrak{o}(\delta_\varepsilon(B^Y))}$ belongs to $V_X^{(\varepsilon)}$. See Appendix C.2.3 for a proof of this property.*

Fix an input triple (X, Y, ε). It is clear that the pair (X, Y) belongs to $V_X^{(\varepsilon)} \times V_Y$. We sort $V_X^{(\varepsilon)}$ and V_Y according to ascending diameter values and denote by $\mathrm{LX}^{(\varepsilon)}$ and LY the respective sorted arrays (with details provided in Appendix A.4). In particular, we require that X and Y are at the end of lists $\mathrm{LX}^{(\varepsilon)}$ and LY, respectively. We devise our DP algorithm **FindCorrDP** (Algorithm 4) so that it maintains a binary variable $\mathrm{DYN}\left(U^X, B^Y\right)$ for each pair $\left(U^X, B^Y\right) \in \mathrm{LX}^{(\varepsilon)} \times \mathrm{LY}$, such that $\mathrm{DYN}\left(U^X, B^Y\right) = 1$ if there exists an ε-correspondence between U^X and B^Y, and $\mathrm{DYN}\left(U^X, B^Y\right) = 0$, otherwise. Now, we elaborate the main idea behind Algorithm **FindCorrDP** (Algorithm 4).

Algorithm 4 starts by looping over all $B^Y \in \mathrm{LY}$. Inside the loop, it computes $\mathrm{DYN}\left(U^X, B^Y\right)$ by looping over all $U^X \in \mathrm{LX}^{(\varepsilon)}$. Most pairs $\left(U^X, B^Y\right)$ fall in the base cases and $\mathrm{DYN}\left(U^X, B^Y\right)$ is determined by comparing diameters. For non base cases, we have the following two situations:

1. If $\mathrm{diam}\left(B^Y\right) > \varepsilon$, we determine $\mathrm{DYN}\left(U^X, B^Y\right)$ by (1) computing the open partition of U^X and B^Y, respectively, to obtain $U^X_{\mathfrak{o}(\delta_\varepsilon(B^Y))} = \left\{U_i^X\right\}_{i\in[N_{U^X}]}$ and $B^Y_{\mathfrak{o}(\delta_0(B^Y))} = \left\{B_j^Y\right\}_{j\in[N_{B^Y}]}$ and (2) by exploiting the precomputed values
$$\left\{\mathrm{DYN}\left(U^X_{\Psi^{-1}(j)}, B_j^Y\right)\right\}_{j\in[N_{B^Y}],\text{surjection } \Psi:[N_{U^X}]\to[N_{B^Y}]}$$
[2]
via the strategy discussed in Section 5.2.1. That $\left(U^X_{\Psi^{-1}(j)}, B_j^Y\right) \in V_X^{(\varepsilon)} \times V_Y$ follows from Remark 38 and that the values $\mathrm{DYN}\left(U^X_{\Psi^{-1}(j)}, B_j^Y\right)$ for all $j \in [N_{B^Y}]$ are pre-computed follows from the fact that $\mathrm{diam}(B_j^Y) < \mathrm{diam}(B^Y)$ and LY is ordered according to increasing diameter values.

2. If $\mathrm{diam}\left(B^Y\right) \le \varepsilon$, we determine $\mathrm{DYN}\left(U^X, B^Y\right)$ directly by applying Algorithm 3 (which arises from Proposition 40 below).

Remark 39 (Interpretation of situation 2)**.** *In order to reduce redundant computations, Algorithm* **FindCorrDP** *(Algorithm 4) only inspects pairs in $V_X^{(\varepsilon)} \times V_Y$ instead of the much larger symmetric set $V_X^{(\varepsilon)} \times V_Y^{(\varepsilon)}$. Due to the asymmetry of $V_X^{(\varepsilon)} \times V_Y$, the exceptional case in item 2 above may arise. This case is dealt with in the recursive algorithm* **FindCorrRec** *(Algorithm 2) by swapping the roles of U^X and B^Y. However, this swapping technique is not feasible for* **FindCorrDP***. We now further elaborate on this point. Suppose we replace line 10 in Algorithm 4 with a swapping between U^X and B^Y. Subsequently, we obtain open partitions of B^Y and U^X as follows:*
$$B^Y_{\mathfrak{o}(\delta_\varepsilon(U^X))} = \left\{B_i^Y\right\}_{i\in[N_{B^Y}]} \text{ and } U^X_{\mathfrak{o}(\delta_0(U^X))} = \left\{U_j^X\right\}_{j\in[N_{U^X}]}.$$
Then, for each surjection $\Psi : [N_{B^Y}] \twoheadrightarrow [N_{U^X}]$, we need to inspect values of $\mathrm{DYN}\left(U_j^X, B^Y_{\Psi^{-1}(j)}\right)$. Being a union of closed balls in B^Y, $B^Y_{\Psi^{-1}(j)}$ does not necessarily belong to V_Y, the set of closed balls in Y. This implies that the value $\mathrm{DYN}\left(U_j^X, B^Y_{\Psi^{-1}(j)}\right)$ does not necessarily exist for which Algorithm 4 may fail to continue.

[2]Here $U^X_{\Psi^{-1}(j)} \coloneqq \cup_{i\in\Psi^{-1}(j)} U_i^X$

Proposition 40. *Let finite ultrametric spaces X and Y and $\varepsilon \geq 0$ be such that the following two conditions hold:*

1. $\operatorname{diam}(X) > \varepsilon$, *and*
2. $\operatorname{diam}(Y) \leq \varepsilon$

Then, there exists an ε-correspondence between X and Y if and only if there exists an injective *map $\varphi : X_{\mathfrak{c}(\varepsilon)} \to Y$ such that $\operatorname{dis}_1(\varphi) \leq \varepsilon$.*

See Appendix C.2.2 for a proof.

Algorithm 3 FindCorrSmall(X, Y, ε)

Assert $\operatorname{diam}(X) > \varepsilon$ **and** $\operatorname{diam}(Y) \leq \varepsilon$
if $|\operatorname{diam}(X) - \operatorname{diam}(Y)| > \varepsilon$ **then return** 0
end if
$X_{\mathfrak{c}(\varepsilon)} =$ **ClosedQuotient**(X, ε)
for *Each injective map* $\Phi : X_{\mathfrak{c}(\varepsilon)} \to Y$ **do**
    Compute $\operatorname{dis}_1(\Phi)$
    if $\operatorname{dis}_1(\Phi) \leq \varepsilon$ **then return** 1
    end if
end for
return 0

Eventually, Algorithm **FindCorrDP** (Algorithm 4) will compute $\operatorname{DYN}(X, Y)$ through a bottom-up approach and thus solve the decision problem **GHDU-dec** with the given input triple (X, Y, ε). The correctness of Algorithm 4 is stated in the following theorem; see Appendix C.2.4 for its proof. Note that, the given pseudocode of Algorithm 4 only determines the existence of ε-correspondence without actually constructing a correspondence. However, it is clear that one can inspect the DYN matrix to produce an ε-correspondence whenever it exists.

Theorem 41 (Correctness of Algorithm **FindCorrDP** (Algorithm 4))**.** *There exists an ε-correspondence between X and Y if and only if* **FindCorrDP**$(X, Y, \varepsilon) = 1$.

Complexity analysis To analyze the complexity of Algorithm **FindCorrDP** (Algorithm 4), we consider the following growth condition in a similar spirit to the FGC:

Definition 42 (Second (ε, γ)-growth condition)**.** *For $\varepsilon \geq 0$, and $\gamma \in \mathbb{N}$, we say that an ultrametric space (X, u_X) satisfies the* second (ε, γ)-growth condition *(SGC) if for all $x \in X$, and $t \geq 2\varepsilon$,*

$$\#\left\{[x']_{\mathfrak{o}(t-2\varepsilon)} : x' \in [x]_{\mathfrak{c}(t)}\right\} \leq \gamma. \tag{14}$$

Note that both closed and open equivalence classes appear on the left-hand side of the relationship above.

Algorithm 4 FindCorrDP(X, Y, ε)

Build $\mathrm{LX}^{(\varepsilon)}$ and LY
$\mathrm{DYN} = \mathrm{zeros}(\#\mathrm{LX}^{(\varepsilon)}, \#\mathrm{LY})$
for $B^Y \in \mathrm{LY}$ **do**
  for $k = 1$ to $\#\mathrm{LX}^{(\varepsilon)}$ **do**
    $U^X = \mathrm{LX}^{(\varepsilon)}(k)$
    if $\left|\mathrm{diam}\left(U^X\right) - \mathrm{diam}\left(B^Y\right)\right| > \varepsilon$ **then**
      $\mathrm{DYN}\left(U^X, B^Y\right) = 0$
    else if $\max\left(\mathrm{diam}\left(U^X\right), \mathrm{diam}\left(B^Y\right)\right) \leq \varepsilon$ **then**
      $\mathrm{DYN}\left(U^X, B^Y\right) = 1$
    else if $\mathrm{diam}\left(U^X\right) > \varepsilon$ **and** $\mathrm{diam}\left(B^Y\right) \leq \varepsilon$ **then**
      $\mathrm{DYN}\left(U^X, B^Y\right) =$ **FindCorrSmall**(U^X, B^Y, ε)
    else
      $\left\{U_i^X\right\}_{i\in[N_{UX}]} =$ **OpenPartition** $\left(U^X, \delta_\varepsilon(B^Y)\right)$
      $\left\{B_j^Y\right\}_{j\in[N_{BY}]} =$ **OpenPartition** $\left(B^Y, \delta_0(B^Y)\right)$
      for *Each surjection* $\Psi : [N_{UX}] \twoheadrightarrow [N_{BY}]$ **do**
        if $\mathrm{DYN}\left(U^X_{\Psi^{-1}(j)}, B^Y\right) = 1, \forall j = 1, \ldots, M$ **then**
          $\mathrm{DYN}\left(U^X, B^Y\right) = 1$
          Continue in line 4
        end if
      end for
    end if
  end for
end for
return DYN(END, END)

We denote by $\mathcal{U}_2(\varepsilon, \gamma)$ the collection of all finite ultrametric spaces satisfying the second (ε, γ)-growth condition. Note that for any $0 \leq \varepsilon' < \varepsilon$, $\mathcal{U}_2(\varepsilon, \gamma) \subseteq \mathcal{U}_2(\varepsilon', \gamma)$.

Remark 43 (Relation with the notion of degree bound from [15])**.** *If we let $\gamma_\varepsilon(X)$ be the supremum over all $x \in X$ and $t \geq 0$ of the quantity on the left-hand side of Equation (14), then for any $\gamma \geq \gamma_\varepsilon(X)$, $X \in \mathcal{U}_2(\varepsilon, \gamma)$. The information captured by γ_ε is in a similar spirit to the concept called degree bound of merge trees as considered in [15]: the ε-degree bound $\tau_\varepsilon(M_X)$ of a merge tree M_X is the largest sum of degrees of all tree vertices inside any closed ε balls in M_X[3]. See Appendix A.1 for a detailed comparison between $\gamma_\varepsilon(X)$ and τ_ε.*

Remark 44 (Interpretation of the SGC and its relation with the FGC)**.** *The second (ε, γ)-growth condition is equivalent to saying for any $x \in X$ and $t > 2\varepsilon$, the number of descendants of $[x]_{\mathfrak{c}(t)}$ at level $t - 2\varepsilon$ is bounded above by γ. Note that if $X \in \mathcal{U}_1(\varepsilon, \gamma)$, then for any $t > \varepsilon$, the number of descendants of any class $[x]_{\mathfrak{c}(t)}$ at $t - \varepsilon$ is bounded above by γ (cf. Remark 36). This implies that $X \in \mathcal{U}_2\left(\frac{\varepsilon}{2}, \gamma\right)$. In other words, $\mathcal{U}_1(\varepsilon, \gamma) \subseteq \mathcal{U}_2\left(\frac{\varepsilon}{2}, \gamma\right)$, which indicates*

[3] In [15], the degree bound is actually defined for two merge trees: for two merge trees M_X and M_Y, the number $\tau_\varepsilon(M_X, M_Y) := \max(\tau_\varepsilon(M_X), \tau_\varepsilon(M_Y))$is called the ε-degree bound of (M_X, M_Y).

*that the second growth condition is **less** rigid than the first growth condition.*

Remark 45 (Relation between the SGC and the doubling constant)**.** *Recall that given $K > 0$, a metric space (X, d_X) is said to be K-doubling if for each $r > 0$, a closed ball with radius r can be covered by at most K closed balls with radius $\frac{r}{2}$. The SGC is related to the doubling constant as follows: (1) if a finite ultrametric space $X \in \mathcal{U}_2(\varepsilon, \gamma)$ for some $\varepsilon > 0$ and $\gamma \geq 1$, then X is $K_{\gamma,\varepsilon}$-doubling for $K_{\gamma,\varepsilon} := \gamma^{\lfloor \frac{\mathrm{diam}(X)}{4\varepsilon} \rfloor + 1}$; (2) conversely, a K-doubling ultrametric space satisfies the second $(0, K)$-growth condition. See Appendix C.2.5 for the proof of the fact. See Lemma 54 for a generalization of the latter fact in the case of finite metric spaces.*

Under the SGC, Algorithm **FindCorrDP** (Algorithm 4) runs in polynomial time and moreover, Algorithm **FindCorrDP** is FPT with respect to parameters in the SGC.

Theorem 46 (Time complexity of Algorithm **FindCorrDP** (Algorithm 4))**.** *Fix some $\varepsilon \geq 0$ and $\gamma \geq 1$. Then, for any $X, Y \in \mathcal{U}_2(\varepsilon, \gamma)$, Algorithm **FindCorrDP**(X, Y, ε) (Algorithm 4) runs in time $O\left(n^2 \log(n) 2^\gamma \gamma^{\gamma+2}\right)$, where $n := \max(\#X, \#Y)$.*

See Appendix C.2.6 for its proof.

5.2.4 Computing the exact value of d_{GH}

Given two finite ultrametric spaces X and Y, we compute the exact value of $d_{\mathrm{GH}}(X, Y)$ in the following way. Define

$$\mathcal{E}(X, Y) := \{|u_X(x, x') - u_Y(y, y')| : \forall x, x' \in X \text{ and } \forall y, y' \in Y\}.$$

Then, for any correspondence R between X and Y, we have $\mathrm{dis}_1(R) \in \mathcal{E}(X, Y)$ by finiteness of X and Y and by Equation (8). Therefore, in order to compute $d_{\mathrm{GH}}(X, Y)$, we first sort the elements in $\mathcal{E}(X, Y)$ in ascending order as $\varepsilon_0 < \varepsilon_1 < \cdots < \varepsilon_M$. If i is the smallest integer such that **FindCorrDP**$(X, Y, \varepsilon_i) = 1$, then $d_{\mathrm{GH}}(X, Y) = \frac{\varepsilon_i}{2}$. We summarize the process in Algorithm 5 and analyze its complexity in Theorem 47.

Theorem 47 (Time complexity of Algorithm **dGH** (Algorithm 5))**.** *Fix some $\varepsilon \geq 0$ and $\gamma \geq 1$. Let $X, Y \in \mathcal{U}_2(\varepsilon, \gamma)$ and assume that $\varepsilon \geq 2\, d_{\mathrm{GH}}(X, Y)$. Then, the algorithm **dGH** (Algorithm 5) with input (X, Y) runs in time $O\left(n^4 \log(n) 2^\gamma \gamma^{\gamma+2}\right)$, where $n = \max(\#X, \#Y)$.*

Remark 48. *Though the complexity in Theorem 47 depends on inherent structures of input spaces, it never means that we to figure out parameters ε and γ beforehand in order to apply our algorithm.*

Proof of Theorem 47. By Proposition 8, we have that $\#\mathcal{E}(X, Y) = O(n^2)$. Then, sorting $\mathcal{E}(X, Y)$ takes time $O(n^2 \log(n^2)) = O(n^2 \log(n))$ in average. For each $\varepsilon_i \in \mathcal{E}(X, Y)$ such that $\varepsilon_i \leq \varepsilon$, we need to invoke once Algorithm **FindCorrDP** (Algorithm 4) with inputs (X, Y, ε_i). For all such ε_is, $X, Y \in \mathcal{U}_2(\varepsilon_i, \gamma)$ and thus Algorithm **FindCorrDP** with inputs

(X, Y, ε_i) runs in time $O\left(n^2 \log(n) 2^\gamma \gamma^{\gamma+2}\right)$ (cf. Theorem 46). Therefore, the total time complexity of Algorithm **dGH** is bounded by

$$O(n^2) \times O\left(n^2 \log(n) 2^\gamma \gamma^{\gamma+2}\right) = O\left(n^4 \log(n) 2^\gamma \gamma^{\gamma+2}\right).$$

□

Algorithm 5 **dGH**(X, Y)

```
1: 𝓔 ← sort(𝓔(X, Y), 'ascend')
2: for i=1 to #𝓔 do
3:     if FindCorrDP(X, Y, 𝓔(i)) then return 𝓔(i)/2
4:     end if
5: end for
```

Remark 49 (Comparison to [15])**.** *Whereas methods from [15] can be adapted to obtain a 2-approximation of d_{GH} between two finite ultrametric spaces, our algorithm* **dGH** *(Algorithm 5) can obtain the exact value* in the same time complexity. *We now elaborate upon this statement.*

As illustrated in Remark 65, each finite ultrametric space naturally maps into a merge tree. In this way, we define the ε-degree bound of an ultrametric space as the ε-degree bound of its corresponding merge tree. By Remark 66, if any ultrametric space X has ε-degree bound τ_ε, it automatically satisfies the second $(\varepsilon, \tau_\varepsilon)$-growth condition.

Now consider the case where two merge trees M_X and M_Y arise from finite ultrametric spaces X and Y such that $d_{\mathrm{GH}}(X, Y) = \frac{\varepsilon}{2}$. In this case, if d_{I} denotes the interleaving distance between merge trees of [32], by [31, Corollary 6.13], then

$$\frac{1}{2} d_{\mathrm{I}}(M_X, M_Y) \leq d_{\mathrm{GH}}(X, Y) \leq d_{\mathrm{I}}(M_X, M_Y). \tag{15}$$

Let $\tau \coloneqq \tau_\varepsilon(M_X, M_Y)$ denote the ε-degree bound of (M_X, M_Y), then by above arguments we have that $X, Y \in \mathcal{U}_2(\varepsilon, \tau)$. Let $\delta \coloneqq d_{\mathrm{I}}(M_X, M_Y)$. Since $\delta = d_{\mathrm{I}}(M_X, M_Y) \leq \varepsilon$, by monotonicity of the degree bound, the δ-degree bound $\tau_\delta \coloneqq \tau_\delta(M_X, M_Y)$ of (M_X, M_Y) satisfies that $\tau_\delta \leq \tau$. Then, it is shown in [15] that one can compute $\delta = d_{\mathrm{I}}(M_X, M_Y)$ in time

$$O\left(n^4 \log(n) 2^{\tau_\delta} \tau_\delta^{\tau_\delta+2}\right) = O\left(n^4 \log(n) 2^\tau \tau^{\tau+2}\right),$$

which by Equation (15) is a 2-approximation of $d_{\mathrm{GH}}(X, Y)$. Note that, in contrast, by Theorem 47, with the same time complexity, our algorithm can compute the ***exact*** *value of $d_{\mathrm{GH}}(X, Y)$.*

Remark 50 (Improved time complexity for computing d_{GH})**.** *Following essentially the same strategy used for proving [15, Theorem 5], we can improve the time complexity for computing d_{GH} to $O\left(n^2 \log^3(n) 2^{2\gamma} (2\gamma)^{2\gamma+2}\right)$ under the same assumptions in Theorem 47. We provide details in Appendix C.2.7.*

5.2.5 Additive approximation of d_{GH} between arbitrary finite metric spaces

For any finite metric space (X, d_X), we introduce the following notion of *absolute ultrametricity* which quantifies how far X is being an ultrametric space:

Definition 51 (Absolute ultrametricity)**.** *For a finite metric space (X, d_X), we define the absolute ultrametricity of X by*

$$\mathrm{ult}^{\mathrm{abs}}(X) = \inf_u \|d_X - u\|_\infty$$

where the infimum is over all possible ultrametrics on X.

Note that $X \in \mathcal{U}^{\mathrm{fin}}$ iff $\mathrm{ult}^{\mathrm{abs}}(X) = 0$.

The notion of absolute ultrametricity is related to a more involved notion simply called *ultrametricity*; see [10] for a detailed study.

One natural ultrametric on X which can be used to approximate d_X is the so-called *single-linkage ultrametric* (or maximal subdominant) u_X^* [7]. The ultrametric u_X^* is defined as follows:

$$u_X^*(x, x') := \inf_{x=x_1,x_2,\ldots,x_n=x'} \max_{i=1,\ldots,n-1} d_X(x_i, x_{i+1}),$$

where the infimum is taken over all finite chains $x_1, x_2, \ldots, x_n \in X$ such that $x_1 = x$ and $x_n = x'$. It turns out that the single-linkage ultrametric is a fairly good ultrametric approximation of d_X:

Proposition 52 ([25, Theorem 3.3])**.** *For any finite metric space (X, d_X),*

$$\|d_X - u_X^*\|_\infty = 2\,\mathrm{ult}^{\mathrm{abs}}(X).$$

Remark 53.

1. *Proposition 52 implies that if (X, u_X) is already an ultrametric space, then $u_X^* = u_X$.*
2. *The time complexity of computing u_X^* from d_X is bounded above by $O(n^2)$ where $n := \#X$, cf. [33].*

For a metric space (X, d_X), its *separation* is defined as $\mathrm{sep}(X) := \inf\{d_X(x, x') | \, x \neq x'\}$. The following result illustrates how to transfer a doubling condition on a given metric space (X, d_X) to a SGC on its corresponding single-linkage ultrametric space (X, u_X^*).

Lemma 54 (Transfer from the doubling property on d_X to the SGC on u_X^*)**.** *Let (X, d_X) be a finite metric space. Let K be a doubling constant for X and let $\delta := 2\,\mathrm{ult}^{\mathrm{abs}}(X)$. Let $s := \mathrm{sep}(X)$ denote the separation of X. Then, for any $\varepsilon \geq 0$, we have that (X, u_X^*) satisfies the second $\left(\varepsilon, \max\left(K, K^{\log_2\left(\frac{2\delta+4\varepsilon}{s}\right)+1}\right)\right)$-growth condition.*

The proof is postponed to Appendix C.2.8. By Remark 53, when (X, u_X) is itself an ultrametric space, $u_X^* = u_X$ and thus $X \in \mathcal{U}_2\left(\varepsilon, \max\left(K, K^{\log_2\left(\frac{4\varepsilon}{s}\right)+1}\right)\right)$. In particular, if $\varepsilon = 0$, we have that $X \in \mathcal{U}_2\,(0, K)$, which coincides with the second claim of Remark 45.

Now, let $\mathcal{M}^{\mathrm{fin}}$ denote the collection of all finite metric spaces. We denote by $\mathfrak{H} : \mathcal{M}^{\mathrm{fin}} \to \mathcal{U}^{\mathrm{fin}}$ the *single-linkage map* sending $(X, d_X) \in \mathcal{M}^{\mathrm{fin}}$ to $\mathfrak{H}(X) := (X, u_X^*) \in \mathcal{U}^{\mathrm{fin}}$.

Proposition 55. *Let $X, Y \in \mathcal{M}^{\mathrm{fin}}$ and let $\delta \coloneqq \max(\mathrm{ult}^{\mathrm{abs}}(X), \mathrm{ult}^{\mathrm{abs}}(Y))$. Then,*

$$d_{\mathrm{GH}}(\mathfrak{H}(X), \mathfrak{H}(Y)) \leq d_{\mathrm{GH}}(X, Y) \leq d_{\mathrm{GH}}(\mathfrak{H}(X), \mathfrak{H}(Y)) + 2\delta.$$

Proof. The leftmost inequality follows directly from the stability result of the single-linkage map, cf. [7, Proposition 2]. For the rightmost inequality, we first have the following obvious observation:

Claim 56. *Given a finite set X and two metrics d_1, d_2 on the set X, we have that*

$$d_{\mathrm{GH}}((X, d_1), (X, d_2)) \leq \|d_1 - d_2\|_\infty.$$

Then, we have that

$$\begin{aligned} d_{\mathrm{GH}}((X, d_X), (Y, d_Y)) &\leq d_{\mathrm{GH}}((X, d_X), (X, u_X^*)) + d_{\mathrm{GH}}((X, u_X^*), (Y, u_Y^*)) + d_{\mathrm{GH}}((Y, d_Y), (Y, u_Y^*)) \\ &\leq \delta + d_{\mathrm{GH}}((X, u_X^*), (Y, u_Y^*)) + \delta \leq d_{\mathrm{GH}}((X, u_X^*), (Y, u_Y^*)) + 2\delta. \end{aligned}$$

This implies that $d_{\mathrm{GH}}(X, Y) \leq d_{\mathrm{GH}}(\mathfrak{H}(X), \mathfrak{H}(Y)) + 2\delta$. □

This proposition indicates that $d_{\mathrm{GH}}(\mathfrak{H}(X), \mathfrak{H}(Y))$ is a 2δ-additive approximation to $d_{\mathrm{GH}}(X, Y)$. Applying Theorem 47 to computing $d_{\mathrm{GH}}(\mathfrak{H}(X), \mathfrak{H}(Y))$, this immediately gives rise to the following time complexity result for computing an additive approximation of $d_{\mathrm{GH}}(X, Y)$.

Corollary 57 (Computing an additive approximation to $d_{\mathrm{GH}}(X, Y)$)**.** *Let X and Y be two K-doubling finite metric spaces for some $K > 0$. Let $\delta \coloneqq 2\max(\mathrm{ult}^{\mathrm{abs}}(X), \mathrm{ult}^{\mathrm{abs}}(Y))$. Let $s \coloneqq \min(\mathrm{sep}(X), \mathrm{sep}(Y))$. Then, the 2δ-additive approximation $d_{\mathrm{GH}}(\mathfrak{H}(X), \mathfrak{H}(Y))$ of $d_{\mathrm{GH}}(X, Y) =: \frac{\varepsilon}{2}$ can be computed in time $O\left(n^4 \log(n) 2^\gamma \gamma^{\gamma+2}\right)$, where $n = \max(\#X, \#Y)$ and $\gamma \coloneqq \max\left(K, K^{\log_2(\frac{2\delta+4\varepsilon}{s})+1}\right)$.*

Proof. By Remark 53, the time complexity of computing u_X^* from d_X and computing u_Y^* from d_Y is bounded above by $O(n^2)$ where $n \coloneqq \max(\#X, \#Y)$.

Note that $\varepsilon = 2d_{\mathrm{GH}}(X, Y)$. Then, by Lemma 54, $\mathfrak{H}(X)$ and $\mathfrak{H}(Y)$ both satisfy the second (ε, γ)-growth condition, where $\gamma \coloneqq \max\left(K, K^{\log_2(\frac{2\delta+4\varepsilon}{s})+1}\right)$. Let $\varepsilon_u \coloneqq 2\, d_{\mathrm{GH}}(\mathfrak{H}(X), \mathfrak{H}(Y))$. Note that $\varepsilon_u \leq \varepsilon$ by Proposition 55. Then, $\mathfrak{H}(X)$ and $\mathfrak{H}(Y)$ both satisfy the second (ε_u, γ)-growth condition. Therefore, by Theorem 47, $\frac{\varepsilon_u}{2} = d_{\mathrm{GH}}(\mathfrak{H}(X), \mathfrak{H}(Y))$, which is a 2δ additive approximation of $\frac{\varepsilon}{2} = d_{\mathrm{GH}}(X, Y)$, can be computed in time $O\left(n^4 \log(n) 2^\gamma \gamma^{\gamma+2}\right)$, where $\gamma \coloneqq \max\left(K, K^{\log_2(\frac{2\delta+4\varepsilon}{s})+1}\right)$.

Therefore, the total time complexity is bounded by

$$O(n^2) + O\left(n^4 \log(n) 2^\gamma \gamma^{\gamma+2}\right) = O\left(n^4 \log(n) 2^\gamma \gamma^{\gamma+2}\right).$$

□

6 Discussion

It is well known that computing d_{GH} between finite metric spaces leads to NP-hard problems. This hardness result holds even in the context of ultrametric spaces, which are highly structured metric spaces appearing in many practical applications.

In contrast to the hardness results for d_{GH}, by exploiting the ultrametric structure of the input spaces we first devised a polynomial time algorithm for computing u_{GH}, an ultrametric variant of d_{GH}, on the collection of all finite ultrametric spaces. Indeed, as a consequence of being more rigid than d_{GH}, we proved that u_{GH} can be computed in $O(n\log(n))$ time via Algorithm 1, which we also extended to the case of ultra-dissimilarity spaces.

From a different angle, but also with the goal of taming the NP-hardness associated to computing d_{GH} on the collection of all finite ultrametric spaces, as a second contribution, we first devised a recursive algorithm (Algorithm 2) and then based on this, a dynamic programming FPT-algorithm (Algorithm 4) for computing d_{GH}.

We provide implementations of Algorithm 1 for u_{GH} and Algorithm 2 for d_{GH} in our GitHub repository [1].

We leave for future work finding extensions of Algorithms 2 and 4 to the case of ultra-dissimilarity spaces and eventually general tree metric spaces.

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A Data structure for ultrametric spaces and implementation details

Whereas dendrograms are helpful for our theoretical development, we found a certain rooted tree structure associated to ultrametric spaces to be extremely helpful for designing our algorithms. In this section, we provide a detailed description of such rooted tree structure.

A.1 Tree structure for ultrametric spaces

Tree structures for ultrametric spaces are thoroughly studied in the literature [36, 24, 13]. Following the *labeled rooted tree* language used in [13], we provide a description of a *weighted rooted tree* representation of any finite ultrametric space.

A node weighted rooted tree is a tuple $T = (V, E, w, r)$ where (V, E) denotes an undirected tree with V being the vertex set and E being the edge set, $w : V \to \mathbb{R}_{\geq 0}$ denotes a node weight function and $r \in V$ is a specified vertex called the root of T. Two weighted rooted trees $T_1 = (V_1, E_1, w_1, r_1)$ and $T_2 = (V_2, E_2, w_2, r_2)$ are said to be *isomorphic*, if there exists a bijection $f : V_1 \to V_2$ such that

1. for every $x, y \in V_1$, $\{x, y\} \in E_1$ iff $\{f(x), f(y)\} \in E_2$;
2. for every $x \in V_1$, $w_1(x) = w_2(f(x))$;
3. $f(r_1) = r_2$.

Remark 58 (Standard terminology for rooted trees)**.** *Given any weighted rooted tree $T = (V, E, w, r)$, we call a collection of distinct vertices $x_0, x_1, \ldots, x_k \in V$ a* path *if for each $i = 0, \ldots, k-1$ we have $\{x_i, x_{i+1}\} \in E$. If for any given distinct $x, y \in V$ there exists a path $x_0 = r, x_1, \ldots, x_k = y$ such that $x = x_i$ for some $i = 0, \ldots, k-1$, then we say that x is an* ancestor *of y and y is a* descendant *of x. If furthermore $\{x, y\} \in E$, then we say that x is the* parent *of y and also that y is a* child *of x.*

The following useful fact will be utilized multiple times in the sequel.

Lemma 59. *Given a weighted rooted tree $T = (V, E, w, r)$, we denote by k_x the number of children of any given $x \in V$. Then,*

$$\sum_{x \in V} k_x = O(\#V).$$

Proof. Note that $\sum_{x \in V} \deg(x) = 2 \cdot \#E$. Since T is a tree, we have that $\#E = \#V - 1$. Moreover, $\deg(x) = k_x + 1 - \delta_{rx}$. Therefore,

$$\sum_{x \in V} k_x = O\left(\sum_{x \in V} \deg(x)\right) = O(\#E) = O(\#V).$$

□

A dendrogram automatically induces a weighted rooted tree as one can deduce from its graphic representation (see Figure 8). We describe this relationship between dendrograms and weighted rooted trees as follows. Let (X, u_X) be a finite ultrametric space and let θ_X be its corresponding dendrogram. Following the notation from Section 5.2.3 we let V_X denote the collection of all closed balls in X. It is then clear that V_X is a finite set. Furthermore, V_X contains X and all singletons $\{x\}$ for $x \in X$. Define a collection E_X of two-element *subsets* of V_X as follows: for any two different $B, B' \in V_X$, $\{B, B'\} \in E_X$ iff B' (resp. B) is the smallest (under inclusion) ball different from but containing B (resp. B'). Then, it is easy to see from the dendrogram that (V_X, E_X) is a combinatorial tree, i.e., a connected graph without cycles (see Figure 8 for an illustration), a fact which we record for later use:

Lemma 60. *(V_X, E_X) is an undirected tree with vertex set V_X and edge set E_X.*

Lemma 61. *Let X be a finite ultrametric space. Then,*

$$\operatorname{spec}(X) = \{\operatorname{diam}(B) : B \in V_X\}.$$

Now, we define a *weighted* rooted tree T_X associated to the ultrametric space X.

Definition 62 (Weighted rooted tree associated to X)**.** *Define $w_X : V_X \to \mathbb{R}_{\geq 0}$ by $w_X(B) \coloneqq \operatorname{diam}(B)$ for each $B \in V_X$. Let $r_X \coloneqq X \in V_X$. Then, we call the tuple $T_X \coloneqq (V_X, E_X, w_X, r_X)$ the weighted rooted tree associated to X.*

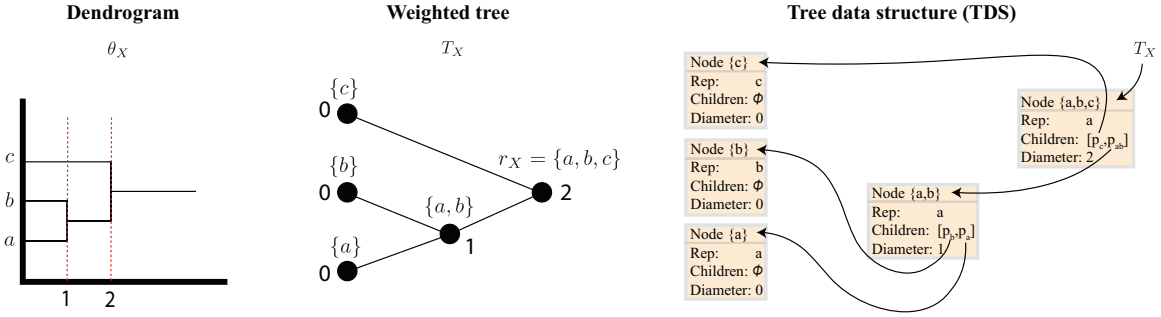


Figure 8: **Tree structure of dendrograms.** The leftmost figure represents the dendrogram θ_X of a three-point ultrametric space X. The middle figure represents the weighted rooted tree T_X associated to X. The numbers beside the nodes represent the weight values given by w_X. Note that the tree structure of T_X is inherited directly from the tree structure of θ_X. The rightmost figure shows the TDS associated to T_X.

Remark 63. *It is obvious that the set of singletons $\{\{x\} \in V_X : x \in X\}$ coincides with the set of leaves of T_X. Since for every rooted tree $\#\text{vertices} = O(\#\text{leaves})$ (indeed, $\#\text{leaves} \leq \#\text{vertices} \leq 2\#\text{leaves}$), we have that $\#V_X = O(n)$ where $n \coloneqq \#X$. Moreover, since $\#E_X = \#V_X - 1 = O(V_X)$, we have that $\#E_X = O(n)$ as well.*

Proposition 64 ([13, Theorem 1.10])**.** *For any two finite ultrametric spaces X and Y, let T_X and T_Y denote their corresponding weighted rooted trees. Then, X is isometric to Y iff T_X is isomorphic to T_Y.*

Remark 65 (Relation to merge trees)**.** *For the precise definition of merge trees, see for example [32]. Let X be a finite ultrametric space and let T_X be its associated weighted rooted tree. We first transform T_X into a topological/metric tree by replacing each edge $\{B, B'\}$ with an interval with length $|\mathrm{diam}(B) - \mathrm{diam}(B')|$. We then attach to the root r_X the half line $[0, \infty)$ to obtain the topological space M_X. We then define the height function $h_X : M_X \to \mathbb{R}$ as follows:*

1. *$h_X(B) = w_X(B) = \mathrm{diam}(B)$ for each $B \in V_X$;*
2. *for each edge, inside its corresponding interval, h_X is defined as the linear interpolation between the function values at the end points;*
3. *for t on the half line $[0, \infty)$, $h_X(t) \coloneqq h_X(r_X) + t$.*

In this way, each finite ultrametric space X canonically induces the merge tree (M_X, h_X).

Remark 66 (Detailed comparison between $\gamma_\varepsilon(X)$ and $\tau_\varepsilon(M_X)$)**.** *Recall that for $\varepsilon \geq 0$, the ε-degree bound $\tau_\varepsilon(M_X)$ of a merge tree M_X is the maximum over all closed ε balls the sum of degrees of all vertices inside the ball (cf. Remark 43). By Remark 65, each finite ultrametric space X canonically induces the merge tree (M_X, h_X). Then, we call $\tau_\varepsilon(X) \coloneqq \tau_\varepsilon(M_X)$ the ε-degree bound of the ultrametric space X. It is easy to see that for any ultrametric space X, we have that*

$$\gamma_\varepsilon(X) \leq \tau_\varepsilon(X) \leq 2\gamma_\varepsilon(X).$$

See Figure 9 for an illustration and a sketch of the proof of this fact. In particular, this implies that $X \in \mathcal{U}_2(\frac{\varepsilon}{2}, \gamma)$ for all $\gamma \geq \tau_\varepsilon(X)$.

A.2 Data structure for ultrametric spaces and algorithms for fundamental operations

Given a finite ultrametric space (X, u_X), let T_X be its corresponding weighted rooted tree as described in Definition 62. We utilize a special self-referential tree structure to represent T_X; for a description of self-referential tree data structures, we refer readers to [22, Chapter 6.5]. In order to represent vertices of T_X, we design a special class `Node` with three fields: `Representative`, `Diameter` and `Children`. Recall that each vertex of T_X represents a closed ball of X. Then, for each vertex/ball $B \in V_X$, its corresponding `Node` object, **which will still be denoted by B,** contains the following fields:

1. `Representative`: an element $x \in B$;[4]
2. `Diameter`: the diameter of B;
3. `Children`: a list of pointers to all `Node` objects representing the children of B in T_X.

In what follows, we will sometimes call a `Node` object simply a *node* and we will for instance write B.`Diameter` to extract the diameter value of a node B.

[4] Any choice of $x \in B$ will do.

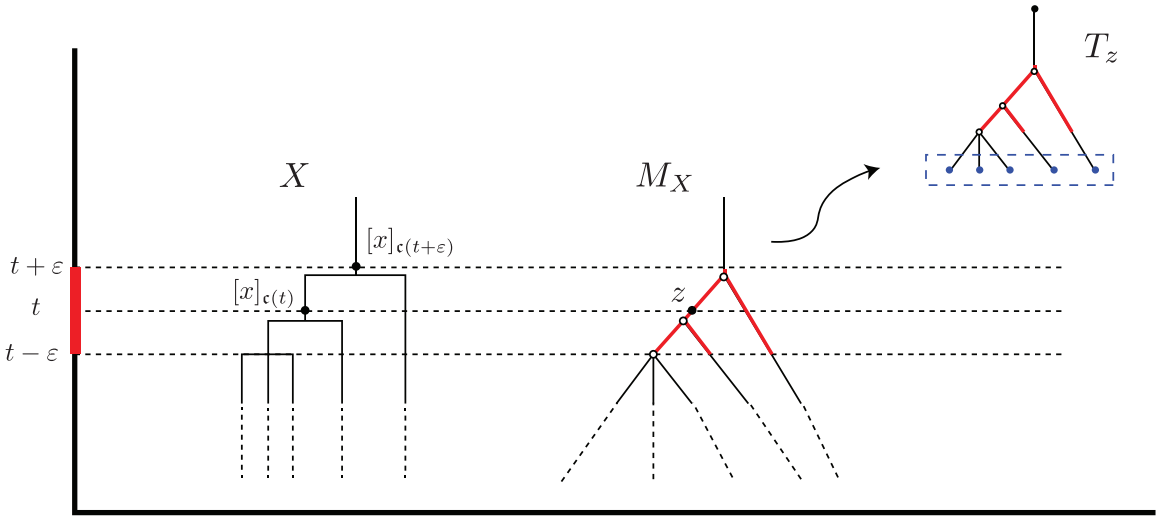


Figure 9: **Illustration of Remark 66.** In this figure, we draw part of the dendrogram of an ultrametric space X and its corresponding merge tree M_X. Pick any $z \in M_X$ with height t. Then, z corresponds to $[x]_{\mathfrak{c}(t)}$ for some point $x \in X$ (cf. Remark 65). Consider the closed ball $B_{\varepsilon}^{M_X}(z) \subseteq M_X$ highlighted in red in the figure. Considering a small closed neighborhood of the highlighted part in M_X, we obtain the rooted tree T_z in the top right corner of the figure. It is obvious that the number of leaves of T_z inside the dotted box equals $\gamma_{\varepsilon}(x, t+\varepsilon) := \#\left\{[x']_{\mathfrak{o}(t-\varepsilon)} : x' \in [x]_{\mathfrak{c}(t+\varepsilon)}\right\}$. On the other hand, the sum of degrees of vertices of the tree M_X contained in the ball $B_{\varepsilon}^{M_X}(z)$, denoted by $\tau_{\varepsilon}(z)$, equals the number of edges in T_z. Therefore, by the standard relationship between the number of leaves and the number of edges in a rooted tree, we have that $\gamma_{\varepsilon}(x, t+\varepsilon) \leq \tau_{\varepsilon}(z) \leq 2\,\gamma_{\varepsilon}(x, t+\varepsilon)$. From this observation, we conclude that $\gamma_{\varepsilon}(X) \leq \tau_{\varepsilon}(X) \leq 2\,\gamma_{\varepsilon}(X)$.

Now, for a given ultrametric space X, the *tree data structure* (TDS) which we will use to represent T_X consists of a collection of nodes stored in memory, one for each vertex $B \in V_X$. This TDS is held by a `Node` pointer referencing the node representing the root $r_X = X$. We call this pointer (to a `Node` object) the *root pointer* of the TDS. The role of the root pointer of a TDS is analogous to the role of the head pointer of a linked list. See Figure 8 for an illustration.

Note. In the sequel, we will overload the symbol T_X and will use it to denote both the weighted rooted tree corresponding to X and to also represent its associated TDS (both in the sense of its root pointer and in the sense of the collection of its nodes). The symbol B for representing any ball $B \in V_X$ is also overloaded to represent its corresponding `Node` object in T_X. In all our algorithms, every ultrametric space is understood as a `Node` object.

Remark 67 (Relationship with the distance matrix data structure)**.** *Starting from the root*

node r_X of T_X, one can progressively trace all nodes in T_X to completely reconstruct the distance matrix of the ultrametric space X. It is clear that the reconstruction process takes time at most $O(n^2)$ where $n := \#X$. Conversely, given the distance matrix of X, one can construct the TDS T_X in the same time complexity $O(n^2)$ (one can use for example the single-linkage algorithm [33] to first obtain the dendrogram induced by the distance matrix (see Figure 8)). If the ultrametric spaces are given in terms of distance matrices, we first need to convert them into TDSs before applying the algorithms described in this paper. The time complexity $O(n^2)$ due to this preprocessing is not counted when analyzing our algorithms.

Remark 68 (Subtree rooted at a ball)**.** *Given the TDS T_X associated to the ultrametric space X, it is easy to induce a TDS T_B for each closed ball $B \in V_X$ (which is itself an ultrametric space). Such T_B consists of all descendant nodes of B in T_X and is held by a pointer to B, i.e., $*(T_B) = B$. Here $*p$ denotes the datum referenced by a pointer p.*

Remark 69 (Finding parent nodes)**.** *Given any non-root node B in T_X, we let $\texttt{Parent}(B, T_X)$ denote its parent node in T_X. To actually find the node $\texttt{Parent}(B, T_X)$ given the node B, one can start from the root node r_X and recursively search T_X for the parent of B. Such a search process takes at most time $O(\#T_X) = O(n)$ where $n := \#X$.*

One major advantage of adopting the above-mentioned TDS is that it allows us to efficiently implement certain fundamental operations on ultrametric spaces. For example, it allows to efficiently obtain the spectrum of an ultrametric space:

Lemma 70. *Given an ultrametric space X, determining and sorting* $\mathrm{spec}(X)$ *can be done in time $O(n \log(n))$ where $n := \#X$.*

Proof. Note that, by Lemma 61, we only need to inspect each node in T_X which takes time $O(n)$. Since $\#\mathrm{spec}(X) = O(n)$ (cf. Proposition 8), the sorting process takes time $O(n \log(n))$. □

Next, we introduce algorithms for other three fundamental operations on ultrametric spaces.

Closed quotient In Algorithm 6 we introduce a recursive algorithm for the t-closed quotient operation. In the algorithm, the notation $\leftarrow$ represents variable assignment and the notation $\&Z$ denotes the memory address of the variable Z. In line 6 of the algorithm, the *one point tree* data structure consists of a single node such that `Diameter` $= 0$ and `Children` is an empty list. Note that in the worst-case scenario, Algorithm 6 inspects all nodes of T_X during the recursion process. At the recursion call with input (B, t) where B is a node of T_X, the main computational cost lies in line 1 for copying the `Node` object B (in the place of X). If we let $k_B := \mathrm{length}(B.\texttt{Children})$, it then costs time $O(k_B)$ to copy B and it takes at most time $O(k_B)$ to update $B.\texttt{Children}$ according to the for-loop in line 2. Therefore, the total time complexity of Algorithm 6 is bounded above by $\sum_{B \in V_X} O(k_B)$. By Lemma 59 and Remark 63, we have that

$$\sum_{B \in V_X} O(k_B) = O(V_X) = O(n),$$

where $n := \#X$. Therefore, the time complexity of Algorithm 6 is bounded above by $O(n)$.

Algorithm 6 ClosedQuotient(X, t)

```
1: Y ← X
2: for Each p ∈ Y.Children do
3:     if ∗p.Diameter > t then
4:         p ← &ClosedQuotient(∗p, t)
5:     else
6:         Z ← the one point tree with the same representative in ∗p
7:         p ← &Z
8:     end if
9: end for
10: return Y
```

Open partition In Algorithm 7 we give pseudocode for constructing the t-open partition of an ultrametric space. In order to implement the 'append' operation (appearing in line 3 and line 6) in constant time, we use a doubly linked list P to represent the open partition. Algorithm 7 will recursively inspect all nodes corresponding to balls in $X_{\mathfrak{o}(t)}$ as well as all of their ancestor nodes. For each inspected node B in T_X, if we let $k_B \coloneqq \text{length}(B.\texttt{Children})$, it then takes time $O(k_B)$ to update the list P. Then, following a similar argument as in the complexity analysis of Algorithm 6, if we let k be the cardinality of P (i.e., $k = \#X_{\mathfrak{o}(t)}$), then Algorithm 7 will inspect at most $O(k)$ nodes in T_X and thus it generates P in at most $O(k)$ steps.

Algorithm 7 OpenPartition(X, t)

```
1: P = [ ]
2: if X.Diameter < t then
3:     P.append(&X)
4: else
5:     for Each p_i ∈ X.Children do
6:         P.append(OpenPartition(∗p_i, t))
7:     end for
8: end if
9: return P
```

Isometry between ultrametric spaces By Proposition 64, two ultrametric spaces X and Y are isometric iff their corresponding weighted rooted trees T_X and T_Y are isomorphic. By adapting the algorithm in [3, Example 3.2], determining isomorphism between rooted trees can be done in time $O(\#\text{vertices})$; see also [3, Theorem 3.3] (and its corollary). Then, by Remark 63, we have the following result:

Lemma 71. *We can determine whether X and Y are isometric in $O(n)$ time.*

A.3 Subspace tree data structure and union of non-intersecting subspaces

In this section, we explain how to utilize the TDS described in the previous section to efficiently perform the union operation (under the conditions specified by Equation (16) below).

First of all, we introduce a TDS for representing subspaces of a given ultrametric space X. We assume that the distance matrix u_X is available and assume that a TDS T_X representing X has already been computed.

Definition 72 (Subspace tree data structure)**.** *For any non-empty subspace $U \subseteq X$, we say that a tree data structure T_U representing the ultrametric space $(U, u_X|_{U\times U})$ is a* subspace tree data structure subordinate to T_X*, if each vertex (i.e., each ball) in $V_U \cap V_X$ is represented by a* `Node` *object belonging to the tree data structure T_X.*

Remark 73 (Construction of subspace TDSs)**.** *For any node $B \in T_X$ (which represents a ball in X), the TDS T_B described in Remark 68 is obviously a subspace TDS subordinate to T_X representing the subspace B. However, for an arbitrary subset U the situation is different from the case of a ball. First, it is easy to verify that any such U can be written as a union of non-intersecting balls $\{B_i\}_{i=1}^k$ satisfying the condition in Equation (16) (see also Lemma 74 below). Then, a subspace TDS representing U subordinate to X can be constructed by applying the union process which we describe below to the set of balls $\{B_i\}_{i=1}^k$.*

Consider a set of non-empty and non-intersecting subspaces $\{U_1, \ldots, U_k\}$ of a given ultrametric space X such that for any distinct $i, j = 1, \ldots, k$, we have

$$\min_{x_i \in U_i, x_j \in U_j} u_X(x_i, x_j) > \max(\mathrm{diam}(U_i), \mathrm{diam}(U_j)). \tag{16}$$

This condition is compatible with the sets obtained by taking a slice of the dendrogram θ_X, which is in turn equivalent to considering open/closed equivalence classes of ultrametric spaces (see the discussion below Definition 14). We assume that each U_i is represented by a subspace TDS T_{U_i} subordinate to T_X.

Now given the above data, we describe how to construct a subspace TDS T_U subordinate to T_X representing the union $U \coloneqq \cup_{i=1}^k U_i$. The whole process is organized through the following three steps.

(I) Constructing a TDS induced by representatives For each $i = 1, \ldots, k$ let x_i be the representative of U_i as given in the TDS T_{U_i} and let $X_k \coloneqq \{x_1, \ldots, x_k\}$. We first consider the ultrametric $u_{X_k} \coloneqq u_X|_{X_k \times X_k}$ on X_k induced by the restriction of u_X to $X_k \times X_k$. Then, we construct a new TDS T_{X_k} to represent (X_k, u_{X_k}) (cf. Remark 67). This construction is possible due to the fact that (X_k, u_{X_k}) is itself an ultrametric space.

It takes time at most $O(k^2)$ to both construct the metric u_{X_k} and create the new TDS T_{X_k} (cf. Remark 67).

(II) Constructing the preliminary union of U_is Recall that up to this point, we have at our disposal the following TDSs: $T_{U_1}, \ldots, T_{U_k}$, and T_{X_k}. Based on these data, we progressively *modify* leaf nodes of T_{X_k} and utilize $T_{U_1}, \ldots, T_{U_k}$ to find a TDS representation for

the union U. For pedagogical reasons, we refer to the outcome TDS as $\tilde{T}_U$. $\tilde{T}_U$ may not be subordinate to T_X, and we thus name it the *preliminary union* of U_is. The modification process can be summarized as simply replacing each leaf node in T_{X_k} with the root node of certain T_{U_i} as shown in Figure 10. More precisely, we traverse all nodes B in T_{X_k} and, if for such a node B there exists an index i_0 such that $*(B.\texttt{Children}(i_0)).\texttt{Diameter} = 0$ (which means B is the parent of a leaf node), then we let j_0 be the index such that $*(B.\texttt{Children}(i_0)).\texttt{Representative} = x_{j_0} \in X_k$, and modify B by assigning $B.\texttt{Children}(i_0) = T_{U_{j_0}}$ (and of course we free the memory used for storing the original node $*(B.\texttt{Children}(i_0))$).

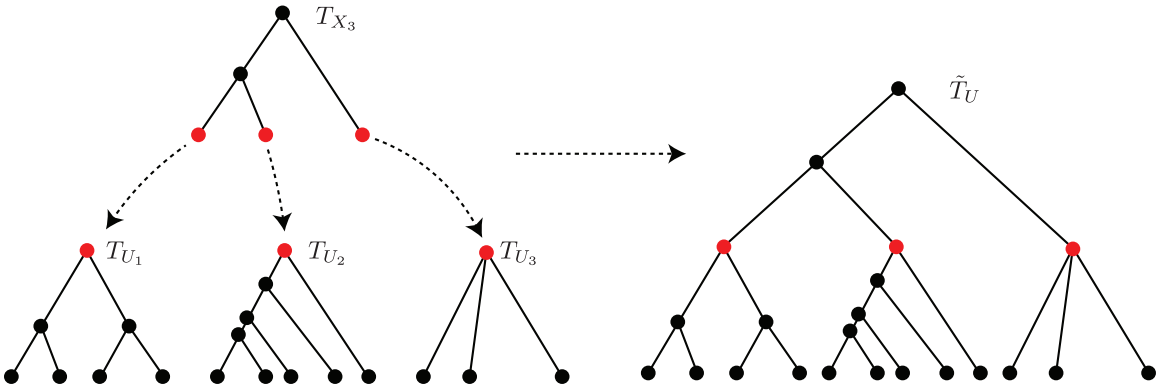


Figure 10: **Illustration of the process for producing the preliminary union $\tilde{T}_U$.**

For the modification process described above, we need to modify at most $O(k)$ pointers, and for each such pointer it takes time $O(k)$ to search for U_{j_0} with the desired representative as described above. So, the time needed for constructing the preliminary union $\tilde{T}_U$ is at most $O(k^2)$.

(III) Taming process for $\tilde{T}_U$ The TDS $\tilde{T}_U$ may not be a subspace TDS subordinate to T_X, i.e., $\tilde{T}_U$ may contain `Node` objects representing balls in V_X which do not belong to the TDS T_X. We will thus further *tame* $\tilde{T}_U$ so that the outcome TDS is subordinate to T_X. For pedagogical reasons, we use the symbol T_U (which is our final TDS representation for U) to refer to the tamed $\tilde{T}_U$.

To accomplish this, we first create an array L consisting of pointers to all nodes in $\tilde{T}_U \setminus \cup_{i=1}^{k} T_{U_i}$, i.e., all modified nodes from T_{X_k}. We sort L according to increasing `Diameter` values of the nodes referenced by its pointers. There are $O(k)$ such nodes and thus building and sorting L takes time $O(k \log(k))$. Let $\ell := \text{length}(\text{L}) = O(k)$. For each $i = 1, \ldots, \ell$, we compare $*(\text{L}(i))$ with each node in T_X as described next. If for some $B \in T_X$, the sets $B.\texttt{Children}$ and $*(\text{L}(i)).\texttt{Children}$ agree, we then replace the node $*(\text{L}(i))$ in $\tilde{T}_U$ with B. There are two cases which can arise during this replacement:

1. if $*(\text{L}(i))$ is not the root node of $*(\tilde{T}_U)$, we find the parent node of $*(\text{L}(i))$ (cf. Remark 69) in $\tilde{T}_U$ and let i_0 be the index such that $\texttt{Parent}\left(*(\text{L}(i)), \tilde{T}_U\right).\texttt{Children}(i_0) = \text{L}(i)$. Then, we replace $\texttt{Parent}\left(*(\text{L}(i)), \tilde{T}_U\right).\texttt{Children}(i_0)$ with a pointer to $B \in T_X$;

2. otherwise if $*(\mathrm{L}(i))$ is the root node $*(\tilde{T}_U)$, we free the memory used for storing $*(\tilde{T}_U)$, and then assign $\tilde{T}_U = \&B$.

For each $i = 1, \ldots, \ell$ let $k_i := \mathrm{length}(*(\mathrm{L}(i)).\texttt{Children})$. Determining whether $B.\texttt{Children} = *(\mathrm{L}(i)).\texttt{Children}$ takes time $O(k_i)$. Therefore, the time complexity of the replacement process mentioned above can be bounded as follows:

$$\sum_{i=1}^{\ell} O(\#T_X) \times k_i = \sum_{i=1}^{\ell} O(nk_i) = O\left(n \sum_{i=1}^{\ell} k_i\right) = O(nk),$$

where $n := \#X$ (we used Lemma 59 in the rightmost equality). The time incurred when finding and accessing the parent node of a given node in L is at most $O(k)$ (cf. Remark 69). So the total time required for taming $\tilde{T}_U$ is at most $O(nk)$.

Therefore by following steps (I), (II) and (III), the total time complexity of the union operation can be bounded by $O(k^2) + O(nk) = O(nk)$.

A.4 Implementation details for Algorithm 4

In this section, we provide details for one possible implementation of Algorithm 4.

A.4.1 Preprocessing

To achieve an actual implementation of Algorithm 4, some preprocessing is needed in order to construct the arrays $\mathrm{LX}^{(\varepsilon)}$ and LY defined in Section 5.2.3. Here, for completeness we describe one possible implementation of these preprocessing steps. We first construct subspace TDSs for subspaces in $\mathrm{LX}^{(\varepsilon)}$ and LY. We then construct arrays $\mathrm{pLX}^{(\varepsilon)}$ and pLY of `Node` pointers referencing subspaces in $\mathrm{LX}^{(\varepsilon)}$ and LY, respectively. For this purpose, we augment the class `Node` by incorporating an integer field called `Order` and a Boolean field called `IsBall`. For each node B in T_X, `Order` is initially set to -1 and `IsBall` is set to `True`. We initialize nodes in T_Y in the same way.

Construction of pLY Given the TDS T_Y, we first create an array pLY containing pointers to all nodes in T_Y. Then, we sort pLY according to increasing `Diameter` values of the `Node` objects referenced by its pointers. This finalizes constructing the array pLY. For each $i = 1, \ldots, \#\mathrm{pLY}$, we set $*(\mathrm{pLY}(i)).\texttt{Order} = i$. In this way, each element in T_Y is such that its field `Order` is different from -1.

By Lemma 91, $\#T_Y = \#V_Y = O(n)$. So building and sorting the list pLY can be done in time $O(n \log(n))$. The above process for setting $B^Y.\texttt{Order}$ for all $B^Y \in \mathrm{pLY}$ can be done in time $O(n)$.

Construction of $\mathrm{pLX}^{(\varepsilon)}$ Unlike the case of LY, the array $\mathrm{LX}^{(\varepsilon)}$ can contain subspaces not belonging to V_X. In order to construct $\mathrm{pLX}^{(\varepsilon)}$, we follow the three steps that we describe next:

1. Create subspace TDSs subordinate to T_X for representing each of the subspaces in $\mathrm{LX}^{(\varepsilon)}\backslash\mathrm{LX}$;
2. Create a *list* $\mathrm{pLX}^{(\varepsilon)}$ of pointers referencing the root nodes representing each of the subspaces in $\mathrm{LX}^{(\varepsilon)}$.
3. Transform the *list* $\mathrm{pLX}^{(\varepsilon)}$ into an *array* (still denoted by $\mathrm{pLX}^{(\varepsilon)}$). In this way, the random access time to elements in $\mathrm{pLX}^{(\varepsilon)}$ is $O(1)$.

Whereas the third step is clear, we will provide a detailed description for the first and second steps. In fact, these two steps are accomplished at the same time: We first construct pLX via a process analogous to the one used above for constructing pLY. This step can be done in time $O(n\log(n))$. Then, we follow two substeps: (a) for each $B^X \in \mathrm{LX}$, we will first construct subspace TDSs subordinate to T_X for all subspaces in $B^X_{(\varepsilon)}\backslash\{B^X\}$; we then construct the list $\mathrm{p}B^X_{(\varepsilon)}$ of pointers referencing all subspaces in $B^X_{(\varepsilon)}$; (b) we will use these constructions from (a) for all B^X to complete step 1 and step 2. Below, we describe substep (a) and substep (b) in detail.

(a) Constructions regarding a single B^X We set $\mathrm{p}B^X_{(\varepsilon)}$ to be an empty list. Applying **OpenPartition** (Algorithm 7) to B^X we obtain the open partition $B^X_{\mathfrak{o}\left(\delta^+_{2\varepsilon}(B^X)\right)} = \{B_i\}_{i=1}^{N_{B^X}}$. By the SGC, $N_{B^X} \le \gamma$ so that the open partition process takes time at most $O(\gamma)$. For each non-empty index set $I \subseteq [N_{B^X}]$ (there are $O(2^\gamma)$ such Is), we apply the union operation (described in Appendix A.3) to obtain the TDS T_{U_I} for the union $U_I := \cup_{i\in I} B_i$ which takes time at most $O(n \times \#I) = O(n\gamma)$. For the new nodes thus created, i.e., for nodes U in $T_{U_I}\backslash T_X$, we set $U.\texttt{IsBall} = \texttt{False}$ and set $U.\texttt{Order} = -1$. If $U_I.\texttt{Diameter} = B^X.\texttt{Diameter}$, we first set $U_I.\texttt{Order} = B^X.\texttt{Order}$ and then update $\mathrm{p}B^X_{(\varepsilon)}$ by appending T_{U_I} (which is a pointer to the node U_I) to it. Otherwise, we delete all nodes in $T_{U_I}\backslash T_X$.

In summary, the time complexity for both constructing subspace TDSs subordinate to T_X for all elements in $B^X_{(\varepsilon)}\backslash\{B^X\}$ and constructing the list $\mathrm{p}B^X_{(\varepsilon)}$ is bounded by $O(n2^\gamma\gamma^2)$.

(b) Completing step 1 and step 2 Finally, we apply the constructions described above to all $B^X \in \mathrm{LX}$ and assemble the corresponding outputs to complete step 1 and step 2 concurrently. More specifically, we first sort LX according to increasing values of $\texttt{Order}$. Then, we apply the above constructions over all $B^X \in \mathrm{LX}$ with respect to this order. After this, subspace TDSs for all elements in $\mathrm{LX}^{(\varepsilon)}\backslash\mathrm{LX}$ have been constructed and stored in memory. Finally, we merge all the resulting lists $\mathrm{p}B^X_{(\varepsilon)}$s together to obtain the list $\mathrm{pLX}^{(\varepsilon)}$. Since $\#\mathrm{LX} = O(n)$ and since for each B^X, $\mathrm{length}(\mathrm{p}B^X_{(\varepsilon)}) = O(2^\gamma)$, the time needed for constructing subspace TDSs for elements in $\mathrm{LX}^{(\varepsilon)}\backslash\mathrm{LX}$ is bounded by $O(n^2 2^\gamma\gamma^2)$, and the merging process takes time at most $O(n2^\gamma)$ (the complexity can be reduced to $O(n)$ if each $\mathrm{p}B^X_{(\varepsilon)}$ is represented by a doubly linked list).

Therefore, step 1 and step 2 together can be accomplished in time

$$O\big(n\log(n) + n^2 2^\gamma\gamma^2 + n2^\gamma\big) = O\big(n^2\log(n)2^\gamma\gamma^2\big).$$

Since $\#\mathrm{pLX}^{(\varepsilon)} = O(n2^{\gamma})$, the time for transforming $\mathrm{pLX}^{(\varepsilon)}$ into an array is bounded by $O(n2^{\gamma})$. So, the total time complexity for building the array $\mathrm{pLX}^{(\varepsilon)}$ is at most $O\big(n^2\log(n)2^{\gamma}\gamma^2\big)$.

Data structure for storing and accessing indices of elements in $\mathrm{LX}^{(\varepsilon)}$ Given any $U^X \in \mathrm{LX}^{(\varepsilon)}$, there exists a unique integer ind such that $U^X = *(\mathrm{pLX}^{(\varepsilon)}(\mathrm{ind}))$ and we refer to ind as the *index* of U^X in $\mathrm{LX}^{(\varepsilon)}$. Now, given any U^X in the form of a subspace TDS subordinate to T_X, in order to access the index of U^X in $\mathrm{LX}^{(\varepsilon)}$ efficiently, we construct a $\underbrace{(1+\#V_X)\times\cdots\times(1+\#V_X)}_{\gamma \text{ terms}}$ dimensional multi-array INDX for storing all such indices. Each dimension of INDX is indexed by integers in the range $\{-1\}\cup\{1,\ldots,\#V_X\}$. The following lemma gives rise to our strategy for indexing this multi-array.

Lemma 74. *For any subset $U^X \subseteq X$, there is a* unique *maximal set of non-intersecting closed balls $\{B_1,\ldots,B_k\}\subseteq V_X$ such that $U^X = \cup_{i=1}^{k} B_i$. Here 'maximal' means that if there exists another set of non-intersecting closed balls $\{B'_1,\ldots,B'_l\}\subseteq V_X$ such that $U^X = \cup_{i=1}^{l} B'_i$, then for each $i=1,\ldots,l$, there exists some $j=1,\ldots,k$ such that $B'_i \subseteq B_j$. We call the unique maximal set $\{B_1,\ldots,B_k\}\subseteq V_X$ the* ball decomposition of U^X.

Now, given $U^X \in \mathrm{LX}^{(\varepsilon)}$, let ind denote its index in $\mathrm{LX}^{(\varepsilon)}$. Let $\{B_1,\ldots,B_k\}$ be the ball decomposition of U^X whose elements are labeled such that

$$B_1.\texttt{Order} < \cdots < B_k.\texttt{Order}.$$

We then store ind, the index of U^X, in INDX as follows

$$\mathrm{INDX}(B_1.\texttt{Order}, B_2.\texttt{Order}, \ldots, B_k.\texttt{Order}, \underbrace{-1,\ldots,-1}_{\gamma-k \text{ terms}}) = \mathrm{ind}.$$

Computation of the ball decomposition Given $U^X \in \mathrm{LX}^{(\varepsilon)}$, we proceed to compute its ball decomposition as follows. If $U^X.\texttt{Order} \neq -1$, then $U^X \in \mathrm{LX}$, i.e., U^X already represents a ball in X. In this case, $\{U^X\}$ is the ball decomposition of U^X. Otherwise, we traverse all nodes of T_{U^X} to create the list $\mathrm{p}U^X$ consisting of pointers to all nodes $\{B_1,\ldots,B_k\}\subseteq T_{U^X}$ such that: for each $i=1,\ldots,k$, $B_i.\texttt{IsBall} = \texttt{True}$ but the parent node of B_i satisfies $\texttt{Parent}\,(B_i, T_{U^X})\,.\texttt{IsBall} = \texttt{False}$. Then, $\{B_1,\ldots,B_k\}$ is the desired ball decomposition of U^X.

The computation of the ball decomposition of $U^X \in \mathrm{LX}^{(\varepsilon)}$ described above can be done in time $O(\gamma\log(\gamma))$ (including the sorting time). Therefore, the total time complexity for storing the indices of all $U^X \in \mathrm{LX}^{(\varepsilon)}$ into INDX is bounded by $O(n2^{\gamma}\gamma\log(\gamma))$ and the time needed for finding the index of U^X into $\mathrm{LX}^{(\varepsilon)}$ is bounded by $O(\gamma\log(\gamma))$. We remark that the space complexity of the multi-array INDX is $O(n^{\gamma})$. The actual size of $\mathrm{LX}^{(\varepsilon)}$ is however $O(n2^{\gamma})$. To reduce the space complexity, one could consider a sparse multi-array data structure or binary search trees.

A.4.2 A refined union operation via an improved taming process

In line 16 of Algorithm 4 we need to construct the union space $U^X_{\Psi^{-1}(j)} := \cup_{i\in\Psi^{-1}(j)} U^X_i$ where all U^X_is are subspaces of U^X. By the SGC, $\#\Psi^{-1}(j) = O(\gamma)$. Therefore, by results in Section A.3, it takes time $O(n\gamma)$ to construct a subspace TDS (subordinate to T_X) for representing $U^X_{\Psi^{-1}(j)}$ given the subspace TDSs (subordinate to T_X) for U^X_is.

Recall from Appendix A.3 that the union operation consists of three steps where the final step, i.e., the taming process, has the leading time complexity. In this section, we provide a *refined union operation* via an *improved taming process* for constructing $U^X_{\Psi^{-1}(j)}$ such that the time complexity of this refined union operation is thus reduced to $O(\gamma^2)$. In the sequel, we use the shorthand $U := U^X_{\Psi^{-1}(j)}$ and also let $k := \#\Psi^{-1}(j) = O(\gamma)$.

Note that in Appendix A.3, the taming process for the preliminary union $\tilde{T}_U$ requires pairwise comparisons between all nodes in $\tilde{T}_U \setminus \cup_{i=1}^k T_{U_i}$ and all nodes in T_X. The fact that $\#T_X = O(n)$ explains the n factor in the time complexity bound $O(nk)$ for the taming process. However, to tame $\tilde{T}_U$, we only need to compare every node in $\tilde{T}_U \setminus \cup_{i=1}^k T_{U_i}$ with a certain *subset* of nodes in T_X. We obtain the improved taming process for $\tilde{T}_U$ by restricting the pairwise comparisons in this way and by keeping the rest of the taming process unchanged.

Now, we explain how to restrict the pairwise comparison. Let $i_0 := U^X.\texttt{Order}$ and let $B^X := \mathrm{LX}(i_0)$. Since $U^X \in \mathrm{LX}^{(\varepsilon)}$, we have that $U^X \in B^X_{(\varepsilon)}$, i.e., the set of ε-maximal unions of closed balls in B^X (cf. Section 5.2.3). Starting from the node B^X, we traverse all of its descendants B in order to identify all those for which $B.\texttt{Diameter} \geq B^X.\texttt{Diameter} - 2\varepsilon$. We let $\mathrm{L}B^X$ denote the set of all such descendants. Then, in the improved taming process of $\tilde{T}_U$, we only compare all nodes in $\tilde{T}_U \setminus \cup_{i=1}^k T_{U_i}$ with all nodes in $\mathrm{L}B^X$.

By the SGC, $\#\mathrm{L}B^X = O(\gamma)$ and thus the time complexity of the improved taming process described above is at most $O(\gamma k) = O(\gamma^2)$. Therefore, constructing T_U via this taming process has cost at most $O(\gamma^2)$.

B Extension of u_{GH} to ultra-dissimilarity spaces

In this section, we will consider the collection $\mathcal{U}^{\mathrm{dis}}$ of finite ultra-dissimilarity spaces, which are generalizations of ultrametric spaces (see also [42] for a more general notion called ultra-network).

Definition 75 (Ultra-dissimilarity space)**.** *An ultra-dissimilarity space is any pair* (X, u_X) *where* X *is a* finite *set and* $u_X : X \times X \to \mathbb{R}_{\geq 0}$ *satisfies, for all* $x, x', x'' \in X$*:*

(1) ***Symmetry:*** $u_X(x, x') = u_X(x', x)$,

(2) ***Strong triangle inequality:*** $u_X(x, x'') \leq \max\left(u_X(x, x'), u_X(x', x'')\right)$,

(3) ***Definiteness:*** $\max\left(u_X(x, x), u_X(x', x')\right) \leq u_X(x, x')$*, and the equality takes place if and only if* $x = x'$.

We refer to u_X *as the* ultra-dissimilarity *on* X*. It is obvious that any finite ultrametric space is an ultra-dissimilarity space. Then,* $\mathcal{U}^{\mathrm{fin}} \subseteq \mathcal{U}^{\mathrm{dis}}$*, where* $\mathcal{U}^{\mathrm{fin}}$ *denotes the collection of finite ultrametric spaces.*

We say two ultra-dissimilarity spaces (X, u_X) *and* (Y, u_Y) *are* isometric *if there exists a bijective function* $f : X \to Y$ *such that for any* $x, x' \in X$

$$u_Y(f(x), f(x')) = u_X(x, x').$$

Such an f *is called an* isometry*.*

Remark 76 (Informal interpretation)**.** *For each* $x \in X$*, the value* $u_X(x, x)$ *is regarded as the 'birth time' of the point* x*; when* u_X *is an actual ultrametric on* X*, all points are born at time* 0*. The value* $u_X(x, x')$ *for different points* x *and* x' *encodes the time when the two points 'merge'. Note that then condition (3) above can be informally interpreted as encoding the property that two points cannot merge before their respective birth times, and that if they merge at the same time they are born, then they are actually the* same *point.*

Given two ultra-dissimilarity spaces X and Y and any correspondence R between them, without any obstacle, we define $\mathrm{dis}_\infty(R)$ in exactly the same way by Equation (8), i.e.,

$$\mathrm{dis}_\infty(R) \coloneqq \sup_{(x,y),(x',y') \in R} \Lambda_\infty(u_X(x, x'), u_Y(y, y')).$$

Definition 77 (u_{GH} between ultra-dissimilarity spaces)**.** *For two ultra-dissimilarity spaces* X *and* Y*, we define* $u_{\mathrm{GH}}(X, Y)$ *by*

$$u_{\mathrm{GH}}(X, Y) := \inf_R \mathrm{dis}_\infty(R). \tag{17}$$

Given an ultra-dissimilarity space X, as we did in Section 2.1, we consider a notion of closed equivalence classes $[x]^X_{\mathfrak{c}(t)} \coloneqq \{x' \in X : u_X(x, x') \leq t\}$ for any $x \in X$ and $t \geq 0$. Furthermore, let

$$[\![x]\!]^X_{\mathfrak{c}(t)} \coloneqq \begin{cases} [x]^X_{\mathfrak{c}(t)} & \text{if } u_X(x, x) \leq t \\ \{x\} & \text{if } u_X(x, x) > t \end{cases} \tag{18}$$

In words, if the 'birth time' of x is no larger than t, i.e., $u_X(x, x) \leq t$, then $[\![x]\!]^X_{\mathfrak{c}(t)}$ is the same as $[x]^X_{\mathfrak{c}(t)}$. Otherwise, $[\![x]\!]^X_{\mathfrak{c}(t)}$ denotes the singleton $\{x\}$. We let $X_{\mathfrak{c}(t)} \coloneqq \{[\![x]\!]^X_{\mathfrak{c}(t)} : \forall x \in X\}$ and define by $u_{X_{\mathfrak{c}(t)}}$ an ultra-dissimilarity on $X_{\mathfrak{c}(t)}$ given by:

$$u_{X_{\mathfrak{c}(t)}}\left([\![x]\!]^X_{\mathfrak{c}(t)}, [\![x']\!]^X_{\mathfrak{c}(t)}\right) \coloneqq \begin{cases} u_X(x, x') & \text{if } [\![x]\!]^X_{\mathfrak{c}(t)} \neq [\![x']\!]^X_{\mathfrak{c}(t)}, \text{ or } x = x' \text{ and } u_X(x, x) > t \\ 0 & \text{otherwise.} \end{cases} \tag{19}$$

Definition 78 (Closed quotient on ultra-dissimilarity spaces)**.** *Given an ultra-dissimilarity space* (X, u_X) *and* $t \geq 0$*, Then, we call* $\left(X_{\mathfrak{c}(t)}, u_{X_{\mathfrak{c}(t)}}\right)$ *the* closed quotient *of* X *at level* t*.*

We are still using the notation $X_{\mathfrak{c}(t)}$ to denote the resulting quotient space as we did in the case of ultrametric spaces (Definition 10) because if (X, u_X) is actually an ultrametric space, then the new definition agrees with the one given previously.

It is obvious that for any ultra-dissimilarity space X and any $t \geq 0$, $X_{\mathfrak{c}(t)}$ is still an ultra-dissimilarity space. Then, the closed quotient gives rise to a map which we call the *t-closed quotient operator* $Q_{\mathfrak{c}(t)} : \mathcal{U}^{\mathrm{dis}} \to \mathcal{U}^{\mathrm{dis}}$ sending $X \in \mathcal{U}^{\mathrm{dis}}$ to $X_{\mathfrak{c}(t)} \in \mathcal{U}^{\mathrm{dis}}$.

Theorem 79 (Structural theorem for u_{GH} on ultra-dissimilarity spaces)**.** *For any two finite ultra-dissimilarity spaces X and Y one has that*

$$u_{\mathrm{GH}}(X,Y) = \min\left\{t \geq 0 : \left(X_{\mathfrak{c}(t)}, u_{X_{\mathfrak{c}(t)}}\right) \cong \left(Y_{\mathfrak{c}(t)}, u_{Y_{\mathfrak{c}(t)}}\right)\right\}.$$

Proof. We first prove a weaker version (with inf instead of min):

$$u_{\mathrm{GH}}(X,Y) = \inf\left\{t \geq 0 : (X_{\mathfrak{c}(t)}, u_{X_{\mathfrak{c}(t)}}) \cong (Y_{\mathfrak{c}(t)}, u_{Y_{\mathfrak{c}(t)}})\right\}. \tag{20}$$

Suppose first that $X_{\mathfrak{c}(t)} \cong Y_{\mathfrak{c}(t)}$ for some $t \geq 0$, i.e. that there exists an isometry $f_t : X_{\mathfrak{c}(t)} \to Y_{\mathfrak{c}(t)}$. Then, we define

$$R_t \coloneqq \left\{(x,y) \in X \times Y : [\![y]\!]^Y_{\mathfrak{c}(t)} = f_t([\![x]\!]^X_{\mathfrak{c}(t)})\right\}.$$

Since f_t is bijective, R_t is a correspondence between X and Y.

Then, we show that $\mathrm{dis}_\infty(R_t) \leq t$, which will imply that $u_{\mathrm{GH}}(X,Y) \leq t$. Choose any $(x,y),(x',y') \in R_t$. If $u_X(x,x') \leq t$, then $u_X(x,x') \leq \max(t, u_Y(y,y'))$. Otherwise, if $u_X(x,x') > t$, we have the following two cases:

1. $x = x'$. Then, $[\![x]\!]^X_{\mathfrak{c}(t)} = [\![x']\!]^X_{\mathfrak{c}(t)} = \{x\}$. Thus, $[\![y]\!]^Y_{\mathfrak{c}(t)} = f_t([\![x]\!]^X_{\mathfrak{c}(t)}) = f_t([\![x']\!]^X_{\mathfrak{c}(t)}) = [\![y']\!]^Y_{\mathfrak{c}(t)}$. Since f_t is isometry, we have that

   $$u_{Y_{\mathfrak{c}(t)}}\left([\![y]\!]^Y_{\mathfrak{c}(t)}, [\![y]\!]^Y_{\mathfrak{c}(t)}\right) = u_{X_{\mathfrak{c}(t)}}\left([\![x]\!]^X_{\mathfrak{c}(t)}, [\![x]\!]^X_{\mathfrak{c}(t)}\right)) = u_X(x,x) > 0.$$

   This implies that

   $$u_Y(y,y) = u_{Y_{\mathfrak{c}(t)}}\left([\![y]\!]^Y_{\mathfrak{c}(t)}, [\![y]\!]^Y_{\mathfrak{c}(t)}\right) = u_X(x,x) > t,$$

   and thus $[\![y]\!]^Y_{\mathfrak{c}(t)} = \{y\}$. Similarly, $[\![y']\!]^Y_{\mathfrak{c}(t)} = \{y'\}$ and thus $\{y\} = [\![y]\!]^Y_{\mathfrak{c}(t)} = [\![y']\!]^Y_{\mathfrak{c}(t)} = \{y'\}$. Then, we have that $y = y'$ and thus $u_Y(y,y') = u_X(x,x')$.

2. $x \neq x'$. Then, $[\![x]\!]^X_{\mathfrak{c}(t)} \neq [\![x']\!]^X_{\mathfrak{c}(t)}$. Since f_t is an isometry, we have that

   $$[\![y]\!]^Y_{\mathfrak{c}(t)} = f_t\left([\![x]\!]^X_{\mathfrak{c}(t)}\right) \neq f_t\left([\![x']\!]^X_{\mathfrak{c}(t)}\right) = [\![y']\!]^Y_{\mathfrak{c}(t)}$$

   and thus $u_Y(y,y') = u_{Y_{\mathfrak{c}(t)}}\left([\![y]\!]^Y_{\mathfrak{c}(t)}, [\![y']\!]^Y_{\mathfrak{c}(t)}\right) = u_{X_{\mathfrak{c}(t)}}\left([\![x]\!]^X_{\mathfrak{c}(t)}, [\![x']\!]^X_{\mathfrak{c}(t)}\right) = u_X(x,x')$.

Therefore, $u_X(x,x') \leq \max(t, u_Y(y,y'))$. Similarly, $u_Y(y,y') \leq \max(t, u_X(x,x'))$. Then, we have that $\mathrm{dis}_\infty(R_t) \leq t$ and thus $u_{\mathrm{GH}}(X,Y) \leq \inf\left\{t \geq 0 : X_{\mathfrak{c}(t)} \cong Y_{\mathfrak{c}(t)}\right\}$.

Conversely, let R be a correspondence between X an Y and let $t := \mathrm{dis}_\infty(R)$. Define a map $f : X \to Y$ by taking $x \in X$ to an arbitrary $y \in Y$ such that $(x,y) \in R$. Consider the induced quotient map $f_t : X_{\mathfrak{c}(t)} \to Y_{\mathfrak{c}(t)}$, defined by $f_t\left([\![x]\!]^X_{\mathfrak{c}(t)}\right) = [\![f(x)]\!]^Y_{\mathfrak{c}(t)}$. We now show that f_t is well-defined. For any $(x,y),(x',y') \in R$ such that $[\![x']\!]^X_{\mathfrak{c}(t)} = [\![x]\!]^X_{\mathfrak{c}(t)}$, we have the following two cases:

1. $u_X(x,x') \leq t$. Then, since $\mathrm{dis}_\infty(R) = t$, we have that $u_Y(y',y) \leq \max\left(t, u_X(x,x')\right) \leq t$.
2. $u_X(x,x) > t$ and $x = x'$. Then, since $\mathrm{dis}_\infty(R) = t$, $u_Y(y,y') = u_X(x,x') = u_X(x,x) > t$. Similarly, $u_Y(y,y) = u_X(x,x) = u_X(x',x') = u_Y(y',y')$. Therefore, $y = y'$ by condition (3) of the definition of ultra-dissimilarity spaces (cf. Definition 75).

Therefore, $[\![y]\!]^Y_{\mathfrak{c}(t)} = [\![y']\!]^Y_{\mathfrak{c}(t)}$, which implies that f_t is well-defined. Similarly, the quotient map $g_t : Y_{\mathfrak{c}(t)} \to X_{\mathfrak{c}(t)}$ induced by a map $g : Y \to X$ such that $g(y) = x$ where $x \in X$ is chosen such that $(x,y) \in R$ is well-defined. It is clear that g_t is the inverse of f_t and thus f_t is bijective. Now we show that f_t is an isometry. Choose $[\![x]\!]^X_{\mathfrak{c}(t)}, [\![x']\!]^X_{\mathfrak{c}(t)} \in X_{\mathfrak{c}(t)}$ and let $y = f(x)$ and $y' = f(x')$. Let $s := u_{X_{\mathfrak{c}(t)}}\left([\![x]\!]^X_{\mathfrak{c}(t)}, [\![x']\!]^X_{\mathfrak{c}(t)}\right)$. If $[\![x]\!]^X_{\mathfrak{c}(t)} \neq [\![x']\!]^X_{\mathfrak{c}(t)}$, then $s > t$ and thus $u_X(x,x') = s$. Since $\mathrm{dis}_\infty(R) = t < s$, $u_Y(y,y')$ is forced to be equal to s and thus

$$u_{Y_{\mathfrak{c}(t)}}\left([\![f(x)]\!]^Y_{\mathfrak{c}(t)}, [\![f(x')]\!]^Y_{\mathfrak{c}(t)}\right) = u_{Y_{\mathfrak{c}(t)}}\left([\![y]\!]^Y_{\mathfrak{c}(t)}, [\![y']\!]^Y_{\mathfrak{c}(t)}\right) = s = u_{X_{\mathfrak{c}(t)}}\left([\![x]\!]^X_{\mathfrak{c}(t)}, [\![x']\!]^X_{\mathfrak{c}(t)}\right).$$

If $[\![x]\!]^X_{\mathfrak{c}(t)} = [\![x']\!]^X_{\mathfrak{c}(t)}$, then we have the following two cases.

1. $u_X(x,x) \leq t$. Then, $[\![x]\!]^X_{\mathfrak{c}(t)} = [\![x']\!]^X_{\mathfrak{c}(t)}$ implies that $u_X(x,x') \leq t$ and $u_{X_{\mathfrak{c}(t)}}\left([\![x]\!]^X_{\mathfrak{c}(t)}, [\![x']\!]^X_{\mathfrak{c}(t)}\right) = 0$. Since $(x,f(x)) \in R$ and $\mathrm{dis}_\infty(R) \leq t$, we have that $u_Y(f(x),f(x)) \leq \max(t, u_X(x,x)) \leq t$. Then,
$$u_{Y_{\mathfrak{c}(t)}}\left([\![f(x)]\!]^Y_{\mathfrak{c}(t)}, [\![f(x')]\!]^Y_{\mathfrak{c}(t)}\right) = u_{Y_{\mathfrak{c}(t)}}\left([\![f(x)]\!]^Y_{\mathfrak{c}(t)}, [\![f(x)]\!]^Y_{\mathfrak{c}(t)}\right) = 0 = u_X\left([\![x]\!]^X_{\mathfrak{c}(t)}, [\![x']\!]^X_{\mathfrak{c}(t)}\right).$$
2. $u_X(x,x) > t$. Then, $x = x'$ and $u_{X_{\mathfrak{c}(t)}}\left([\![x]\!]^X_{\mathfrak{c}(t)}, [\![x']\!]^X_{\mathfrak{c}(t)}\right) = u_X(x,x) > t$. Since $(x,f(x)) \in R$ and $\mathrm{dis}_\infty(R) \leq t$, we have $\Lambda_\infty(u_X(x,x), u_Y(f(x),f(x))) \leq t$, which implies that $u_Y(f(x),f(x)) = u_X(x,x) > t$. Then,
$$\begin{aligned} &u_{Y_{\mathfrak{c}(t)}}\left([\![f(x)]\!]^Y_{\mathfrak{c}(t)}, [\![f(x')]\!]^Y_{\mathfrak{c}(t)}\right) = u_{Y_{\mathfrak{c}(t)}}\left([\![f(x)]\!]^Y_{\mathfrak{c}(t)}, [\![f(x)]\!]^Y_{\mathfrak{c}(t)}\right) = u_X(x,x) \\ =&u_{X_{\mathfrak{c}(t)}}\left([\![x]\!]^X_{\mathfrak{c}(t)}, [\![x]\!]^X_{\mathfrak{c}(t)}\right) = u_{X_{\mathfrak{c}(t)}}\left([\![x]\!]^X_{\mathfrak{c}(t)}, [\![x']\!]^X_{\mathfrak{c}(t)}\right). \end{aligned}$$

Therefore, f_t is an isometry and thus $u_{\mathrm{GH}}(X,Y) \geq \inf\left\{t \geq 0 : X_{\mathfrak{c}(t)} \cong Y_{\mathfrak{c}(t)}\right\}$.

Now, since X is finite, it is obvious that for each $t \geq 0$, there exists $\varepsilon > 0$ such that whenever $s \in [t, t+\varepsilon]$, we have that $X_{\mathfrak{c}(t)} \cong X_{\mathfrak{c}(s)}$. Therefore, the infimum of Equation (20) is attained which concludes the proof. □

Analogously to the case of ultrametric spaces, the structural theorem (Theorem 79) for u_{GH} on the collection $\mathcal{U}^{\mathrm{dis}}$ of all finite ultra-dissimilarity spaces allows us to devise an algorithm similar to Algorithm 1 for computing u_{GH} between ultra-dissimilarity spaces. The argument for the complexity analysis of Algorithm 1 can be adapted to show that the time complexity of computing u_{GH} on $\mathcal{U}^{\mathrm{dis}}$ is still $O(n\log(n))$.

Graphical representations of ultra-dissimilarity spaces As shown in Theorem 15, ultrametric spaces are equivalent to dendrograms. Analogously, ultra-dissimilarity spaces can be viewed as certain objects named *treegrams*. To define treegrams, we first introduce a notion called subpartitions: given a set X, a partition P' of a subset $X' \subseteq X$ is called a *subpartition*. We denote by $\mathrm{SubPart}(X)$ the collection of all subpartitions of X. For any subpartitions P_1 and P_2, we say P_1 is *coarser* than P_2 if any block in P_2 is contained in some block in P_1.

Example 80 (Examples of subpartitions)**.**

1. *Note that the empty set $P = \emptyset$ is a subpartition of any set X.*
2. *Given a finite set X, let $P = \{B_1, \ldots, B_n\}$ be a partition. Then, for any non-empty subset $X' \subseteq X$, we obtain a subpartition $P|_{X'}$ by restricting P to X' as follows: $P|_{X'} = \{B_1 \cap X', \ldots, B_n \cap X'\} \backslash \{\emptyset\}$.*

Definition 81 (Treegrams)**.** *A treegram θ_X over a finite set X is a function $\theta_X : [0,\infty) \to \mathrm{SubPart}(X)$ satisfying the following conditions:*

(1) For $0 \leq s < t$, $\theta_X(t)$ is coarser than $\theta_X(s)$.

(2) There exists $t_X > 0$ such that $\theta_X(t_X) = \{X\}$.

(3) For any $r \geq 0$, there exists $\varepsilon > 0$ such that $\theta_X(r) = \theta_X(t)$ for $t \in [r, r+\varepsilon]$.

(4) For each $x \in X$, there exists $t \geq 0$ such that $\{x\} \in \theta_X(t)$ is a block.

Our definition is a slight modification of treegrams defined in [42, 23] where the domain of treegrams is the entire real line $\mathbb{R}$ instead of $\mathbb{R}_{\geq 0}$.

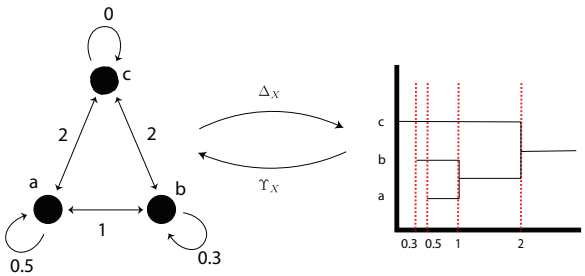


Figure 11: **Treegrams and ultra-dissimilarity spaces**.

Fix a finite set X, denote by $\mathcal{U}^{\mathrm{dis}}(X)$ the collection of all ultra-dissimilarities over X and denote by $\mathcal{T}(X)$ the collection of all treegrams over X. We define a map Δ_X :

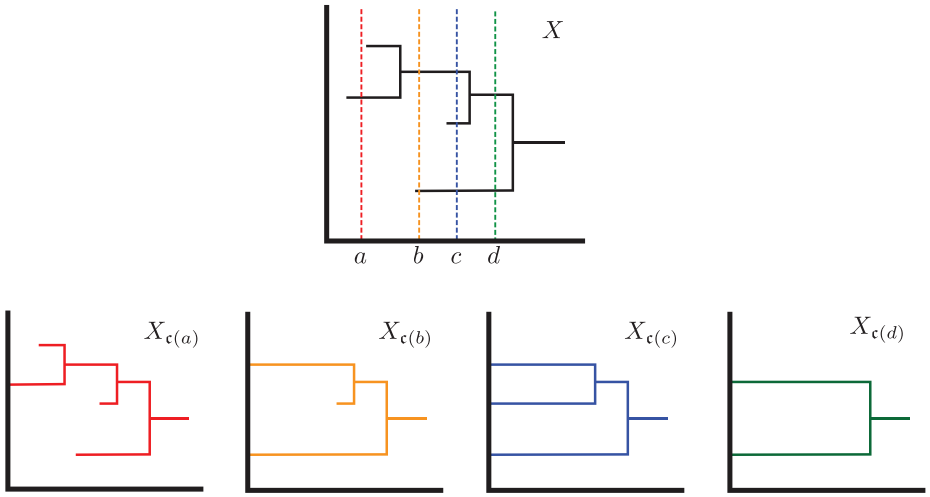


Figure 12: **Illustration of Definition 78.** We represent a 4-point ultra-dissimilarity space X as a treegram in the first row of the figure. The second row shows the treegrams corresponding to $X_{\mathfrak{c}(t)}$ at different levels t.

$\mathcal{U}^{\mathrm{dis}}(X) \to \mathcal{T}(X)$ by sending u_X to a treegram θ_X as follows: for each $t \geq 0$, let $\hat{X}_t := \{x \in X : u_X(x,x) \leq t\}$ and let $\theta_X(t) := \left\{[x]^X_{\mathfrak{c}(t)} : x \in \hat{X}_t\right\}$. Note in particular that θ_X satisfies condition (4) of Definition 81 due to the definiteness of u_X (cf. condition (3) in Definition 75). Conversely, we define a map $\Upsilon_X : \mathcal{T}(X) \to \mathcal{U}^{\mathrm{dis}}(X)$ as follows. Let $\theta_X \in \mathcal{T}(X)$. Then, we define an ultra-dissimilarity $u_X := \Upsilon_X(\theta_X)$ on X by:

$$u_X(x,x') := \inf\left\{t \geq 0 : [x]^{\theta_X}_t = [x']^{\theta_X}_t\right\}, \quad \forall x, x' \in X,$$

where $[x]^{\theta_X}_t \in \theta_X(t)$ denotes the block containing x. Note that definiteness of u_X follows from condition (4) of Definition 81. Then, in analogy to Theorem 15, we have the following theorem. See Figure 11 for an illustration.

Theorem 82. *Given any finite set X, $\Delta_X : \mathcal{U}^{\mathrm{dis}}(X) \to \mathcal{T}(X)$ is bijective with inverse $\Upsilon_X : \mathcal{T}(X) \to \mathcal{U}^{\mathrm{dis}}(X)$.*

It is obvious that the collection of dendrograms $\mathcal{D}(X)$ over X is a proper subset of $\mathcal{T}(X)$ and that the collection of ultrametrics $\mathcal{U}(X)$ over X is a proper subset of $\mathcal{U}^{\mathrm{dis}}(X)$. Then, Theorem 82 is actually an extension/generalization of Theorem 15.

Via Theorem 82, one can easily represent an ultra-dissimilarity space via a treegram. In particular, see Figure 12 for an illustration of the closed quotient operator via treegrams.

C Relegated proofs

C.1 Proofs from Section 3

Proof of Proposition 23. For any correspondence R between X and Y, we have

$$\begin{aligned}\mathrm{dis}_1(R,u_X,u_Y) &= \sup_{(x,y),(x',y')\in R} |u_X(x,x') - u_Y(y,y')| \\ &= \sup_{(x,y),(x',y')\in R} \left| \left((u_X)^{\frac{1}{p}}(x,x')\right)^p - \left((u_Y)^{\frac{1}{p}}(y,y')\right)^p \right| \\ &= \left(\mathrm{dis}_p\left(R,(u_X)^{\frac{1}{p}},(u_Y)^{\frac{1}{p}}\right)\right)^p\end{aligned}$$

Therefore, by Equation (1) and Equation (9), we have

$$d_{\mathrm{GH}}(X,Y) = \left(d_{\mathrm{GH}}^{(p)}\left(S_{\frac{1}{p}}(X), S_{\frac{1}{p}}(Y)\right)\right)^p.$$

Similarly, $\mathrm{dis}_p\left(R,u_X,u_Y\right) = \left(\mathrm{dis}_1\left(R,(u_X)^p,(u_Y)^p\right)\right)^{\frac{1}{p}}$ and thus $d_{\mathrm{GH}}^{(p)}(X,Y) = \left(d_{\mathrm{GH}}(S_p(X),S_p(Y))\right)^{\frac{1}{p}}$. □

Proof of Theorem 29. Let Z be an ultrametric space such that there exist isometric embeddings $\varphi_X : X \hookrightarrow Z$ and $\varphi_Y : Y \hookrightarrow Z$. Let $\eta \coloneqq d_{\mathrm{H}}^Z(X,Y)$, where we identify X and Y with $\varphi_X(X), \varphi_Y(Y) \subseteq Z$, respectively. Define $R \coloneqq \{(x,y) \in X \times Y : u_Z(x,y) \leq \eta\}$. That R is a correspondence between X and Y follows from the condition that $d_{\mathrm{H}}^Z(X,Y) = \eta$ and compactness of X and Y. Now, consider any $(x,y),(x',y') \in R$. Without loss of generality, we assume that $u_X(x,x') \geq u_Y(y,y')$. If $u_X(x,x') = u_Y(y,y')$, then $\Lambda_\infty(u_X(x,x'),u_Y(y,y')) = 0$. So we further assume that $u_X(x,x') > u_Y(y,y')$. Then,

$$\begin{aligned}&\Lambda_\infty(u_X(x,x'),u_Y(y,y')) = u_X(x,x') = u_Z(x,x') \\ \leq &\max(u_Z(x,y),u_Z(y,y'),u_Z(y',x')) \\ \leq &\max(u_Z(x,y),u_Z(x',y')) \leq \eta.\end{aligned}$$

In the second inequality we used the assumption that $u_X(x,x') > u_Y(y,y')$. Thus, by taking supremum over all pairs $(x,y),(x',y') \in R$, one has $\mathrm{dis}_\infty(R) \leq \eta$. Then, we obtain that $u_{\mathrm{GH}}(X,Y) = \inf_R \mathrm{dis}_\infty(R) \leq d_{\mathrm{GH}}^{(\infty)}(X,Y)$.

For the reverse inequality, let R be an arbitrary correspondence between X and Y. Let $\eta \coloneqq \mathrm{dis}_\infty(R)$. Define a function $u : X \sqcup Y \times X \sqcup Y \to \mathbb{R}_{\geq 0}$ as follows:

1. $u|_{X\times X} \coloneqq u_X$ and $u|_{Y\times Y} \coloneqq u_Y$;
2. for any $(x,y) \in X \times Y$, $u(x,y) \coloneqq \inf_{(x',y')\in R} \max(u_X(x,x'),u_Y(y',y),\eta)$;
3. for any $(y,x) \in Y \times X$, $u(y,x) \coloneqq u(x,y)$.

Now we show that u is an ultrametric on the disjoint union $X \sqcup Y$. Because of the symmetric roles of X and Y, we only need to verify the following two cases:

Case 1: $\forall x, x' \in X$ and $\forall y \in Y$, $u(x,y) \leq \max(u(x,x'), u(x',y))$;

Case 2: $\forall x, x' \in X$ and $\forall y \in Y$, $u(x,x') \leq \max(u(x,y), u(x',y))$.

For Case 1,

$$\begin{aligned}\max(u(x,x'),u(x',y)) &= \max\left(u(x,x'), \inf_{(x_1,y_1)\in R} \max(u_X(x',x_1), u_Y(y_1,y), \eta)\right)\\ &= \inf_{(x_1,y_1)\in R} \max\left(u(x,x'), u_X(x',x_1), u_Y(y_1,y), \eta\right)\\ &\geq \inf_{(x_1,y_1)\in R} \max\left(u_X(x,x_1), u_Y(y_1,y), \eta\right)\\ &= u(x,y).\end{aligned}$$

For Case 2,

$$\begin{aligned}&\max\left(\inf_{(x_1,y_1)\in R} \max\left(u_X(x,x_1), u_Y(y_1,y), \eta\right), \inf_{(x_2,y_2)\in R} \max\left(u_X(x_2,x'), u_Y(y_2,y), \eta\right)\right)\\ =&\inf_{(x_1,y_1),(x_2,y_2)\in R} \max\left(u_X(x,x_1), u_Y(y_1,y), \eta, u_X(x_2,x'), u_Y(y_2,y), \eta\right)\\ \geq&\inf_{(x_1,y_1),(x_2,y_2)\in R} \max\left(u_X(x,x_1), u_X(x_2,x'), u_Y(y_1,y_2), \eta\right)\\ \geq&\inf_{(x_1,y_1),(x_2,y_2)\in R} \max\left(u_X(x,x_1), u_X(x_2,x'), u_X(x_1,x_2)\right) \geq u_X(x,x') = u(x,x').\end{aligned}$$

The second inequality above follows from the fact that $\Lambda_\infty(u_X(x_1,x_2), u_Y(y_1,y_2)) \leq \mathrm{dis}_\infty(R) = \eta$ and Equation (7).

Note that $u(x,y) = \eta$ for every $(x,y) \in R$. Therefore, $d_{\mathrm{GH}}^{(\infty)}(X,Y) \leq d_{\mathrm{H}}^{(X \sqcup Y, u)}(X,Y) = \eta$. This implies that $d_{\mathrm{GH}}^{(\infty)}(X,Y) \leq \inf_R \mathrm{dis}_\infty(R) = u_{\mathrm{GH}}(X,Y)$. □

C.2 Proofs from Section 5

C.2.1 Proof of Theorem 37 (Time complexity of Algorithm FindCorrRec (Algorithm 2))

Lemma 83 (Inheritance of the FGC)**.** *Let X and Y be finite ultrametric spaces such that $X, Y \in \mathcal{U}_1(\varepsilon,\gamma)$. Assume that $0 \leq \mathrm{diam}(Y) - \mathrm{diam}(X) \leq \varepsilon$ and that $\mathrm{diam}(Y) > \varepsilon$. Write[5] $X_{\mathfrak{o}(\delta_\varepsilon(Y))} = \{X_i\}_{i\in[N_X]}$ and $Y_{\mathfrak{o}(\delta_0(Y))} = \{Y_j\}_{j\in[N_Y]}$. Then, given any surjection $\Psi : [N_X] \twoheadrightarrow [N_Y]$, for each $j \in [N_Y]$, we have that $X_{\Psi^{-1}(j)} \in \mathcal{U}_1(\varepsilon,\gamma)$ and $Y_j \in \mathcal{U}_1(\varepsilon,\gamma)$.*

Proof. We first prove that $Y_j \in \mathcal{U}_1(\varepsilon,\gamma)$. For any $y \in Y_j$, note that $Y_j = [y]^Y_{\mathfrak{o}(\delta_0(Y))}$. Given any $t \geq \varepsilon$, if $t \leq \delta_0(Y)$, then we easily see that $[y]^{Y_j}_{\mathfrak{o}(t-\varepsilon)} = [y]^Y_{\mathfrak{o}(t-\varepsilon)}$. Then,

$$\#[y]^{Y_j}_{\mathfrak{c}(t)} \leq \underbrace{\#[y]^Y_{\mathfrak{c}(t)} \leq \gamma \cdot \#[y]^Y_{\mathfrak{o}(t-\varepsilon)}}_{\text{FGC}} = \gamma \cdot \#[y]^{Y_j}_{\mathfrak{o}(t-\varepsilon)}.$$

[5] Recall that $\delta_\varepsilon(Y) := \mathrm{diam}(Y) - \varepsilon$ and in particular, $\delta_0(Y) = \mathrm{diam}(Y)$.

Otherwise, if $t > \delta_0(Y)$, we then have that $[y]^{Y_j}_{\mathfrak{c}(t)} = Y_j = [y]^{Y_j}_{\mathfrak{c}(\delta_0(Y))}$. Then,

$$\#[y]^{Y_j}_{\mathfrak{c}(t)} = \#[y]^{Y_j}_{\mathfrak{c}(\delta_0(Y))} \leq \gamma \cdot \#[y]^{Y_j}_{\mathfrak{o}(\delta_0(Y)-\varepsilon)} \leq \gamma \cdot \#[y]^{Y_j}_{\mathfrak{o}(t-\varepsilon)},$$

where the first inequality follows from the previous case $t \leq \delta_0(Y)$. Therefore, $Y_j \in \mathcal{U}_1(\varepsilon, \gamma)$.

We then prove that $X_{\Psi^{-1}(j)} \in \mathcal{U}_1(\varepsilon, \gamma)$. For $X_{\Psi^{-1}(j)}$, we choose any $x \in X_{\Psi^{-1}(j)}$ and any $t \geq \varepsilon$. If $t \leq \delta_0(X)$, then it is easy to see that $t - \varepsilon < \delta_\varepsilon(Y)$. Therefore, $[x]^{X_{\Psi^{-1}(j)}}_{\mathfrak{o}(t-\varepsilon)} = [x]^{X}_{\mathfrak{o}(t-\varepsilon)}$ and thus

$$\#[x]^{X_{\Psi^{-1}(j)}}_{\mathfrak{c}(t)} \leq \underbrace{\#[x]^{X}_{\mathfrak{c}(t)} \leq \gamma \cdot \#[x]^{X}_{\mathfrak{o}(t-\varepsilon)}}_{\text{FGC}} = \gamma \cdot \#[x]^{X_{\Psi^{-1}(j)}}_{\mathfrak{o}(t-\varepsilon)}.$$

If $t > \delta_0(X)$, then we have that $[x]^{X_{\Psi^{-1}(j)}}_{\mathfrak{c}(t)} = X_{\Psi^{-1}(j)} = [x]^{X_{\Psi^{-1}(j)}}_{\mathfrak{c}(\delta_0(X))}$. Then,

$$\#[x]^{X_{\Psi^{-1}(j)}}_{\mathfrak{c}(t)} = \#[x]^{X_{\Psi^{-1}(j)}}_{\mathfrak{c}(\delta_0(X))} \leq \gamma \cdot \#[x]^{X_{\Psi^{-1}(j)}}_{\mathfrak{o}(\delta_0(X)-\varepsilon)} \leq \gamma \cdot \#[x]^{X_{\Psi^{-1}(j)}}_{\mathfrak{o}(t-\varepsilon)},$$

where the first inequality follows from the previous case $t \leq \delta_0(X)$. Therefore, $X_{\Psi^{-1}(j)} \in \mathcal{U}_1(\varepsilon, \gamma)$.

□

In this way, the inputs to each subproblem encountered while running Algorithm 2 will satisfy the first (ε, γ)-growth condition. This justifies the analysis presented below regarding the number and the size of subproblems.

Proof of Theorem 37. We are going to invoke the master theorem [12] in order to analyze the complexity of our recursive algorithm (Algorithm 2).

Number of subproblems By the FGC (or Remark 36), we have that $\max(N_X, N_Y) \leq \gamma$. There will be at most γ^γ surjections $\Psi : [N_X] \twoheadrightarrow [N_Y]$. For each such a surjection, Algorithm 2 inspects N_Y $(\leq \gamma)$ subproblems. Therefore, there are at most $\gamma^{\gamma+1}$ subproblems.

Size of a subproblem Fix a surjection $\Psi : [N_X] \twoheadrightarrow [N_Y]$. For each $k \in [N_Y]$, we write $Y_k \coloneqq [y_k]^{Y}_{\mathfrak{o}(\delta_0(Y))}$ for some $y_k \in Y$. Then, for a fixed $j \in [N_Y]$, we have that

$$\#Y = \#[y_j]^{Y}_{\mathfrak{o}(\delta_0(Y))} + \sum_{k \in [N_Y] \setminus \{j\}} \#[y_k]^{Y}_{\mathfrak{o}(\delta_0(Y))} \geq \#Y_j + \frac{M-1}{\gamma} \cdot \#Y,$$

where the last inequality follows from the FGC and the fact that $[y_k]^{Y}_{\delta_0(Y)+\varepsilon} = Y$ for each $k \in [N_Y]$. Therefore, $\#Y_j \leq \left(1 - \frac{M-1}{\gamma}\right) \cdot \#Y$. Now, regarding $X_{\Psi^{-1}(j)}$, since Ψ is a surjection and $M \geq 2$, there exists $i \notin \Psi^{-1}(j)$ such that $X_i \cap X_{\Psi^{-1}(j)} = \emptyset$. Assume $X_i = [x_i]^{X}_{\mathfrak{o}(\delta_\varepsilon(Y))}$ for some $x_i \in X$. Then,

$$\#[x_i]^{X}_{\mathfrak{o}(\delta_\varepsilon(Y))} \geq \frac{\#[x_i]^{X}_{\mathfrak{c}(\delta_\varepsilon(Y)+\varepsilon)}}{\gamma} \geq \frac{\#[x_i]^{X}_{\mathfrak{o}(\delta_\varepsilon(Y)+\varepsilon)}}{\gamma} \geq \frac{\#[x_i]^{X}_{\mathfrak{c}(\delta_\varepsilon(Y)+2\varepsilon)}}{\gamma^2} = \frac{\#X}{\gamma^2},$$

where the last equality follows from the fact that $\delta_\varepsilon(Y) + 2\varepsilon = \operatorname{diam}(Y) + \varepsilon > \operatorname{diam}(X)$. Therefore,

$$\#X_{\Psi^{-1}(j)} \leq \#X - \#X_i \left(1 - \frac{1}{\gamma^2}\right) \#X.$$

When $\gamma \geq 1$, we have that $1 - \frac{1}{\gamma^2} \geq 1 - \frac{M-1}{\gamma}$. So, $\max(\#X_{\Psi^{-1}(j)}, \#Y_j) \leq \left(1 - \frac{1}{\gamma^2}\right) n$ and thus the size of a subproblem is bounded above by $\left(1 - \frac{1}{\gamma^2}\right) n$.

Base case complexity and other work If (X, Y, ε) is one of the base cases, it takes time at most $O(n^2)$ in order to either directly generate a correspondence or 0. In order to utilize the results from the subproblems, we need at most $O(n^2)$ time to construct distance matrices u_X and u_Y from TDSs in order to implement the unions $X_{\Psi^{-1}(j)}$ (cf. Appendix A). It then takes time at most $O(n^2)$ in total to construct all the unions $X_{\Psi^{-1}(j)}$, R_j and R, and to transpose R when/if needed.

Conclusion Denote by $W(n)$ the time complexity of the algorithm where $n = \max(\#X, \#Y)$. Then,

$$W(n) \leq \gamma^{\gamma+1} \cdot W\left(\frac{n}{\gamma^2/(\gamma^2-1)}\right) + O\left(n^2\right).$$

Since by assumption that $\gamma \geq 2$, the critical exponent $\log_{\frac{\gamma^2}{\gamma^2-1}} \gamma^{\gamma+1}$ is strictly greater than 2. Therefore, by the master theorem we have that $W(n) = O\left(n^{(\gamma+1)\log_{\mathrm{b}(\gamma)} \gamma}\right)$. This concludes the proof. □

C.2.2 Proof of Proposition 40

In the following lemmas, *we will always assume that X and Y are finite ultrametric spaces and that there exists $\varepsilon \geq 0$ such that* $\operatorname{diam}(X) > \varepsilon$ *and* $\operatorname{diam}(Y) \leq \varepsilon$.

Lemma 84. *There exists an ε-correspondence between X and Y if and only if there exists an ε-correspondence between $X_{\mathfrak{c}(\varepsilon)}$ and Y.*

Proof. Suppose that R is an ε-correspondence between X and Y. Then, we define the set $R_\varepsilon \subseteq X_{\mathfrak{c}(\varepsilon)} \times Y$ as follows:

$$R_\varepsilon \coloneqq \left\{ \left([x]^X_{\mathfrak{c}(\varepsilon)}, y \right) \in X_{\mathfrak{c}(\varepsilon)} \times Y : (x, y) \in R \right\}.$$

It is easy to verify that R_ε is a correspondence between $X_{\mathfrak{c}(\varepsilon)}$ and Y. Now, for any two pairs

$\left([x]^X_{\mathfrak{c}(\varepsilon)}, y\right), \left([x']^X_{\mathfrak{c}(\varepsilon)}, y'\right) \in R_\varepsilon$, we have the following:

$$\left|u_{X_{\mathfrak{c}(\varepsilon)}}\left([x]^X_{\mathfrak{c}(\varepsilon)}, [x']^X_{\mathfrak{c}(\varepsilon)}\right) - u_Y(y,y')\right| = \begin{cases} |u_X(x,x') - u_Y(y,y')| & \text{if } [x]^X_{\mathfrak{c}(\varepsilon)} \neq [x']^X_{\mathfrak{c}(\varepsilon)} \\ u_Y(y,y') & \text{if } [x]^X_{\mathfrak{c}(\varepsilon)} = [x']^X_{\mathfrak{c}(\varepsilon)} \end{cases}$$
$$\leq \begin{cases} \mathrm{dis}_1(R) & \text{if } [x]^X_{\mathfrak{c}(\varepsilon)} \neq [x']^X_{\mathfrak{c}(\varepsilon)} \\ \mathrm{diam}(Y) & \text{if } [x]^X_{\mathfrak{c}(\varepsilon)} = [x']^X_{\mathfrak{c}(\varepsilon)} \end{cases} \leq \varepsilon$$

Therefore, $\mathrm{dis}_1(R_\varepsilon) \leq \varepsilon$ and thus R_ε is an ε-correspondence between $X_{\mathfrak{c}(\varepsilon)}$ and Y.

Now for the converse, suppose that there exists an ε-correspondence R_ε between $X_{\mathfrak{c}(\varepsilon)}$ and Y. Then, we define $R \subseteq X \times Y$ as follows:

$$R \coloneqq \left\{(x,y) \in X \times Y : \left([x]^X_{\mathfrak{c}(\varepsilon)}, y\right) \in R\right\}.$$

It is easy to verify that R is a correspondence between X and Y. Then, for any $(x,y),(x',y') \in R$, if $[x]^X_{\mathfrak{c}(\varepsilon)} \neq [x']^X_{\mathfrak{c}(\varepsilon)}$, we have

$$|u_X(x,x') - u_Y(y,y')| \leq \left|u_{X_{\mathfrak{c}(\varepsilon)}}\left([x]^X_{\mathfrak{c}(\varepsilon)}, [x']^X_{\mathfrak{c}(\varepsilon)}\right) - u_Y(y,y')\right| \leq \mathrm{dis}_1(R_\varepsilon) \leq \varepsilon.$$

If $[x]^X_{\mathfrak{c}(\varepsilon)} = [x']^X_{\mathfrak{c}(\varepsilon)}$, then $u_X(x,x') \leq \varepsilon$. Moreover, $u_Y(y,y') \leq \mathrm{diam}(Y) \leq \varepsilon$. Then, $|u_X(x,x') - u_Y(y,y')| \leq \varepsilon$. Therefore, $\mathrm{dis}_1(R) \leq \varepsilon$ and thus R is an ε-correspondence between X and Y. This concludes the proof. □

Lemma 85. *For any ε-correspondence R_ε between $X_{\mathfrak{c}(\varepsilon)}$ and Y, there exists a surjection ψ from Y to $X_{\mathfrak{c}(\varepsilon)}$ such that $R_\varepsilon = \{(\psi(y), y) : y \in Y\}$.*

Proof. We first show that if $\left([x]^X_{\mathfrak{c}(\varepsilon)}, y\right), ([x']^X_{\mathfrak{c}(\varepsilon)}, y) \in R_\varepsilon$, then $[x]^X_{\mathfrak{c}(\varepsilon)} = [x']^X_{\mathfrak{c}(\varepsilon)}$. Otherwise, suppose that $[x]^X_{\mathfrak{c}(\varepsilon)} \neq [x']^X_{\mathfrak{c}(\varepsilon)}$, which is equivalent to the condition that $u_X(x,x') > \varepsilon$. Then,

$$\varepsilon \geq \mathrm{dis}_1(R_\varepsilon) \geq \left|u_{X_{\mathfrak{c}(\varepsilon)}}\left([x]^X_{\mathfrak{c}(\varepsilon)}, [x']^X_{\mathfrak{c}(\varepsilon)}\right) - u_Y(y,y)\right| = u_X(x,x') > \varepsilon,$$

which is a contradiction! Then, R_ε naturally induces a well-defined map $\psi : Y \to X_{\mathfrak{c}(\varepsilon)}$ taking $y \in Y$ to $[x]^X_{\mathfrak{c}(\varepsilon)}$ such that $\left([x]^X_{\mathfrak{c}(\varepsilon)}, y\right) \in R_\varepsilon$. It is easy to check that ψ is surjective and that $R_\varepsilon = \{(\psi(y), y) : y \in Y\}$. □

Recall from Section 5.2.5 that $\mathrm{sep}(X) \coloneqq \min\{d_X(x,x') : x, x' \in X \text{ and } x \neq x'\}$ denotes the *separation* of a finite metric space (X, d_X).

Lemma 86. *Assume that $\mathrm{sep}(X) > \varepsilon$. Then any injective map $\varphi : X \to Y$ with $\mathrm{dis}_1(\varphi) \leq \varepsilon$ induces an ε-correspondence between X and Y.*

Proof. Suppose $X = \{x_1, \ldots, x_n\}$ and $\mathrm{im}(\varphi) = \{y_1, \ldots, y_n\}$ where $y_i = \varphi(x_i)$ for every $i = 1, \ldots, n$. For any $y \in Y$, define

$$i_y \coloneqq \min\left\{\operatorname*{argmin}_{j=1,\ldots,n} u_Y(y, y_j)\right\}.$$

Obviously, $i_{y_j} = j$. Then, we define $R \subseteq X \times Y$ as follows:

$$R \coloneqq \{(x_{i_y}, y) \in X \times Y : \forall y \in Y\}.$$

It is easy to check that R is a correspondence. Now, we verify that $\mathrm{dis}_1(R) \leq \varepsilon$. Let $(x_i, y), (x_j, y') \in R$, where $i = i_y$ and $j = i_{y'}$. If $i = j$, then $|u_X(x_i, x_i) - u_Y(y, y')| = u_Y(y, y') \leq \mathrm{diam}(Y) \leq \varepsilon$. Now assume $i \neq j$. It is obvious that $(x_i, y_i), (x_j, y_j) \in R$ and thus $u_X(x_i, x_j) - u_Y(y_i, y_j) \leq \mathrm{dis}_1(\varphi) \leq \varepsilon$. Since $u_X(x_i, x_j) - u_Y(y, y') \geq \mathrm{sep}(X) - \mathrm{diam}(Y) \geq 0$, the inequality $|u_X(x_i, x_j) - u_Y(y, y')| \leq \varepsilon$ follows from the following observation:

Claim 87. *For $y, y' \in Y$, if $i_y \neq i_{y'}$, then $u_Y(y, y') \geq u_Y(y_i, y_j)$ where $i \coloneqq i_y$ and $j \coloneqq i_{y'}$.*

Proof of Claim 87. Suppose otherwise that $u_Y(y, y') < u_Y(y_i, y_j)$. If $u_Y(y_i, y) \leq u_Y(y, y')$, then

$$u_Y(y', y_i) \leq \max(u_Y(y, y'), u_Y(y, y_i)) \leq u_Y(y, y').$$

By definition of $j = i_{y'}$, we have that $u_Y(y_j, y') \leq u_Y(y_i, y') \leq u_Y(y, y')$. Then, $u_Y(y_i, y_j) \leq \max(u_Y(y_i, y'), u_Y(y', y_j)) \leq u_Y(y, y')$, which is a contradiction. Therefore, $u_Y(y_i, y) > u_Y(y, y')$ and similarly $u_Y(y_j, y') > u_Y(y, y')$. Then, by the strong triangle inequality we have that $u_Y(y, y_i) = u_Y(y', y_i)$ and $u_Y(y, y_j) = u_Y(y', y_j)$. By definition of $i = i_y$ and $j = i_{y'}$, we have that

$$u_Y(y, y_j) = u_Y(y', y_j) \leq u_Y(y', y_i) = u_Y(y, y_i),$$

which implies that $j \in \mathrm{argmin}_{k=1,\ldots,n} u_Y(y, y_k)$ and thus $j > i$. However, we can similarly prove that $i > j$, which is a contradiction. Therefore, $u_Y(y, y') \geq u_Y(y_i, y_j)$. □

□

Proof of Proposition 40. By Lemma 84, we only need to prove that there exists an ε-correspondence between $X_{\mathfrak{c}(\varepsilon)}$ and Y if and only if there exists an injective map $\varphi : X_{\mathfrak{c}(\varepsilon)} \to Y$ with $\mathrm{dis}_1(\varphi) \leq \varepsilon$.

Assuming the existence of such a correspondence R_ε, then by Lemma 85, there exists a surjection $\psi : Y \twoheadrightarrow X_{\mathfrak{c}(\varepsilon)}$ such that $R_\varepsilon = \{(\psi(y), y) : y \in Y\}$. Then, we construct an injective map $\varphi : X_{\mathfrak{c}(\varepsilon)} \to Y$ by mapping each $[x]^X_{\mathfrak{c}(\varepsilon)} \in X_{\mathfrak{c}(\varepsilon)}$ to y, where y is arbitrarily chosen from $\psi^{-1}\left([x]^X_{\mathfrak{c}(\varepsilon)}\right)$. Then, $\mathrm{dis}_1(\varphi) \leq \mathrm{dis}_1(\psi) \leq \varepsilon$.

Now, assume that there exists an injective map $\varphi : X_{\mathfrak{c}(\varepsilon)} \to Y$ with $\mathrm{dis}_1(\varphi) \leq \varepsilon$. Obviously, we have $\mathrm{sep}(X_{\mathfrak{c}(\varepsilon)}) > \varepsilon$, and thus, by Lemma 86, there exists a correspondence R_ε between $X_{\mathfrak{c}(\varepsilon)}$ and Y with $\mathrm{dis}_1(R_\varepsilon) \leq \varepsilon$. □

C.2.3 Proof of Remark 38

We first establish the following characterization of elements in $V_X^{(\varepsilon)}$.

Lemma 88. *Let X be a finite ultrametric space and let $\varepsilon \geq 0$. Then, a subset $U^X \subseteq X$ belongs to $V_X^{(\varepsilon)}$ if and only if U^X contains all $x \in X$ such that $u_X\left(x, U^X\right) < \mathrm{diam}\left(U^X\right) - 2\varepsilon$, where $u_X\left(x, U^X\right) \coloneqq \min\left\{u_X(x, x') : x' \in U^X\right\}$.*

Proof. If $U^X \in V_X^{(\varepsilon)}$, then U^X is an ε-maximal union of closed balls in $B^X \subseteq X$ (cf. Section 5.2.3): write $B^X_{\mathfrak{o}\left(\delta^+_{2\varepsilon}(B^X)\right)} = \{B_1^X, \ldots, B_N^X\}$, where $\delta^+_{2\varepsilon}\left(B^X\right) \coloneqq \max\left(\operatorname{diam}\left(B^X\right) - 2\varepsilon, 0\right)$; then, $U^X = \bigcup_{i\in I} B_i^X$ for some non-empty $I \subseteq \{1, \ldots, N\}$ and U^X satisfies that $\operatorname{diam}\left(U^X\right) = \operatorname{diam}\left(B^X\right)$. Without loss of generality, we assume that $\operatorname{diam}\left(U^X\right) - 2\varepsilon > 0$. Given any $x \in X$ such that $u_X\left(x, U^X\right) < \operatorname{diam}\left(U^X\right) - 2\varepsilon$, there exists $x' \in U^X$ such that $u_X(x, x') < \operatorname{diam}\left(U^X\right) - 2\varepsilon = \operatorname{diam}(B^X) - 2\varepsilon$. Since $x' \in U^X \subseteq B^X$, $x' \in B_i^X$ for some $i \in I$. Then,

$$B_i^X = [x']^X_{\mathfrak{o}\left(\delta^+_{2\varepsilon}(B^X)\right)} = \{x'' \in X : u_X(x', x'') < \operatorname{diam}(B^X) - 2\varepsilon\}.$$

Therefore, $x \in B_i^X \subseteq U^X$.

Now, let $U^X \subseteq X$ be a subset containing all $x \in X$ such that $u_X\left(x, U^X\right) < \operatorname{diam}\left(U^X\right) - 2\varepsilon$. For any $x \in U^X$, consider the ball $B^X \coloneqq [x]^X_{\mathfrak{c}(\delta)}$, where $\delta \coloneqq \operatorname{diam}\left(U^X\right)$. It is obvious that $U^X \subseteq B^X$ and $\operatorname{diam}(U^X) = \operatorname{diam}(B^X)$. If $\operatorname{diam}\left(U^X\right) \leq 2\varepsilon$, then $\delta^+_{2\varepsilon}(B^X) = 0$. Therefore, $B^X_{\mathfrak{o}\left(\delta^+_{2\varepsilon}(B^X)\right)} = \{\{x\} : x \in B^X\}$ and thus obviously, $U^X \in B^X_{(\varepsilon)} \subseteq V_X^{(\varepsilon)}$. Now, assume that $\operatorname{diam}\left(U^X\right) > 2\varepsilon$. Then, $\delta^+_{2\varepsilon}(B^X) = \operatorname{diam}(B^X) - 2\varepsilon = \operatorname{diam}\left(U^X\right) - 2\varepsilon$. By assumption we have that for each $x \in U^X$

$$[x]^X_{\mathfrak{o}\left(\delta^+_{2\varepsilon}(B^X)\right)} = \{x' \in X : u_X(x, x') < \operatorname{diam}\left(U^X\right) - 2\varepsilon\} \subseteq U^X.$$

This implies that

$$U^X = \bigcup_{x\in U^X} [x]^X_{\mathfrak{o}\left(\delta^+_{2\varepsilon}(B^X)\right)}.$$

So U^X is the union of some elements in $B^X_{\mathfrak{o}\left(\delta^+_{2\varepsilon}(B^X)\right)}$ and thus $U^X \in B^X_{(\varepsilon)} \subseteq V_X^{(\varepsilon)}$. □

Proof of Remark 38. Let $U^X \in V_X^{(\varepsilon)}$ and let $B^Y \in V_Y$ be such that $|\operatorname{diam}\left(U^X\right) - \operatorname{diam}\left(B^Y\right)| \leq \varepsilon$ and $\operatorname{diam}\left(B^Y\right) > \varepsilon$. Let $U^X_{\mathfrak{o}(\delta_\varepsilon(B^Y))} = \left\{U_1^X, \ldots, U^X_{N_{U^X}}\right\}$. For any subset $I \subseteq [N_{U^X}]$ (which can be a singleton), we will prove next that the union $U_I^X \coloneqq \bigcup_{i\in I} U_i^X$ belongs to $V_X^{(\varepsilon)}$. For any $x \in X$, suppose that there is $x' \in U_i^X \subseteq U_I^X$ such that $u_X(x, x') < \operatorname{diam}(U_I^X) - 2\varepsilon \leq \operatorname{diam}\left(U^X\right) - 2\varepsilon$. Then, $x \in U^X$ since $U^X \in V_X^{(\varepsilon)}$ (cf. Lemma 88). Now, $u_X(x, x') < \operatorname{diam}(U_I^X) - 2\varepsilon \leq \operatorname{diam}\left(U^X\right) - 2\varepsilon \leq \delta_\varepsilon\left(B^Y\right)$. So x and x' belong to the same block in $U^X_{\mathfrak{o}(\delta_\varepsilon(B^Y))}$ and thus $x \in U_i^X$. Therefore, $x \in U_I^X$ and thus by Lemma 88 we have that $U_I^X \in V_X^{(\varepsilon)}$. □

C.2.4 Proof of Theorem 41 (correctness of Algorithm FindCorrDP (Algorithm 4))

Proof. We prove a more general result, namely that for any $\left(U^X, B^Y\right) \in V_X^{(\varepsilon)} \times V_Y$, $\operatorname{DYN}\left(U^X, B^Y\right) = 1$ if and only if there exists an ε-correspondence between U^X and B^Y.

If $\left(U^X, B^Y\right)$ belongs to one of the base cases, the statement is obviously true. For non-base cases, we prove the claim by induction on $\operatorname{diam}\left(B^Y\right) \in \operatorname{spec}(Y)$. For this, we

exploit the fact that the spectrum $\mathrm{spec}(Y) = \{0 = t_0 < \cdots < t_M = \mathrm{diam}(Y)\}$ is a finite set. When $\mathrm{diam}\left(B^Y\right) = t_0 = 0$, for any U^X, $\left(U^X, B^Y\right)$ belongs to one of the base cases, so the statement holds true trivially. Let $1 < i_0 \leq M$ and suppose that the claim holds true for all t_i when $i < i_0$ and all $\mathrm{DYN}\left(U^X, B^Y\right)$ are known whenever $\mathrm{diam}\left(B^Y\right) < t_{i_0}$. Then, the induction step follows directly from Theorem 5 and Proposition 40. We elaborate this via the two cases described in page 27 as follows:

1. If $\mathrm{diam}\left(B^Y\right) > \varepsilon$, Algorithm **FindCorrDP** partitions U^X and B^Y to obtain $U^X_{\mathfrak{o}(\delta_\varepsilon(B^Y))} = \{U_i^X\}_{i\in[N_{U^X}]}$ and $B^Y_{\mathfrak{o}(\delta_0(B^Y))} = \{B_j^Y\}_{j\in[N_{B^Y}]}$, respectively. It is obvious that $B_j^Y \in V_Y$ for each $j \in [N_{B^Y}]$, and by Remark 38 we know that $U_i^X \in V_X^{(\varepsilon)}$ for each $i \in [N_{U^X}]$. Since $\mathrm{diam}(B_j^Y) < \mathrm{diam}\left(B^Y\right) = t_{i_0}$ for each $j \in [N_{B^Y}]$, by the induction assumption, the value $\mathrm{DYN}\left(U^X_{\Psi^{-1}(j)}, B_j^Y\right)$ has already been computed for any surjection $\Psi : [N_{U^X}] \to [N_{B^Y}]$ so we already know whether or not there exists any ε-correspondence between $U^X_{\Psi^{-1}(j)}$ and B_j^Y. $\mathrm{DYN}\left(U^X, B^Y\right)$ is then determined via Theorem 5: this value is 1 if there exists an ε-correspondence between U^X and B^Y, and is 0 otherwise.
2. If $\mathrm{diam}\left(B^Y\right) \leq \varepsilon$, Algorithm **FindCorrDP** assigns the value $\mathbf{FindCorrSmall}(U^X, B^Y, \varepsilon)$ to $\mathrm{DYN}\left(U^X, B^Y\right)$. Then, due to Proposition 40, $\mathrm{DYN}\left(U^X, B^Y\right) = 1$ if and only if there exists an ε-correspondence between U^X and B^Y.

Since we know that X and Y are at the end of the arrays $\mathrm{LX}^{(\varepsilon)}$ and LY, respectively, then $\mathrm{DYN}(\mathrm{END}, \mathrm{END}) = 1$ if and only if there exists an ε-correspondence between X and Y. □

C.2.5 Proof of Remark 45

Proof. Let X be a finite ultrametric space such that $X \in \mathcal{U}_2(\varepsilon, \gamma)$ for some $\varepsilon \geq 0$ and $\gamma \geq 1$. For any $x \in X$ and any $r > \mathrm{diam}(X)$, we have that $B_r(x) = X = B_{\mathrm{diam}(X)}(x)$. So if $B_{\mathrm{diam}(X)}(x)$ can be covered by K many balls with radius $\frac{\mathrm{diam}(X)}{2}$, then it is obvious that $B_r(x)$ can also be covered by K balls with radius $\frac{r}{2} > \frac{\mathrm{diam}(X)}{2}$. Therefore, to determine the doubling constant of X, we only need to consider a radius r within the range $(0, \mathrm{diam}(X)]$. Let $k = \lfloor \frac{r}{4\varepsilon} \rfloor + 1$, then k is the unique integer such that $r - 2\varepsilon \cdot k < \frac{r}{2} \leq r - 2\varepsilon \cdot (k-1)$. We assume that $r - 2\varepsilon \cdot k \geq 0$ (the case when $r - 2\varepsilon \cdot k < 0$ can be proved similarly and we omit the details). Then, by the SGC we have that

$$\begin{aligned}
\#\left\{[x']_{\mathfrak{c}\left(\frac{r}{2}\right)} : x' \in [x]_{\mathfrak{c}(r)}\right\} &\leq \#\left\{[x']_{\mathfrak{c}(r-2\varepsilon\cdot k)} : x' \in [x]_{\mathfrak{c}(r)}\right\} \leq \#\left\{[x']_{\mathfrak{o}(r-2\varepsilon\cdot k)} : x' \in [x]_{\mathfrak{c}(r)}\right\} \\
&\leq \gamma \cdot \#\left\{[x']_{\mathfrak{c}(r-2\varepsilon\cdot(k-1))} : x' \in [x]_{\mathfrak{c}(r)}\right\} \leq \gamma \cdot \#\left\{[x']_{\mathfrak{o}(r-2\varepsilon\cdot(k-1))} : x' \in [x]_{\mathfrak{c}(r)}\right\} \\
&\leq \cdots \leq \gamma^{k-1} \cdot \#\left\{[x']_{\mathfrak{c}(r-2\varepsilon)} : x' \in [x]_{\mathfrak{c}(r)}\right\} \leq \gamma^k.
\end{aligned}$$

Since $B_r(x) = [x]_{\mathfrak{c}(r)}$ and $B_{\frac{r}{2}}(x') = [x']_{\mathfrak{c}\left(\frac{r}{2}\right)}$, we have that $B_r(x)$ can be covered by a union of at most γ^k balls with radius $\frac{r}{2}$: $[x]_{\mathfrak{c}(r)} = \cup_{x'\in[x]_{\mathfrak{c}(r)}}[x']_{\mathfrak{c}\left(\frac{r}{2}\right)}$. Since $k \leq \lfloor \frac{\mathrm{diam}(X)}{4\varepsilon} \rfloor + 1$ for each $r \in (0, \mathrm{diam}(X)]$, we have that X is $\gamma^{\lfloor \frac{\mathrm{diam}(X)}{4\varepsilon} \rfloor + 1}$-doubling.

Conversely, suppose that X is K-doubling. Then, for any $x \in X$ and $t > 0$, there exist $x_1, \ldots, x_n$ such that $n \leq K$ and $B_t(x) \subseteq \cup_{i=1}^n B_{\frac{t}{2}}(x_i)$. Without loss of generality, we assume that $B_{\frac{t}{2}}(x_i) \cap B_t(x) \neq \emptyset$ for each $i = 1, \ldots, n$. Then, by Proposition 8, we have that $[x_i]_{\mathfrak{c}\left(\frac{t}{2}\right)} = B_{\frac{t}{2}}(x_i) \subseteq B_t(x) = [x]_{\mathfrak{c}(t)}$. Therefore, $[x_i]_{\mathfrak{c}\left(\frac{t}{2}\right)} \subseteq [x_i]_{\mathfrak{o}(t)} \subseteq [x]_{\mathfrak{c}(t)}$. Since $[x]_{\mathfrak{c}(t)} \subseteq \cup_{i=1}^n [x_i]_{\mathfrak{c}\left(\frac{t}{2}\right)} \subseteq \cup_{i=1}^n [x_i]_{\mathfrak{o}(t)}$, we have that

$$\#\left\{[x']_{\mathfrak{o}(t)} : x' \in [x]_{\mathfrak{c}(t)}\right\} \leq n \leq K.$$

This implies that $X \in \mathcal{U}_2(0, K)$. □

C.2.6 Proof of Theorem 46 (Time complexity of Algorithm FindCorrDP (Algorithm 4))

Lemma 89 (Time complexity of Algorithm **FindCorrSmall** (Algorithm 3))**.** *Algorithm 3 runs in time $O(n^2 n^n)$ where $n := \max(\#X, \#Y)$.*

Proof. **ClosedQuotient**(X, ε) runs in time $O(n)$ (cf. Appendix A). There are at most n^n injective maps and for each injective map $\Phi : X_{\mathfrak{c}(\varepsilon)} \to Y$, we need $O(n^2)$ time to compute $\mathrm{dis}_1(\Phi)$. Therefore, Algorithm 3 runs in time bounded by $O(n^2 n^n)$. □

Lemma 90 (Inheritance of the SGC)**.** *If X satisfies the second (ε, γ)-growth condition, then so does each $U^X \in V_X^{(\varepsilon)}$ and in particular, so does each ball $B^X \in V_X$.*

Proof. Let $U^X \in V_X^{(\varepsilon)}$. Fix a $t \geq 2\varepsilon$ and $x \in U^X$. Note that for two distinct points $x', x'' \in [x]^{U^X}_{\mathfrak{c}(t)}$, $[x']^{U^X}_{\mathfrak{o}(t-2\varepsilon)} \neq [x'']^{U^X}_{\mathfrak{o}(t-2\varepsilon)}$ if and only if $u_{U^X}(x', x'') = u_X(x', x'') \geq t - 2\varepsilon$. This is then also equivalent to the condition $[x']^X_{\mathfrak{o}(t-2\varepsilon)} \neq [x'']^X_{\mathfrak{o}(t-2\varepsilon)}$. Then, we have that

$$\#\left\{[x']^{U^X}_{\mathfrak{o}(t-2\varepsilon)} : x' \in [x]^{U^X}_{\mathfrak{c}(t)}\right\} \leq \#\left\{[x']^X_{\mathfrak{o}(t-2\varepsilon)} : x' \in [x]^X_{\mathfrak{c}(t)}\right\} \leq \gamma.$$

This concludes the proof that $U^X \in \mathcal{U}_2(\varepsilon, \gamma)$. □

Lemma 91. *Let X and Y be two finite ultrametric spaces. Then, $\#V_X = O(\#X)$ and $\#V_Y = O(\#Y)$. If $X \in \mathcal{U}_2(\varepsilon, \gamma)$, then $\#V_X^{(\varepsilon)} = O(\#X \cdot 2^\gamma)$.*

Proof. By Remark 63, we have that $\#V_X = O(\#X)$ and $\#V_Y = O(\#Y)$.

$V_X^{(\varepsilon)}$ is defined in Section 5.2.3 as: $V_X^{(\varepsilon)} := \bigcup_{B^X \in V_X} B^X_{(\varepsilon)}$. For notational simplicity, we let $\delta^+_{2\varepsilon} := \delta^+_{2\varepsilon}(B^X)$. Each $B^X_{(\varepsilon)}$ is a collection of unions of elements in $B^X_{\mathfrak{o}\left(\delta^+_{2\varepsilon}\right)}$ and thus $B^X_{(\varepsilon)}$ is a subset of the power set $2^{B^X_{\mathfrak{o}\left(\delta^+_{2\varepsilon}\right)}}$. Since B^X is a closed ball in X, $B^X = [x]^X_{\mathfrak{c}(\delta_0(B^X))}$ for some $x \in X$. By the second (ε, γ)-growth condition, we have that

$$\#B^X_{\mathfrak{o}\left(\delta^+_{2\varepsilon}\right)} = \#\left\{[x']^X_{\mathfrak{o}\left(\delta^+_{2\varepsilon}\right)} : x' \in [x]^X_{\mathfrak{c}(\delta_0(B^X))}\right\} \leq \gamma$$

and thus $\#2^{B^X_{\mathfrak{o}\left(\delta^+_{2\varepsilon}\right)}} \leq 2^\gamma$. Then, by $\#V_X = O(\#X)$, we have $\#V_X^{(\varepsilon)} = O(\#X \cdot 2^\gamma)$. □

Proof of Theorem 46. **Preprocessing.** In order to implement the union operation (cf. Appendix A.3) efficiently, we will reconstruct the distance matrices u_X and u_Y from the TDSs T_X and T_Y, respectively. This process takes time at most $O(n^2)$ (cf. Remark 67). $\mathrm{LX}^{(\varepsilon)}$ and LY can be constructed in time $O\left(n^2 \log(n) 2^\gamma \gamma^2\right)$ (cf. Appendix A.4). We create an all-zero matrix DYN of size $\#\mathrm{LX}^{(\varepsilon)} \times \#\mathrm{LY}$ in time $O(n^2 2^\gamma)$.

Main part of the algorithm. For each $B^Y \in \mathrm{LY}$, we have the following cases for $U^X \in \mathrm{LX}^{(\varepsilon)}$:

1. $|\operatorname{diam}\left(U^X\right) - \operatorname{diam}\left(B^Y\right)| > \varepsilon$ or $\max(\operatorname{diam}\left(U^X\right), \operatorname{diam}\left(B^Y\right)) \leq \varepsilon$: It takes constant time to assign either 0 or 1 to $\mathrm{DYN}(U^X, B^Y)$ based on this.

2. $\operatorname{diam}\left(B^Y\right) \leq \varepsilon < \operatorname{diam}\left(U^X\right)$: In this case, both $\operatorname{diam}\left(U^X\right)$ and $\operatorname{diam}\left(B^Y\right)$ are bounded above by 2ε (since the pair $\left(U^X, B^Y\right)$ does not satisfy the condition in the first case). Then, by the SGC and Lemma 90, it is easy to check that $\#U^X, \#B^Y \leq \gamma$. Thus, by Lemma 89, Algorithm **FindCorrSmall** with input (U^X, B^Y, ε) runs in time $O(\gamma^2 \gamma^\gamma)$.

3. $\operatorname{diam}\left(B^Y\right) > \varepsilon$: In this case, by the SGC and Lemma 90, it takes at most $O(\gamma)$ time to partition both U^X and B^Y via Algorithm **OpenPartition** (Algorithm 7) into at most γ blocks, respectively. We then have at most γ^γ surjections to consider. Given any such surjection Ψ, for each $j \in [N_Y]$ let $k_j \coloneqq \#\{U_i^X\}_{i \in \Psi^{-1}(j)}$. Then, it takes time at most $O(k_j^2 + \gamma k_j) = O(\gamma k_j)$ to construct the union $U^X_{\Psi^{-1}(j)}$ via the refined union operation discussed in Appendix A.3 (see also Appendix A.4.2). It takes time at most $O(\gamma \log(\gamma))$ to find the index of $U^X_{\Psi^{-1}(j)}$ in $\mathrm{LX}^{(\varepsilon)}$ and constant time to find the index of B_j^Y in LY (cf. Appendix A.4.1). Therefore, accessing the value $\mathrm{DYN}\left(U^X_{\Psi^{-1}(j)}, B_j^Y\right)$ has cost at most $O(\gamma \log(\gamma))$. Then, the time complexity for accessing values in DYN for the surjection Ψ is at most

$$\sum_j O(\gamma k_j \log(\gamma)) = O(\gamma^2 \log(\gamma)),$$

where we use the fact that $\sum_j k_j = O(\gamma)$. Therefore, the total time complexity of this case is bounded by $O(\gamma^{\gamma+2} \log(\gamma))$

Therefore for a single $B^Y \in \mathrm{LY}$, completing all the operations taking place between line 4 and line 22 of Algorithm 4 requires at most time $O(n 2^\gamma) \times O(\gamma^{\gamma+2} \log(\gamma)) = O\left(n 2^\gamma \gamma^{\gamma+2} \log(\gamma)\right)$. Then, completing the for-loop in line 3 of Algorithm 4 requires requires at most time

$$O(n) \times O\left(n 2^\gamma \gamma^{\gamma+2} \log(\gamma)\right) = O\left(n^2 2^\gamma \gamma^{\gamma+2} \log(\gamma)\right)$$

operations to fill out the matrix DYN.

Total time complexity. By combining the time complexity of the preprocessing part, we have that the total time complexity of Algorithm 4 is bounded by

$$O\left(n^2 \log(n) 2^\gamma \gamma^2\right) + O(n^2 2^\gamma) + O\left(n^2 2^\gamma \gamma^{\gamma+2} \log(\gamma)\right) = O\left(n^2 \log(n) 2^\gamma \gamma^{\gamma+2}\right).$$

This concludes the proof. □

C.2.7 Proof of Remark 50

Let $\mathcal{E}(X,Y) = \{\varepsilon_0 < \varepsilon_1 < \ldots < \varepsilon_M\}$. Observe that for each ε_i, since the number of all vertices in T_X or T_Y are bounded above by $2n$, $\gamma_{\varepsilon_i}(X,Y) := \max(\gamma_{\varepsilon_i}(X), \gamma_{\varepsilon_i}(Y))$ takes values in $\{1,2,\ldots,2n\}$. It is obvious that for each $k \in \{1,2,\ldots,2n\}$, the set of ε_is such that $\gamma_{\varepsilon_i}(X,Y) = k$ is an interval, i.e., a consecutive subsequence of $\mathcal{E}(X,Y)$, denoted by $[\varepsilon_{\ell_k}, \ldots, \varepsilon_{r_k}]$. Note that $\varepsilon := 2d_{\mathrm{GH}}(X,Y) \in \mathcal{E}(X,Y)$. Then, there exists $i \in \{0,1,\ldots,M\}$ such that $\varepsilon_i = \varepsilon$. It is obvious that i is the smallest index such that there exists a ε_i-correspondence between X and Y. In order to find the index i (and thus to compute $d_{\mathrm{GH}}(X,Y)$), we apply the same procedure as in the proof of [15, Theorem 5] to search in $\{1,\ldots,2n\}$ for the smallest number k^* such that $\{\varepsilon_{\ell_{k^*}}, \ldots, \varepsilon_{r_{k^*}}\} \cap [\varepsilon, \infty)$ is non-empty. This k^* satisfies the condition $k^* = \gamma_\varepsilon(X,Y)$ and $\ell_{k^*} \leq i \leq r_{k^*}$. Then, we apply binary search to find the index i. Via an argument similar to the one stated in [15, Theorem 5], this whole process for finding i can be completed in time $O\left(n^2 \log^3(n) 2^{2k^*} (2k^*)^{2k^*+2}\right)$. Since $X, Y \in \mathcal{U}_2(\varepsilon, \gamma)$, we have that $\gamma \geq \gamma_\varepsilon(X,Y) = k^*$. Therefore, we conclude that the exact value $d_{\mathrm{GH}}(X,Y)$ can be computed in time complexity at most $O\left(n^2 \log^3(n) 2^{2\gamma} (2\gamma)^{2\gamma+2}\right)$.

C.2.8 Proof of Lemma 54

Proof. Pick any positive real number $t \geq 2\varepsilon$. By Proposition 52 we have that $\|d_X - u_X^*\|_\infty = \delta$. Then, $B_t^{u_X^*}(x) \subseteq B_{t+\delta}^{d_X}(x)$ for any $x \in X$ and $t \geq 0$. Here $B_t^d(x) := \{x' \in X : d(x,x') \leq t\}$ represents the closed ball centered at x with radius t with respect to the metric d. For later use, we let $B_{\mathfrak{o}(t)}^d(x) := \{x' \in X : d(x,x') < t\}$ denote an open ball. Since $u_X^* \leq d_X$, we have that $B_t^{d_X}(x) \subseteq B_t^{u_X^*}(x)$.

We first assume that $\frac{t+\delta}{2} < t - 2\varepsilon$. Since (X, d_X) is K-doubling, there exist $x_1, \ldots, x_K \in X$ such that $B_{t+\delta}^{d_X}(x) \subseteq \bigcup_{i=1}^K B_{\frac{t+\delta}{2}}^{d_X}(x_i)$. For each x_i, we have that

$$B_{\frac{t+\delta}{2}}^{d_X}(x_i) \subseteq B_{\frac{t+\delta}{2}}^{u_X^*}(x_i) \subseteq B_{\mathfrak{o}(t-2\varepsilon)}^{u_X^*}(x_i),$$

where the last inclusion follows from the assumption that $\frac{t+\delta}{2} < t - 2\varepsilon$. Then, we have that

$$B_t^{u_X^*}(x) \subseteq B_{t+\delta}^{d_X}(x) \subseteq \bigcup_{i=1}^K B_{\frac{t+\delta}{2}}^{d_X}(x_i) \subseteq \bigcup_{i=1}^K B_{\mathfrak{o}(t-2\varepsilon)}^{u_X^*}(x_i).$$

Using the notation for open and closed equivalence relations for ultrametric spaces, we conclude that $[x]_{\mathfrak{c}(t)} \subseteq \bigcup_{i=1}^K [x_i]_{\mathfrak{o}(t-2\varepsilon)}$ which implies that

$$\#\left\{[x']_{\mathfrak{o}(t-2\varepsilon)} : x' \in [x]_{\mathfrak{c}(t)}\right\} \leq K.$$

Now, we assume that $\frac{t+\delta}{2} \geq t - 2\varepsilon$ (equivalently $t \leq \delta + 4\varepsilon$). First note that $s := \mathrm{sep}(X, d_X) = \mathrm{sep}(X, u_X)$. Then, if $t + \delta < s$, we have that $B_t^{u_X^*}(x) = B_t^{d_X}(x) = \{x\}$. Hence,

$$\#\left\{[x']_{\mathfrak{o}(t-2\varepsilon)} : x' \in [x]_{\mathfrak{c}(t)}\right\} \leq 1.$$

Otherwise, we assume that $t+\delta \geq s$. Then, we let $k \in \mathbb{N}$ be such that $\frac{t+\delta}{2^k} < s \leq \frac{t+\delta}{2^{k-1}}$. Equivalently, we have

$$k-1 \leq \log_2\left(\frac{t+\delta}{s}\right) < k.$$

By the K-doubling property of (X, d_X), we have that $B^{d_X}_{t+\delta}(x)$ can be covered by at most K^k many balls with radius $\frac{t+\delta}{2^k}$. Since $\frac{t+\delta}{2^k} < s$, these balls are singletons and thus $\#B^{d_X}_{t+\delta}(x) \leq K^k$. Therefore,

$$\#B^{u^*_X}_t(x) \leq \#B^{d_X}_{t+\delta}(x) \leq K^k \leq K^{\log_2(\frac{t+\delta}{s})+1} \leq K^{\log_2(\frac{2\delta+4\varepsilon}{s})+1},$$

where we use the assumption $t \leq \delta + 4\varepsilon$ in the last inequality. Consequently,

$$\#\left\{[x']_{\mathfrak{o}(t-2\varepsilon)} : x' \in [x]_{\mathfrak{c}(t)}\right\} \leq K^{\log_2(\frac{2\delta+4\varepsilon}{s})+1}.$$

In conclusion, for any $x \in X$ and any $r \geq 2\varepsilon$ we have that

$$\#\left\{[x']_{\mathfrak{o}(t-2\varepsilon)} : x' \in [x]_{\mathfrak{c}(t)}\right\} \leq \max\left(K, K^{\log_2(\frac{2\delta+4\varepsilon}{s})+1}\right),$$

and thus $(X, u^*_X) \in \mathcal{U}_2\left(\varepsilon, \max\left(K, K^{\log_2(\frac{2\delta+4\varepsilon}{s})+1}\right)\right)$. □